Interfacing circuits for Closed-Loop Suspended Gate FET(CLIP-SGFET) based accelerometer

A dissertation submitted in partial fulfillment of
the requirements for the degree of

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by

Ganga K M
(Roll No. EE21MS010)

Under the guidance of
Dr. Naveen Kadayinti
and
Dr. Seena V

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भा. प्रौ. सं. धारवाड
I.I.T. DHARWAD

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Ganga K M
Roll No. EE21MS010

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Abstract

Suspended Gate FETs (SGFETs) have been explored for various applications in recent years as a better alternative to existing inertial sensors. Extensive research is being conducted and reported on device physics, fabrication techniques, and characterization. This work advances the idea, discussing the design of interface electronics and closed-loop system for SGFET-based accelerometers.

SGFETs have a suspended gate attached to a proof mass that moves in response to the applied external acceleration, modulating the current flowing between the source and drain. A simple current-biased differential amplifier is designed for interfacing SGFETs, generating an output voltage proportional to the current in the SGFETs. This constitutes an open-loop accelerometer, an example design of which shows an open-loop sensitivity of 927 mV/g. This offers direct transduction from mechanical input to electrical output with in-built amplification.

To achieve high sensitivity and a large measuring range, linearity, etc., accelerometer is operated in closed-loop. This work proposes a Closed-Loop In-Plane movable Suspended Gate FET(CLIP-SGFET) based accelerometer, implemented using In-Plane movable Suspended Gate FETs(IP-SGFETs). The loop uses force-feedback mechanism where the actuator, in feedback, generates the electrostatic force required to bring back gate to its nominal position. The output of the differential amplifier through an integral controller, with an actuator driven by dynamically reconfigurable charge pump in feedback, realizes the closed-loop system.

The charge pump in the feedback path, generates high-voltage differential signals that drive the actuator. One of the important properties of the charge pumps is that they allow current flow in one direction. A multi-level converter-based discharge stage is proposed, which is added at the output node of the charge pump to facilitate the discharge of the output node when input to the charge pump reduces. It is seen that the latency of charge pumps is dependent on the input and its efficiency, which is a source of error in its output; especially when two charge pumps are used for differential operation. A bootstrap switch-based charge pump is designed,

which maintains constant latency at all efficiency settings improving its accuracy. In this work, the bootstrap-switch based dynamically reconfigurable charge pump, with the discharge stage, is designed to generate differential signals up to 10.5±5.12 V. Using the actuator and charge pump, the closed-loop accelerometer with IP-SGFETs as input pair, is designed and simulated in UMC180 nm technology for a dynamic acceleration range of ±4g at a supply voltage of 3.3 V.

Contents

List of Tables

List of Figures

List of Abbreviations

AFE Analog Front End.

CCCP Cross Coupled Charge Pump.

CLIP-SGFET Closed Loop In-Plane SGFET.

CMFB Common Mode FeedBack.

CMOS Complementary Metal Oxide Semiconductor.

CP-II Cross-coupled Charge Pump type-II.

CSV Comma Seprated Values.

DFF D Flip Flop.

DRC Design Rule Check.

EMI ElectroMagnetic Interference.

FA Full Adder.

FCC Flying Capcitor Converter.

FET Field Effect Transistor.

GDSII Graphic Design System II.

HDL Hardware Description Language.

IC Integrated Circuit.

IEEE-CASS IEEE Circuits And Systems Society.

IP-SGFET In-Plane Suspended Gate Field Effect Transistor.

LMGFET Laterally Movable Gate Field Effect Transistor.

LUT Look-Up Table.

LVS Layout v/s Schematic.

MEMS MicroElectroMechanical Systems.

MLC Mutli Level Converter.

MSE Mean Squared Error.

NMOS N-channel Metal Oxide Semicondutor.

NN Neural Network.

OP-SGFET Out-of-Plane Suspended Gate Field Effect Transistor.

PMOS P-channel Metal Oxide Semicondutor.

ReLU Rectified Linear Unit.

RTL Register Transfer Level.

SGFET Suspended Gate Field Effect Transistor.

SPICE Simulation Program with Integrated Circuit Emphasis.

STA Static Timing Analysis.

TCAD Technology Computer Aided Design.

TDNN Time Delayed Neural Network.

UNIC-CASS Universalization of IC Design from CASS.

VMGFET Vertically Movable Gate Field Effect Transistor.

Chapter 1

Introduction

MicroElectroMechanical Systems (MEMS) are miniature systems comprising mechanical structures and interface electronics, with sizes ranging from a few micrometers to millimeters. The mechanical structure or device has movable parts that sense and move in response to the external stimulus (physical), changing the physical parameters (like distance, effective area, etc.) of the structure. This mechanical quantity is then converted into an electrical signal, which can be detected by the circuits. Some common commercial applications of MEMS include pressure sensors, barometers, actuators, and Inertial Measurement systems (IMUs) like MEMS accelerometers, MEMS gyroscopes, MEMS magnetometers, etc. In this thesis, we explore MEMS accelerometers, mainly novel sensor structures, and their interface circuits.

An accelerometer is an electromechanical device whose primary purpose is to detect and measure the acceleration applied to the object to which it is attached. They rely on various transduction mechanisms to convert the mechanical quantity to an electrical signal. Acceleration measurement has various industrial and commercial applications, including crash detection for air-bag deployment, vibration analysis in industrial machinery, buildings, commercial vehicles, seismic activity, etc. The most commonly used sensing or transduction techniques are piezoelectric, piezoresistive, and capacitive. Capacitive transduction is widely used in accelerometers.

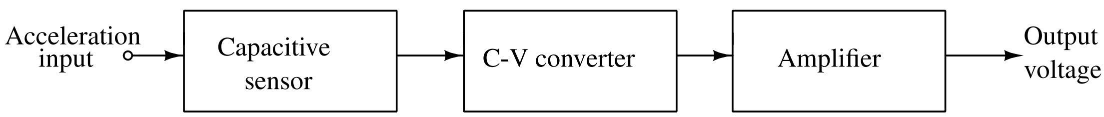


Figure 1.1: Block diagram of capacitive accelerometer and interface electronics.

Figure 1.1 shows the complete block diagram of MEMS capacitive accelerometer system. It mainly consists of two parts: a mechanical sensor and an electronic (or interfacing) circuit for detection and measurement.

Figure 1.2(a) shows an isometric view of the mechanical sensor. A micro-machined proof mass suspended between two parallel electrodes which constitutes the sensing element or the mechanical sensor. This structure forms two air-gap capacitors between the proof mass and fixed plates. The proof mass is free to move in response to an applied acceleration, as shown in Figure 1.2(b), changing the air gap between the electrodes. The proof mass forms two air gaps

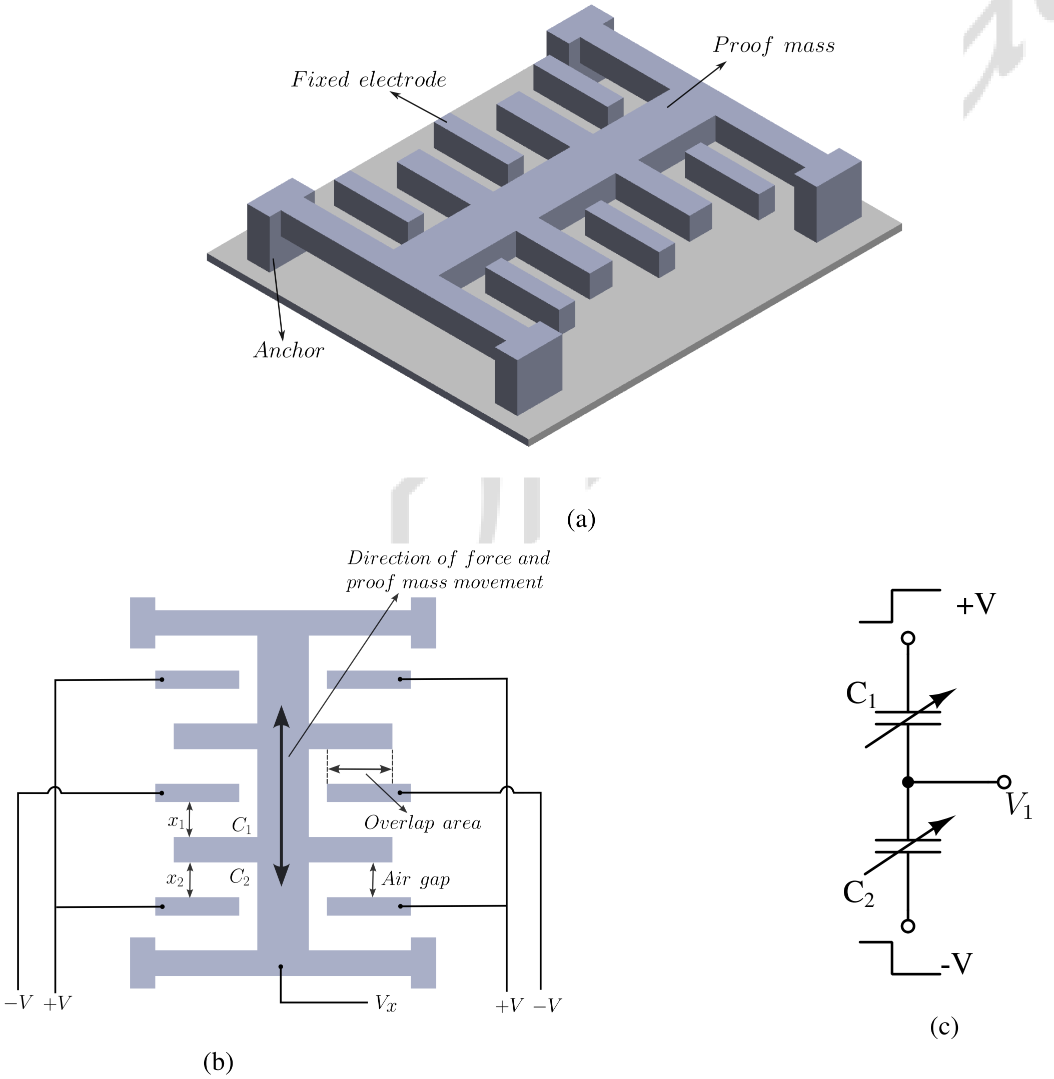


Figure 1.2: (a) Isometric view of mechanical sensor of a MEMS capacitive accelerometer. (b) Top-view of sensor. The proof mass is free to move between electrodes, resulting in change the air-gap between electrodes. (c) Circuit equivalent of the sensor.

with fixed electrodes. The circuit equivalent for the structure is shown in Figure 1.2(c).

When proof mass gets displaced, one of the air gaps reduces and the other increases. This changes the effective capacitance, which can be used to measure applied acceleration by interface circuits, which will be discussed next.

The interfacing electronics consisting of a C-V converter followed by an amplifier, converts this change in capacitance to an equivalent electrical signal. Two basic types of circuit configurations, namely analog and switched-capacitor type are reported in [1]. These circuits use the charge-voltage relationship of the capacitor which is given by (1.1), for measuring the input acceleration.

$$Q = CV \tag{1.1}$$

where, Q is the total charge on capacitor plates, C is the effective capacitance, and V is the voltage applied across the plates of capacitor.

The first configuration uses differential capacitors. Figure 1.2(c) shows the circuit equivalent of the sensor. A differential DC voltage is applied across the capacitor plates. At zero input, the differential capacitors cancel each other effects, and the output is zero. When the structure sees an external acceleration, the movable plate is displaced. Based on the direction of movement, one capacitor value increases, and the other reduces as the gap changes, changing the output voltage. This changing capacitance leads to current, which is governed by,

$$i_c = \frac{\partial Q}{\partial t} = C\frac{\partial V}{\partial t} + \frac{\partial C}{\partial t}V \tag{1.2}$$

A trans-impedance amplifier can capture this current flowing through the capacitor. This needs very accurate matching between differential capacitors at input. If sinusoidal voltage is applied instead of DC across the capacitor plates, the amplitude of the output voltage is directly proportional to the capacitance change. The second configuration of the C-V converter is a switched capacitor type. The differential capacitor is connected to a switched-capacitor integrator. The output is passed through a low-pass filter, providing an averaged value.

These sensors have limitations. Some of them are discussed in the following. The mechanical structures are large, and downsizing reduces their sensitivity, as the capacitance is proportional to the area of the plates. Some techniques, like lowering the air gap, can help improve sensitivity and performance but reduce the dynamic operating range and make it challenging to fabricate the sensors. They are also prone to ElectroMagnetic Interference (EMI) due to the high impedance nodes at the input of C-V converter circuits. Also, parasitic or stray capacitance affects the measurement accuracy because they are in the same range as sensing capacitance. The overall performance is deterioted due mismatch due to parasitics between electrodes (input dependent), including in the package and circuit, Moreover, the interface electronics and signal detection circuits, which consist of sophisticated capacitance-voltage converter, amplification stages, and filter circuits limit the noise performance of the system, affecting the signal-to-noise ratio [1, 2]. In recent years, researchers have explored Suspended Gate Field Effect Transistor (SGFET) based MEMS sensors which could potentially overcome these limitations of capacitive sensing.

1.1 SGFET based sensors

In SGFET based sensors, gate is suspended over the channel, which is displaced by external acceleration, changing the position of the movable gate as shown in Figure 1.3 (adapted from [3]). SGFET based sensors have been demonstrated in literature for resonator [4], position sensors [5], gas sensors [6], force sensors, pressure sensors [7] etc.

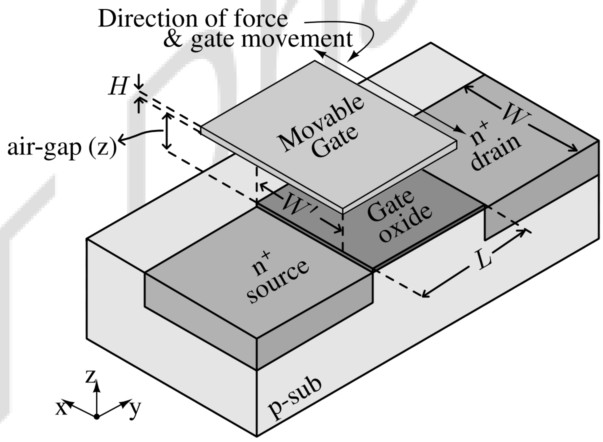


Figure 1.3: Isometric view of the SGFET transistor. The gate is free to move along the width direction (in x-axis).

The gate motion of the Field Effect Transistor (FET) modulates the current flowing between the source and drain. This change in current could be read out with the help of a simple current-biased, high-gain differential amplifier with SGFETs as input transitors, offering direct transduction with amplification. This eliminates the need for capacitance-to-voltage converter, amplifier and filter circuits. Figure 1.4 is a block diagram of SGFET based accelerometer.

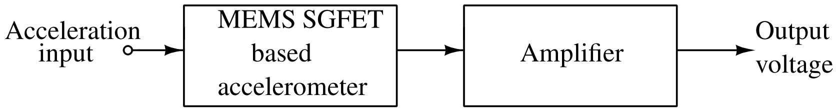


Figure 1.4: Suspended Gate FET based accelerometer and it's interface electronics.

1.2 Major contributions

This work extensively studies accelerometers using SGFETs, designing interface circuits for the sensor, and optimizing system design. The main contributions of this thesis are:

- The design of analog front-end interface circuits for SGFETs.
- The design of various blocks involved in the design of a closed-loop accelerometer.
  - The design of a bootstrap switch-based reconfigurable charge pump, which reduces latency variations.
  - Circuit design techniques to control the output voltage generated by varying the efficiency of the charge pump.
  - The design of a multi-level converter-based discharge stage and Common Mode FeedBack (CMFB) circuit for high voltage circuits for setting the common-mode voltage of charge pumps.
- An accurate way of modeling electrostatic actuators, which can be used in simulations.
- Techniques to reduce the simulation time of closed-loop systems by modeling the dynamically reconfigurable charge pump as a black box.
- System design and simulation of a closed-loop accelerometer using the aforementioned techniques.

1.3 Dissertation outline

The thesis is organized as follows.

- Chapter 2 review the literature for reported SGFET, actuator, and charge pump architectures.

- Chapter 3 discusses the modeling of SGFET and actuator, the design of various circuits involved in the design of closed loop SGFET-based accelerometers, and presents simulated results of the closed-loop system.

- Chapter 4 in the report discusses circuit-level optimization techniques, mainly focusing on reconfigurable charge pumps to improve measurement accuracy.

- Chapter 5 presents system-level optimization techniques, including accurate ways to model electromechanical actuators and ways to reduce closed-loop system simulation time.

- Chapter 6 discusses the system design using open-source CAD tools in Skywater 130nm technology with a supply of 1.8 V and presents post-layout simulation results.

- Finally, Chapter 7 discusses the conclusion and future direction of this work.

Chapter 2

Literature Survey

This chapter reviews reported charge pump architectures, SGFETs, and actuator structures for MEMS applications. It is a comparative study on SGFETs, actuators, and charge pumps considering various constraints such as sensitivity, linearity, size, the effect of equivalent parasitics, etc., which play an essential role in designing the proposed closed-loop accelerometer. These results provide a reference in selecting the suitable device structure and circuit architecture for our application.

2.1 Survey of SGFETs

As discussed Chapter 1, passive accelerometer systems are prone to parasitics, EMI, and noise. Active transduction techniques involve suspended gate FETs. Many FET-based sensors have been reported in the literature. Based on the direction of gate movement, they are 2 types of SGFET configurations. The suspended gate can move in the lateral direction along the width of the FET channel OR in the vertical direction to the plane of the substrate varying the effective air gap. The former is called a Laterally Movable Gate Field Effect Transistor (LMGFET), and the latter is a Vertically Movable Gate Field Effect Transistor (VMGFET). In both the methods mentioned above, gate movement results in a change in the drain current of transistor, governed by (2.1).

$$I_D = \frac{\mu_n C_{eff} W_{eff}}{2L} \left(V_{GS} - V_{th,eff}\right)^2 \tag{2.1}$$

where C_{eff} is the effective gate capacitance (which generally is, a series combination of air-gap capacitance and oxide capacitance for SGFETs), W_{eff} is the effective width of the transistor and $V_{th,eff}$ is the effective threshold voltage of the transistor.

2.1.1 Out-of-Plane Suspended Gate Field Effect Transistor (OP-SGFET)

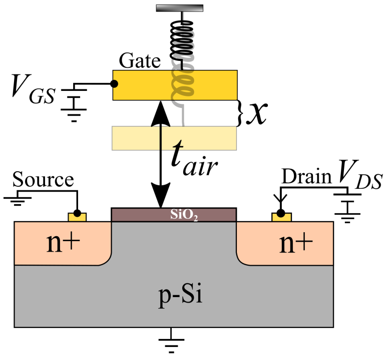


Figure 2.1: Out-of-Plane Suspended Gate FET

Figure 2.1 shows an OP-SGFET. The external acceleration changes the air gap and, thus, the total current flowing in the transistor. These sensors are highly sensitive as the threshold voltage and the effective capacitance change when external acceleration is applied. VMGFET or OP-SGFET based sensors are reported in [8, 9].

It should be noted that, the stable displacement of the suspended gate in the normal direction is limited to one-third of the air gap [1]. The sensor is susceptible to the electrostatic force due to gate voltage, due to which, the gate (moving electrode) will be pulled into contact with the fixed electrode. This phenomenon is called pull-in instability. Moreover, when the gate moves away from the channel, the air gap is large, and this weakens the gate control over the channel. The effect of drain potential becomes more significant, resulting in pseudo-short channel effects. This limits the operating range of SGFET in this configuration. In addition, various parameters affect threshold voltage like air gap, etc., as reported in [8]. As the total current in the transistor depends on the threshold voltage of the SGFET, which varies, making the device behavior non-linear.

Some techniques to eliminate these drawbacks in VMGFETs or OP-SGFETs and improve their reliability are reported in the literature. Authors of [10] discuss pseudo-short channel effects when the air gap is large and propose U-channel FET to mitigate these effects. Structural changes in the device to improve dynamic range and pull-in voltage are reported in [11–13]. Despite having the advantage of high sensitivity and various design techniques to improve device reliability, non-linear behaviour of the device makes it unfit for most application.

2.1.2 In-Plane Suspended Gate Field Effect Transistor (IP-SGFET)

LMGFET or IP-FET(in-plane SGFET), on the other hand, keeps the air gap constant, which is shown in Figure 2.2. When external acceleration is applied, the gate is displaced along the plane

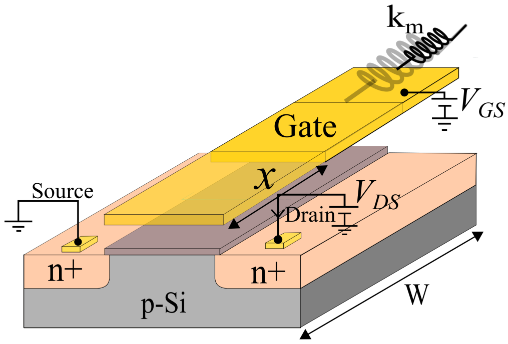


Figure 2.2: In-plane Suspended Gate FET

of the channel, varying the effective width, keeping the air gap constant. This configuration has reduced sensitivity compared to OP-SGFET, but they have a much wider dynamic range and linearity. Authors of [5, 14] have demonstrated IP-SGFET based sensors. Ways to improve the sensitivity of the device are demonstrated, which will be discussed next.

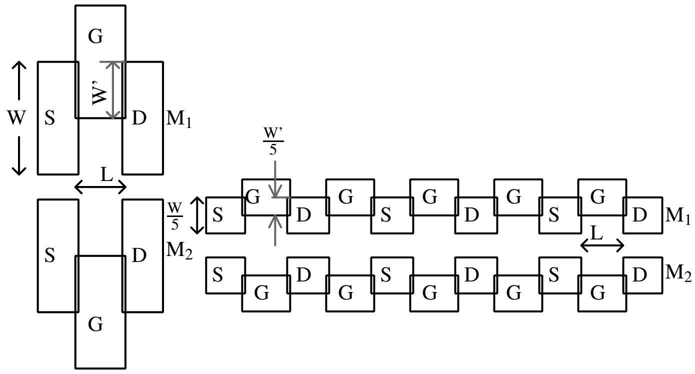
One of the reported techniques uses a differential sensing method to improve sensitivity [5]. This technique uses two LMGFETs or IP-SGFETs, arranged differentially, to design a sensor as shown in Figure 2.3(a). The gate movement increases the effective width of one of the transistors and reduces that of the other, doubling the change in capacitance and thus improving sensitivity. In addition, this increases the linearity of the sensor.

To further improve the sensitivity of the sensor, a transistor with multiple fingers can be used as shown in Figure 2.3(b) [15, 16]. For an acceleration acting along the prime axis, the sensitivity of the device (when biased in saturation region), can be derived as,

$$S \propto \frac{\delta I_{DS}}{\delta W'} = \frac{1}{2}\mu_n C' \frac{1}{L}\frac{(V_{GS} - V_{TH})^2}{m} \tag{2.2}$$

The drain current sensitivity of the device is independent of the gate width. Hence, the sensitivity can be increased by increasing V_{GS} or by reducing L. When a fingered transistor is used (Figure. 2.3(b)), where instead of a single transistor of width W', n transistors of width $W_T = \frac{W'}{n}$ are connected in parallel, the sensitivity will be

$$S \propto \frac{\delta I_{DS}}{\delta W_T} = \frac{\delta I_{DS}}{\delta \frac{W'}{n}} = n \times \frac{\delta I_{DS}}{\delta W'} \tag{2.3}$$



(a) (b)

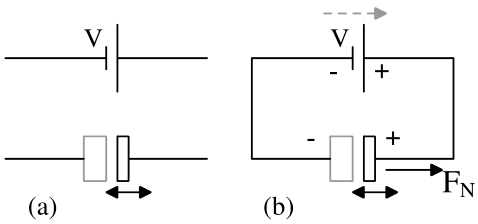
Figure 2.3: (a) Differentially arranged SGFET, (b) shows the fingered version of the same. The sensitivity of the SGFET gets multiplied by the number of fingers n, as any displacement affects all the fingers equally.

Hence, the sensitivity is directly proportional to the number of fingers used. The suspended gate is mechanically arranged so that the gate motion covers (or uncovers) all the fingers simultaneously when subjected to acceleration.

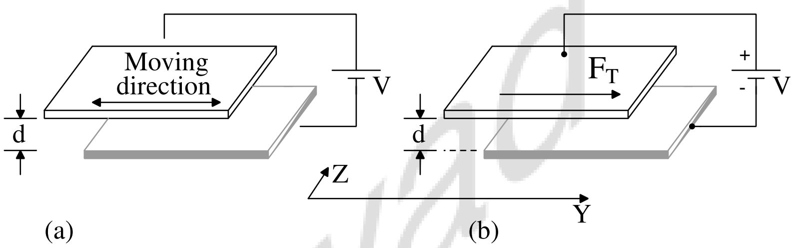
2.2 Survey of MEMS actuators

Transducers convert mechanical energy to electrical energy, whereas actuators cause mechanical motion when they are driven by electrical signals. Various actuation mechanisms have been used in MEMS structures [17–19]. These are broadly classified into: electrothermal [20], electrostatic [21] and electromagnetic [22]. Electrothermal actuation uses a thermally sensitive material that heats and expands when supplied with an electric current, resulting in the movement of electrode attached to it. This actuator can be operated at very low driving voltages. However, these have very slow responses, and temperature should be accurately maintained so that the heat produced will not cause permanent deformation. The force in electromagnetic actuators is due to the current and magnetic field interaction. These actuators have very fast response, but are very large in size due to the use of inductors for generating magnetic field.

Electrostatic actuation which uses the attractive force between the plates of capacitors with opposite charges for actuation. These are easy to fabricate and respond faster response. Figure 2.4 shows an electrostatic actuator. The plates of the capacitor are supplied with voltage. Due to this, the moving and the fixed plates acquire opposite polarities. The electrostatic force generated can be normal or tangential. Normal electrostatic force varies the air gap between and tangential movement changes overlap area. The force balance between applied external stimuli on the moving plate and the electrostatic force by the voltage driving determines the equilibrium position of the mass. However, non-linearity and very high drive voltage are major drawbacks of the actuator. Techniques to improve linearity and drive voltage are reported in literature, which is discussed later.



(a) Normal electrostatic force on a parallel-plate actuator (a) before connection (b) after connection.



(b) Tangential electrostatic force on a parallel-plate actuator (a) before connection (b) after connection.

Figure 2.4: Electrostatic force (a) Normal (b) Tangential

2.2.1 Types of mechanical actuators

The Figure 2.5 shows a parallel plate actuator. The movable electrode or plate displaces in the normal direction and is pulled toward the fixed electrode when voltage is applied across the electrodes, as shown in Figure 2.4(a). Based on the direction of the mass due to external stimuli, the polarity of the voltage applied across the plates can be chosen to balance the moving electrode. One of the major drawbacks of the parallel plate actuator is the collapse of the structure due to pull-in instability. The total displacement is limited to one-third of the original distance between the plates to eliminate the pull-in effect. Various techniques to eliminate pull-in and improve the dynamic range are reported in [12, 13, 23].

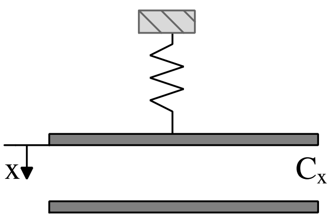


Figure 2.5: Parallel-plate electrostatic actuator.

The Figure 2.6 shows a comb-drive actuator. Due to lower overlap capacitance between the fixed and moving plate of the actuator, comb drive actuators are weaker than parallel plate actuators. These use tangential electrostatic force. However, comb actuators are preferred as they allow a larger displacement range (a few micrometers). To increase the electrostatic force, the air gap can be reduced, or the number of fingers in the comb drive can be increased. This also reduces the driving voltage. Various driving mechanisms are reported, out of which alternate voltage or push-pull driving is the most commonly used driving method [1]. Alternate signals between the fixed and movable electrodes are shown in Figure 2.6.

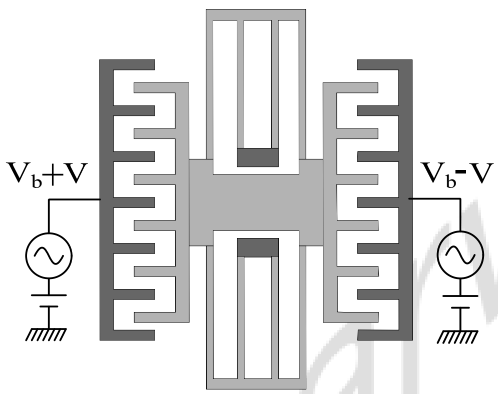


Figure 2.6: Differentially driven comb-drive actuator.

The force can be applied from both sides to keep the proof mass in the desired position. Push-pull driving is best suited for comb-drive actuators to control the position of the proof mass and also improves the linearity of the actuator [1].

2.3 Survey of charge pump circuits

As discussed earlier, an electrostatic force feedback actuation technique is used in capacitive accelerometers. The electrostatic actuator structures commonly used were discussed in the previous section. We saw that the force generated is proportional to the driving voltage applied across the plates. We intend to achieve an operating range of $\pm$ 4g in our proposed design, for which, the actuator must be driven at high voltages (up to 15 V) to generate sufficient counterbalance electrostatic force to keep the suspended gate at its nominal position. Charge pump circuits can be used to generate these high voltages. In this section, we compare recently reported charge pump architectures based on various parameters like conversion ratio, the number of components present (size), equivalent parasitic capacitance, etc.

Charge pumps are DC-DC converter circuits used in systems that require a voltage higher than the supply voltage to operate. They use capacitors as storage elements and generate higher voltage(positive or negative) through charge sharing. Multiple stages can be cascaded to generate higher voltages. The relation between the output voltage and the number of cascaded stages (n) is called the conversion ratio (M), based on which these circuits are classified into three categories, namely, linear [24–29], Fibonacci [30] and exponential [31] charge pumps.

1. Linear : $M = n$
2. Fibonacci : $M = F_n$, where F_n is the n^{th} Fibonacci number in the Fibonacci series.
3. Exponential : $M = 2^n$

Fibonacci and Exponential charge pumps have a higher voltage conversion ratio than linear charge pump architectures. But, in practical circuits, output impedance and the gain reduction due to the effect of equivalent parasitic capacitance is higher in these two types, as reported in [32], [33]. The output impedance can be reduced by sizing the capacitors appropriately(in Fibonacci charge pump [32]). Still, effect of parasitic capacitance on the charge pump voltage pumping gain is significant compared to the linear charge pump. Different linear charge pump structures will be discussed in upcoming sections of this chapter.

2.3.1 Linear charge pumps

This section discusses various linear charge pump architectures as listed below:

1. Heap charge pump
2. Cockcroft-Walton charge pump
3. Dickson charge pump

Heap charge pump

The concept of heap charge pump was presented in [34]. Figure 2.7 shows the schematic of a 4 stage heap pump in set-up state. The charge pump operates in 2 states: set-up and pump-up. In set-up state, all the capacitors, C_1 - C_4 are charged to V_{DD} and in pump-up state, S_1 is turned off and all the capacitors are connected in series.

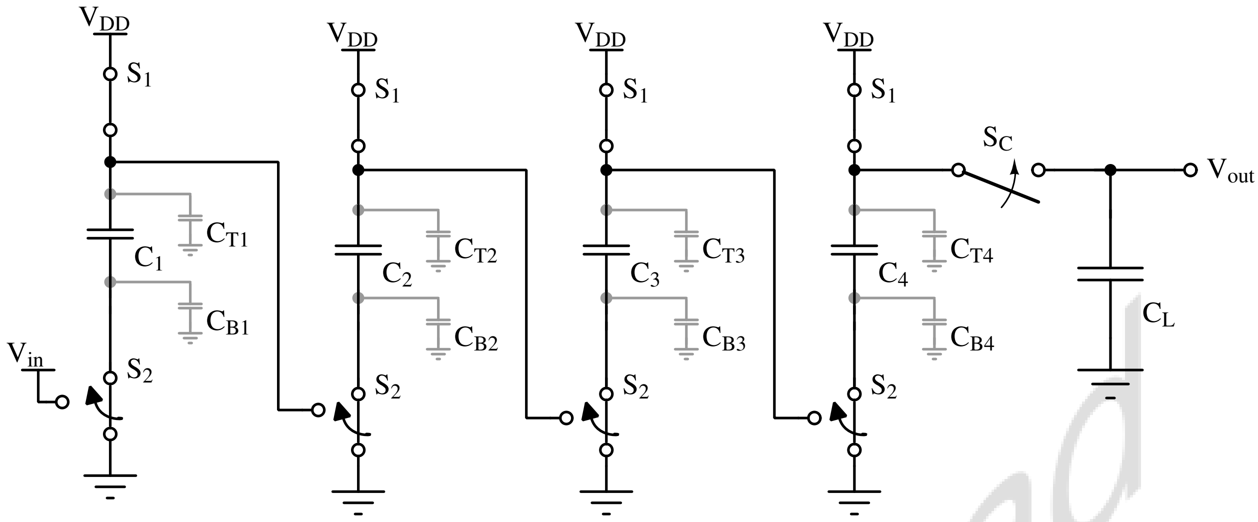


Figure 2.7: Schematic of heap charge pump with parasitics in set-up phase.

The ideal condition assumes no loss of current due to switching and parasitics. But when parasitics are considered, the stray capacitances on the top and bottom plates of the coupling capacitors shown in Figure 2.7 reduce the gain. During the set-up phase, the top and bottom plate stray capacitances are connected to V_{DD} and ground, respectively. In the pump-up phase, these stray capacitances draw current, which reduces the voltage across the coupling capacitors and, hence, the final output voltage. A design technique to eliminate the charge loss due to stray capacitance is reported in [35]. This architecture uses different-sized capacitors, resulting in a 30 $\%$ improvement in voltage gain but increases the circuit complexity.

Cockcroft-Walton charge pump

This charge pump uses diodes and serially connected capacitors, as shown in Figure 2.8. It is used as a voltage boosting circuit and generates voltage higher than input [25]. In the first clock phase, the capacitors are charged and then connected in series in the next phase. The limitation

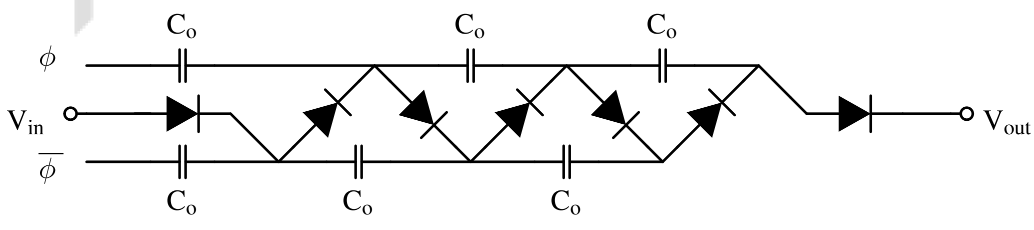


Figure 2.8: Schematic of Cockcroft-Walton charge pump. ϕ and $\overline{\phi}$ are non-overlapping clock signals.

of this configuration is same as the heap charge pump. As the number of stages increases, the output impedance of the circuit increases as the capacitors are connected in series.

Dickson charge pump

Figure 2.9 shows the conventional Dickson charge pump. This circuit operates in 2 phases.

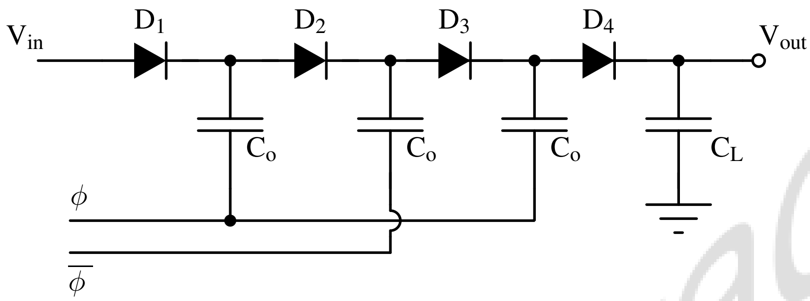


Figure 2.9: Schematic of conventional Dickson charge pump.

First is the charging phase; the nodes are charged to the input voltage level. The second phase begins when the clock goes high. It involves pumping node voltage, which was previously charged to input potential by clock amplitude. The pumped voltage is transferred in the next clock cycle, and the adjacent node charges up to this voltage level. The clock signals are non-overlapping [36] and are applied to the negative plate of the coupling capacitors.

The output voltage of charge pump is given by Eq.2.4:

$$V_{\text{out}} = V_{\text{in}} + N\left(V_{\phi} - V_{th}\right) \tag{2.4}$$

where, V_{out} is the output voltage of the charge pump, V_{in} is the input applied, N is the number of stages, V_{th} is the threshold voltage of diode and V_{ϕ} is the amplitude of clock signal. It should be noted that charge flow is unidirectional, i.e., from input to output.

Dickson charge pump lower output impedance compared to heap or Cockcroft-Walton charge as the capacitors are connected in parallel and also effect of parasitics is lower in this architecture compared to heap charge pump [37]. Due to these advantages, Dickson charge pump is the one of the widely used charge pumps. In the next subsection, we will review Complementary Metal Oxide Semiconductor (CMOS) implementation of Dickson charge pumps.

2.3.2 Dickson charge pump : CMOS implementations

As discussed earlier, the Dickson charge pump has fewer parasitic effects and lower output impedance compared to other linear charge pump architectures. Due to these advantages, most of the reported literature [24,26,34] discusses, analyzes, and optimizes the design of the Dickson charge pump. This section discusses the Dickson charge pump structure and its reported variants in detail.

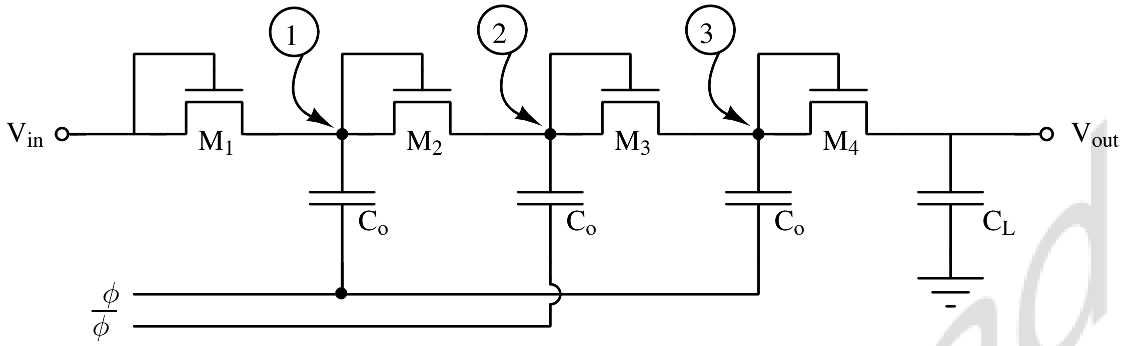


Figure 2.10: Schematic showing implementation of Dickson charge pump using diode-connected transistors. ϕ and $\overline{\phi}$ are non-overlapping clock signals.

The simplest CMOS Dickson charge pump is implemented using diode-connected transistors [26], in place of diodes as shown in Figure 2.10. The output voltage can be given by 2.5:

$$V_{\text{out}} = V_{\text{in}} + N\left(\frac{C_o}{C_o + C_p}V_\phi - V_{th}\right) - V_{th} \tag{2.5}$$

where, V_{out} is the output voltage of the charge pump, V_{in} is the input applied, C_o is the coupling capacitor, C_p is the parasitic capacitance in parallel with coupling capacitor, N is the number of stages, V_{th} is the forward voltage drop across the diode-connected N-channel Metal Oxide Semicondutor (NMOS) FET and V_ϕ is the amplitude of clock signal.

When ϕ goes from low to high and $\overline{\phi}$ goes from high to low, the voltage at node 1 settles to $V_1 + \Delta\, V$ and node 2 settles at V_2. $\Delta\, V$ is the voltage added to the initial voltage, V_1 of node 1 when the clock transitioned. It becomes a necessary condition that $\Delta\, V$ is greater than threshold voltage, V_{th} of the pass transistor, then pumping gain, G_v of the transistor can be expressed as :

$$G_v = V_2 - V_1 = \Delta\, V - V_{tn} \tag{2.6}$$

The above equation ignores the variations in threshold voltage, V_{tn} due to body effect. An increase in threshold voltage can reduce the voltage gain. The high output voltage obtained by cascading multiple stages saturates when V_{tn}(due to body effect) equals $\Delta\, V$, reducing the overall pumping gain.

To summarize, implementing the Dickson charge pump using diode-connected transistors (NMOS) has a few limitations:

1. The input voltage should be greater than the threshold voltage of the switch (NMOS). This becomes a necessary condition.

2. If the circuit uses N-well technology, threshold voltage drops across the diode-connected transistor increases and worsens at higher voltages (as the number of cascaded stages increases) due to the body effect.

CTS based charge pumps

CTSs use precise gate and body potential control to increase voltage gain as shown in Figure 2.11. Gate biasing circuits adjust the gate voltage and transfer input to the next stage without forward voltage drop. Body bias circuits eliminate the threshold voltage variations due to the body effect. Some of the most commonly used bias techniques are discussed next.

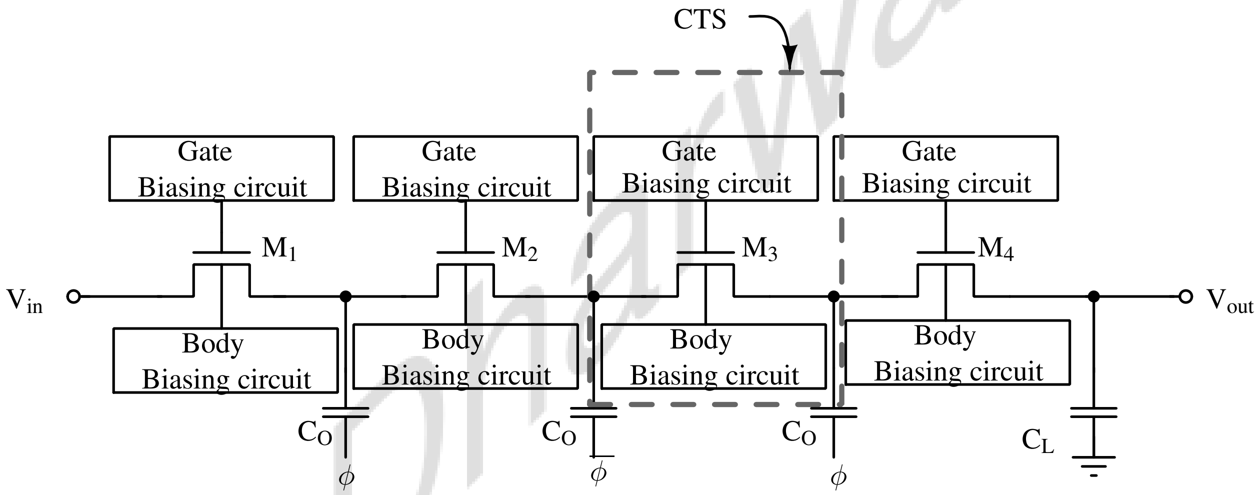


Figure 2.11: Schematic showing implementation of Dickson charge pump using NMOS based Charge Transfer Switches(CTS).

Body biasing circuits

The body effect becomes more significant as the number of stages increases. The input voltage is applied at the source of the switch. When the substrates are tied to the ground, the difference between the source and substrate potential results in a body effect, which becomes significant as the number of stages increases. To alleviate the threshold voltage variations due to body effect, body biasing methods such as floating well, adaptive biasing, or triple-well can be implemented as shown in Figure 2.12 [26, 38, 39].

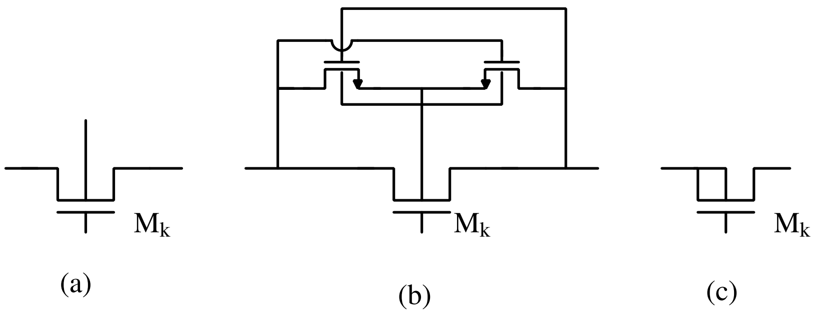


Figure 2.12: Schematic showing implementation of high voltage body bias circuitry to eliminate effects of body effect of the output voltage of N-type CTS based Dickson charge pump shown in Figure 2.11. (a) Floating-well method (b) Adaptive body-biasing scheme (c) triple-well process.

In the floating well method, the body terminal of the transistor is left open (not connected to the ground). Although this might be the easiest way to eliminate the body effect, the current leakage may degrade performance. An adaptive biasing scheme connects the substrate to the lowest potential, but this needs extra transistors. The last method uses triple well transistors and connects the source and substrate of MOS, eliminating the body effect without extra transistors and reducing the leakage as the substrate is not left open.

Gate biasing circuits

Various architectures of gate biasing circuits like static CTS-based charge pump [26, 28, 29], dynamic inverters [26, 38], gate bootstrapping [26, 38] are reported in literature describing ways to precisely controlling the gate voltage to increase voltage gain as shown in Figure 2.13. This section discusses these biasing circuits in detail.

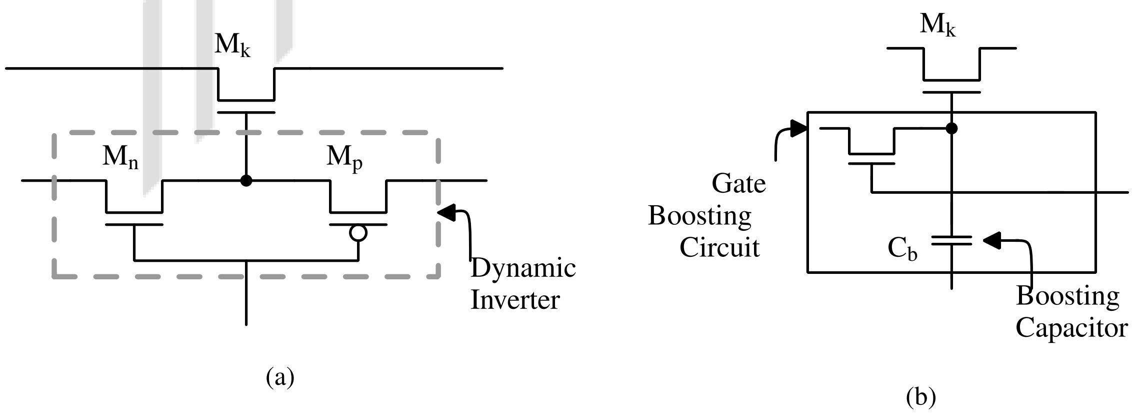


Figure 2.13: Schematic showing some of the gate biasing circuits for N-type CTS based Dickson charge pump shown in Figure 2.11.

Figure 2.13(a) uses dynamic inverters. In this scheme, each diode-connected switch is accompanied by two transistors which connect the gate to the highest potential among source and drain. On the other hand, the gate control scheme shown in Figure 2.13(b) boosts the input applied and then drives the gate. The gate potential is higher than the input, eliminating forward voltage drop.

1. Static CTS based charge pump

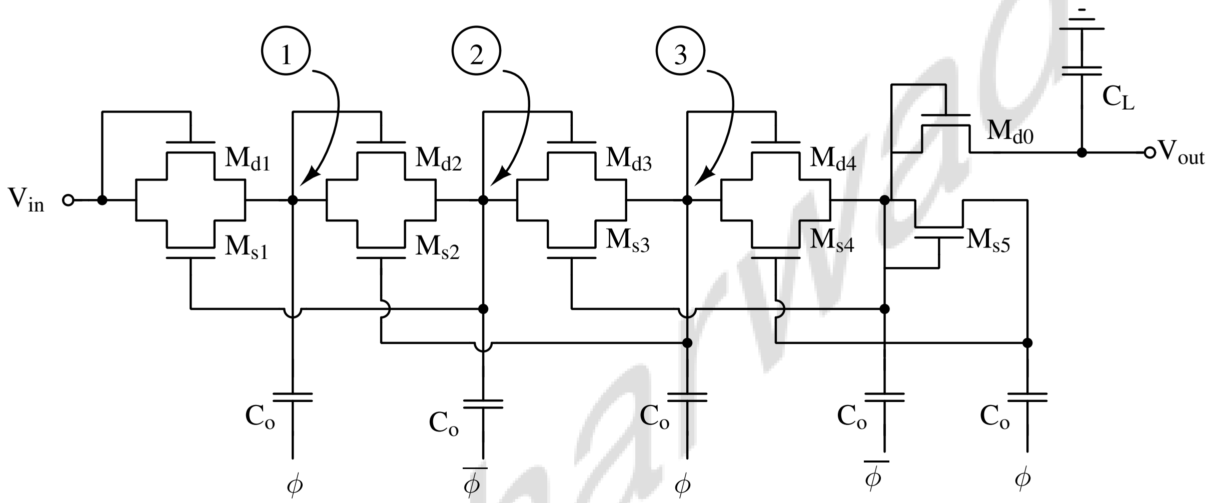


Figure 2.14: Schematic of a 4-stage static CTS based Dickson charge pump. M_{sk} are auxiliary switches with static backward control that direct the flow of charge during pumping operation.

Each diode-connected switch in the Dickson charge pump, shown in Figure 2.10, is accompanied by an auxiliary switch that is connected in parallel, which is involved in pumping operation as shown in Figure 2.14. The primary motivation is eliminating the forward threshold voltage drop due to diode-connected transistors in the Dickson charge pump.

The operation of static CTS-based Dickson charge can be explained as follows: When ϕ is high and $\overline{\phi}$ is low, voltage V_3 is pumped, which turns on transistor M_{s2}. The gate-to-source voltage of M_{s2} is given by:

$$V_{GS,M_{s2}} = V_3 - V_1 = 2\,\Delta\,V \tag{2.7}$$

As V_{GS} is greater than the threshold voltage of the transistor, the voltage on node V_1 is pushed to node V_2 without any forward voltage drop [27, 28]. The voltage gain of the charge pump, ignoring threshold voltage variations due to body effect, is given by:

$$G_v = V_2 - V_1 = \Delta\,V \tag{2.8}$$

We can see that the voltage gain is no longer affected by threshold voltage. On the other hand, when ϕ is low and $\overline{\phi}$ is high, ideally M_{s2} should be turned off completely. The gate-to-source voltage of the transistor should be smaller than the threshold. But this condition is not satisfied. Therefore, M_{s2} is never completely off, leading to reverse charge sharing between node 1 and node 2, reducing the charge pump final output.

Static CTS-based charge pump eliminated the forward threshold voltage drop through backward gate control voltage scheme, but reverse charge sharing affects the voltage gain in these circuits, [27–29].

2. Dynamic inverters/CTS based charge pump

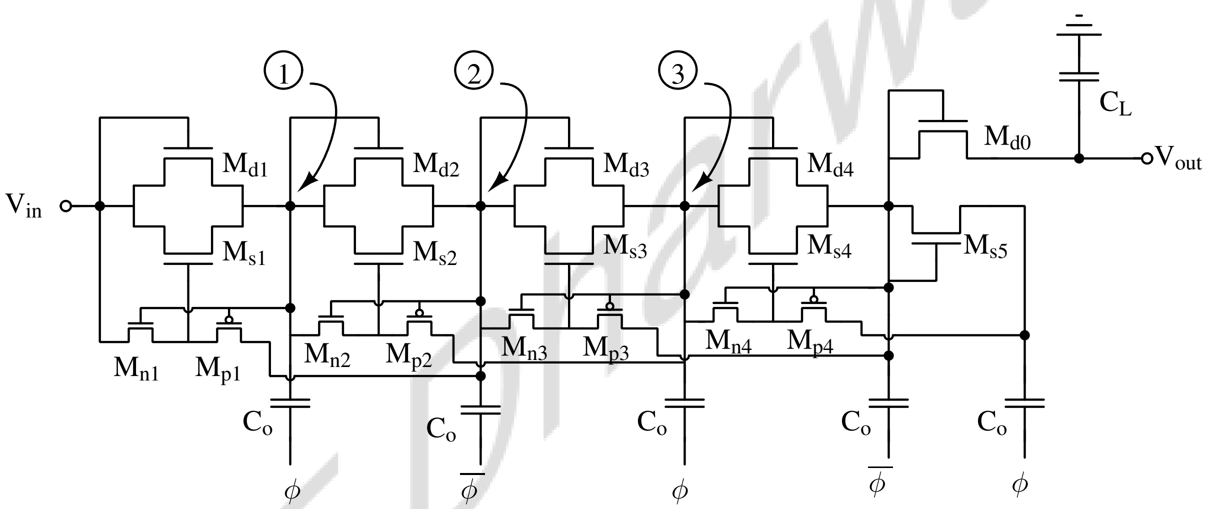


Figure 2.15: Schematic of 4-stage dynamic inverter based charge pump. Pass transistors, M_{nk} and M_{pk} are added which eliminates reverse charge sharing and provides accurate gate control to reduce the forward voltage drop.

Pumping efficiency can be improved by eliminating the reverse charge-sharing phenomenon of static CTS-based charge pumps. Figure 2.15 shows new charge pump architecture with pass transistors M_{n} and M_{p} added to charge pump allowing dynamic control the gate voltage of auxiliary transistor M_{s} allowing complete turn off of transistor eliminating reverse charge sharing and also can be turned on easily as in static CTS based charge pump shown in Figure 2.14.

The operation of dynamic CTS based charge pump can be explained as follows: When ϕ is high and $\overline{\phi}$ is low, voltage at node 3 is higher. M_{p2} is turned on, so the transistor, M_{s2} is completely on due to voltage at node 3, passing the voltage at node 1 to node 2 without any

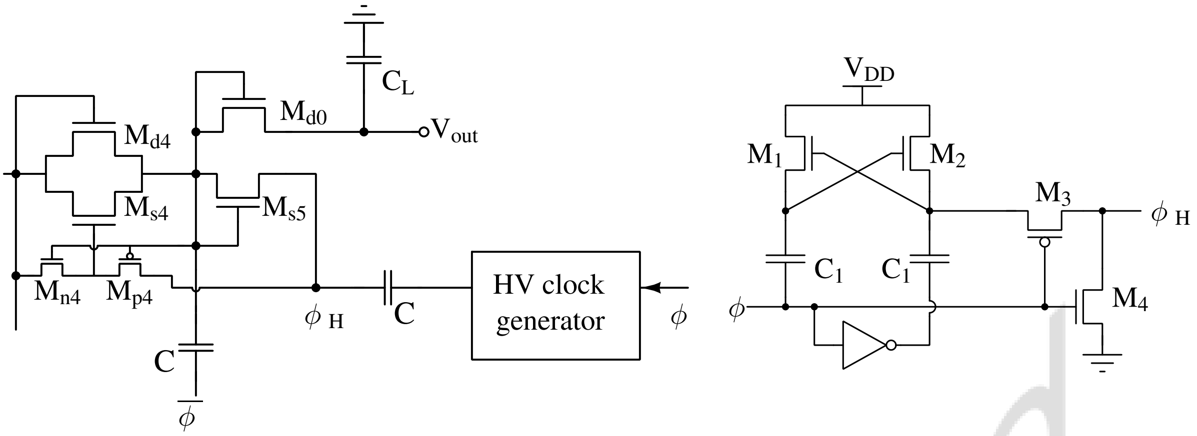


Figure 2.16: Schematic of dynamic CTS based Dickson charge pump with output stage NMOS diode-connected transistor, M_{d0} driven by enhanced high voltage clock to eliminate threshold voltage drop. ϕ_{H} is the enhanced clock.

voltage drop. On the other hand, When ϕ is low and $\overline{\phi}$ is high, M_{n} is turned on, M_{n} is on and gate-to-source voltage of M_{s} is zero preventing reverse charge sharing [26, 27, 40].

Output stage of CTS based charge pump

Charge pumps utilizing CTS based architecture provide better voltage gain but the one of the limitation in these circuits is not being able to use CTS in output stage. Diode-connected transistor is used as shown in Figure 2.15 which limits the maximum available output voltage. M_{s5} in Figure 2.15 is used to generate voltage at the drain of M_{p4}.

[27] proposes a high voltage clock generator to drive the NMOS pass transistor in the output stage. The high voltage clock signal that drives the gate of the NMOS pass transistor eliminating the threshold voltage drop and improving the voltage gain as shown in Figure 2.16.

[41, 42] reports a different approach to eliminate the threshold voltage drop at the output stage of the charge pump. The working is same as the charge pump discussed in Figure 2.15, instead the transistor, M_{d0} that transfers charge to the output node is replaced by a PMOS transistor. This change leads to elimination of threshold voltage drop and loss of gain. Figure 2.17 shows the schematic of charge pump without threshold voltage drop. M_{b} connects the bulk of transistor M_{d0} to higher potential.

As discussed earlier, M_{s5} in Figure 2.15 is used to generate voltage at the drain of M_{p4} in

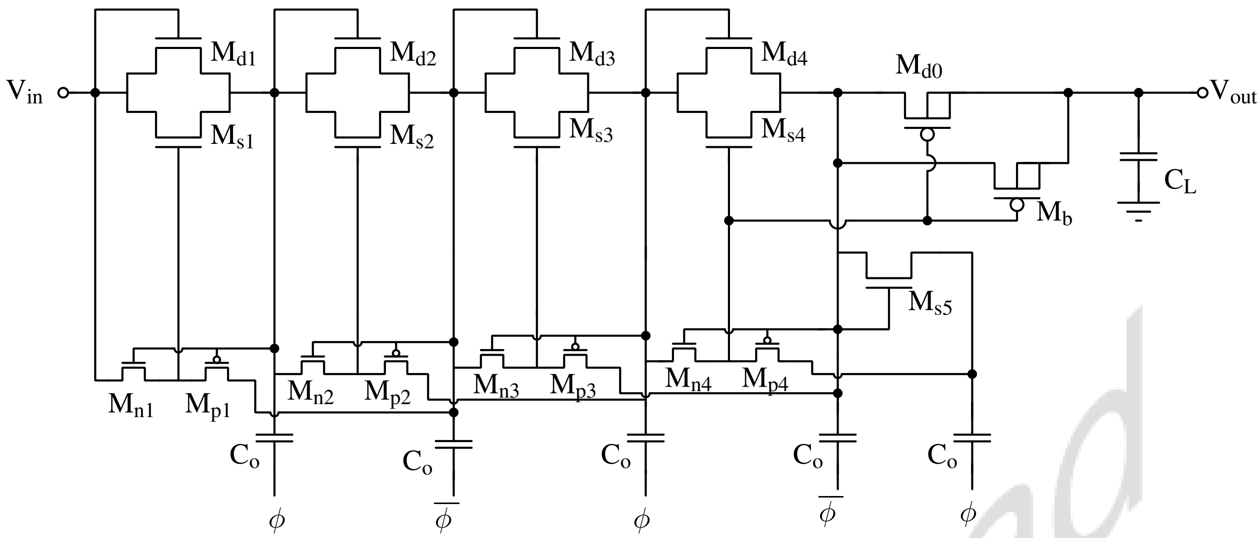


Figure 2.17: Dynamic CTS based Dickson charge pump modified to improve voltage gain. NMOS diode-connected transistor in the output stage is replaced by P-channel Metal Oxide Semicondutor (PMOS) transistor resulting in higher output voltage.

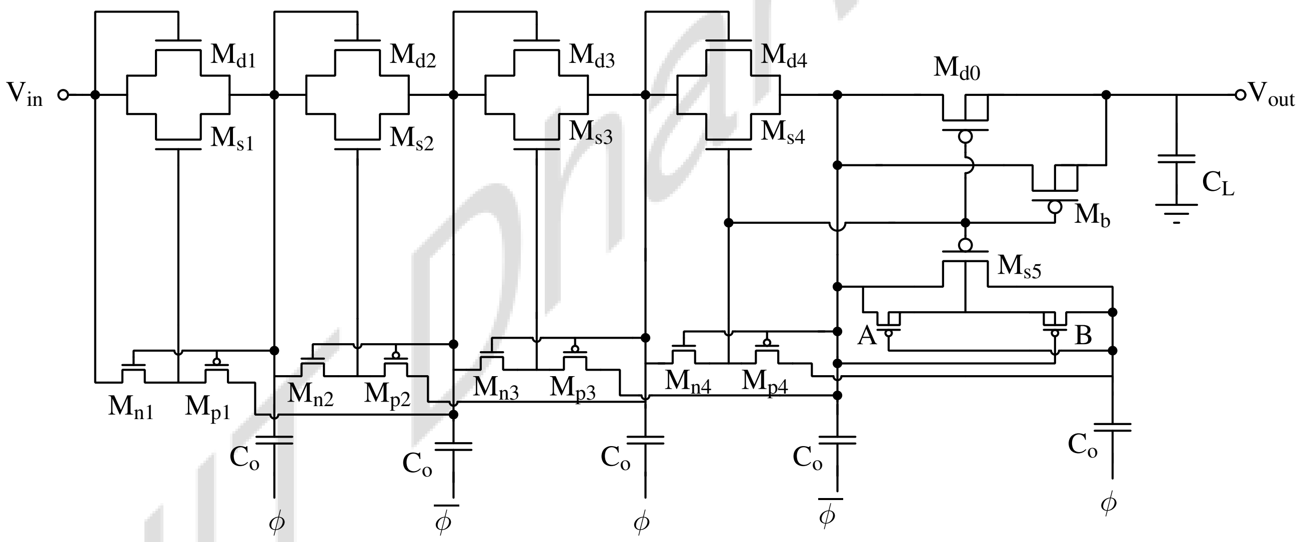


Figure 2.18: Output stage of dynamic CTS based Dickson charge pump modified to improve pumping speed.

Figure 2.15. There is threshold voltage drop due NMOS. Using PMOS pass transistor here will eliminate the threshold voltage drop and also this will improve the speed of operation of charge pump, i.e., the output of charge pump will settle faster. Figure 2.18 shows the schematic of fast Dickson charge pump where diode-connected NMOS is replaced by PMOS (M_{s5}). The transistors, A and B connect the bulk to highest potential [42]. This reported architecture has both higher pump-up speed and higher voltage gain compared to all other reported charge

pumps.

3. Gate boosting circuits

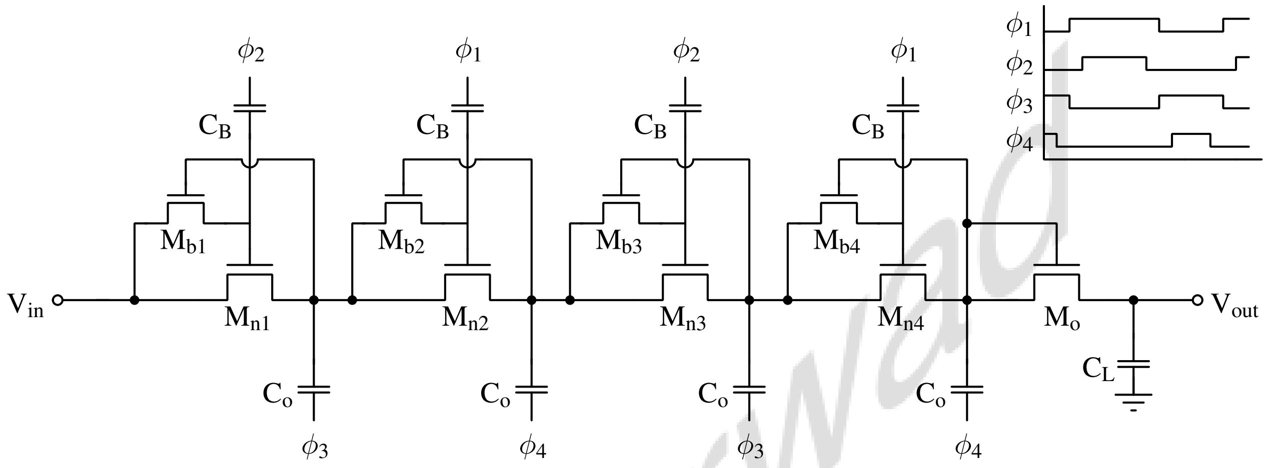


Figure 2.19: Schematic of gate boosting charge pump. It uses 4-phase clocking scheme. ϕ_1, ϕ_2, ϕ_3 and ϕ_4 are the clock signals.

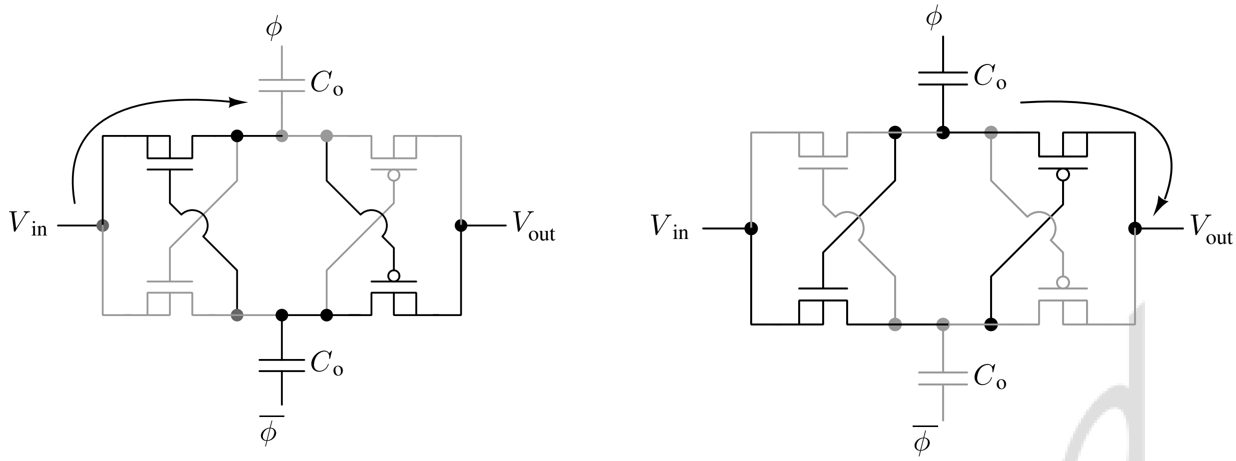
Gate boosting circuits consists of small boosting capacitor, C_{B} connected to the gate of the pass transistor, M_{nk}. The transistor, M_{bk} sets the gate voltage when clock,ϕ_3 is applied to the gate terminal. Boosting capacitor boosts the voltage when ϕ_2 is applied. The voltage is passed to next node without any voltage drop [26].

Although, the gate boosting capacitor and the extra transistor are small in size, this charge pump structure uses 4-phase clocking scheme as shown in Figure 2.19, increasing it's complexity and area.

4. Cross Coupled Charge Pump (CCCP)

By connecting two oppositely-operated inverters in parallel, the gate of all the CTSs in a charge pump circuit can be controlled effectively by the signals of its complementary section without any extra circuits. CCCP structure is based on this idea and also has the highest voltage conversion ratio.

CCCPs are implemented using two cross-coupled inverters and coupling capacitors discussed in [43]. The gates of the transistors are precisely controlled by non-overlapping clock signals and boosted by the coupling capacitors on the counter path as shown in Figure 2.20.



(a) Phase I: Voltage transfer mode

(b) Phase II: Voltage pumping mode

Figure 2.20: Schematic showing the two phase of charge pump operation using CCCP

When the clock is not applied, the charge pumps conducts through leakage currents and output is equal to the input applied.

Charge pumps operate in two phases: During the voltage transfer phase, as shown in Figure 2.20(a), the intermediate node is charged to the input voltage. In the pump-up/voltage pump phase, as shown in Figure 2.20(b), the intermediate node is pumped up by the ratio $\frac{C_O}{C_O+C_P}$, and the PMOS switch transfers charge to the output node of the charge pump, eliminating the forward voltage drop. CCCP is implemented in triple-well technology to eliminate the body effect by shorting substrate and source, as it allows biasing substrates at a voltage other than ground.

These are best suited for our application due to fewer components, simple design, and better voltage gain. They are also least prone to body effect.

2.4 Summary

To summarize, SGFET sensors are primarily of in-plane or out-of-plane type, of which, in-plane movable gate SGFETs have high dynamic range and good linearity, but lower sensitivity than out-of-plane movable SGFET. It is seen that, there is a gap in the literature with respect to interface circuits and system design for SGFET sensors.

In this thesis, we explore circuits and system design with IP-SGFET sensors. To increase sensitivity, we explore closed-loop operation with these sensors. For closed-loop operation, we need actuators and we have reviewed different types of MEMS actuators. Among the different reported actuators, comb-drive type actuators are suitable as they can be designed to have high dynamic range.

In order to realize closed-loop system with actuators, we need high voltage to drive the actuator. In this chapter, we have reviewed different types of charge pump circuits which could be used for generating these high voltages. The important characteristics of charge pumps, like parasitic capacitance effect, output impedance, conversion ratio, component size and number, etc., were discussed. Moreover, the high voltage generated by the charge pumps for our system mainly demands linear dependence between output and input voltage. Therefore, linear charge pumps, mainly the cross-coupled charge pump, are a better choice to generate the necessary high voltage to perform electrostatic actuation. It also has more straightforward design, smaller component size, number of components(switches and capacitors), lower output impedance, and lower parasitic effect than other linear charge pumps.

Chapter 3

Modeling and design of Closed Loop In-Plane SGFET (CLIP-SGFET) based accelerometer

This chapter discusses design interface circuits for IP-SGFET. In addition to that, we discuss the design of closed-loop accelerometer when IP-SGFETs as input pair. Figure 3.1 shows the block diagram of the proposed closed-loop accelerometer [44, 45]. The gate of IP-SGFET, interfaced using differential amplifier, is displaced when external acceleration is applied. The controller

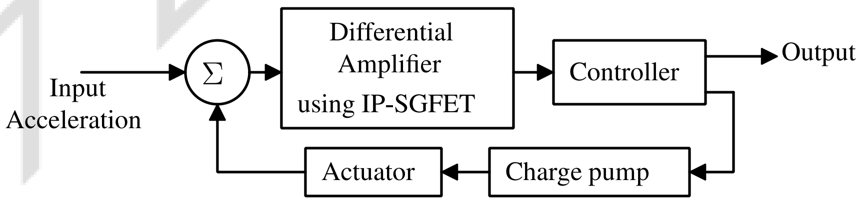


Figure 3.1: Block diagram of closed-loop accelerometer system [44, 45].

through charge pump drives the actuator which counterbalances the external acceleration and keeps gate in mean position. Closed-loop operation is necessary to meet the specifications of high dynamic operating range and high sensitivity.

The IP-SGFET and front-end circuits are designed in UMC180 nm technology to operate at a supply of 3.3 V for an operating range of ± 4g. The simulated results are presented.

3.1 Modeling of mechanical structures

To design, simulate, and ensure the reliable operation of the proposed accelerometer system, it is necessary to model the mechanical structures accurately considering all the phenomena. Various techniques have been reported. This section discusses the modeling of the structures used in our system using the Look-Up Table (LUT) method [44, 46].

3.1.1 IP-SGFET device

Figure 3.2 shows an IP-SGFET transistor which is used in sensing acceleration. The gate is free to move in x-direction. The gate overlaps half of the total channel width at the nominal

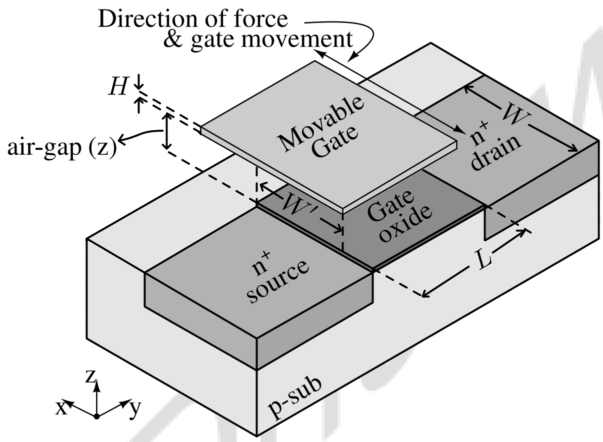


Figure 3.2: Isometric view of the IP-SGFET transistor. The gate is free to move along the width direction (in x-axis).

position. When external acceleration is applied, the gate of the IP-SGFET moves from $W/2$ to $W/2 + x$, where x is the displacement due to acceleration. Depending on the direction of acceleration, x can be negative or positive. This way, the effective width of the transistor changes and modulates the drain current of the transistor, which is governed by 3.1. These current variations are read-out by interface circuits, which are discussed in the next section.

$$I_D = \frac{\mu_n C_{eff} W_{eff}}{2L} (V_{GS} - V_{TH})^2 \tag{3.1}$$

where, C_{eff} is the effective gate capacitance which is series combination of air-gap capacitance and oxide capacitance, W_{eff} is the effective channel width of the transistor.

Developing SGFET sensor involves designing MEMS structure and FET, which interacts with multiple domains. To predict the dimensions of the transistor (here, effective width), which depend on input acceleration, the mechanical behavior of the device must be analyzed, which involves solving Newtonian mechanics, in addition to the electrostatics of the actuator. Followed by this, the simulation of FET behavior needs solving Schrödinger's, Poission's, and Gauss's law equations. A simple and time-efficient, LUT based approach is used to simulate this complex-multidomain, SGFET-based sensor in Spectre.

The IP-SGFET is a five fingered transistor. The total channel length is 10 μm and the width varies between 6 μm to 14 μm depending on the input acceleration. LUT for SGFET of this dimension is generated using Technology Computer Aided Design (TCAD) and MEMS+. The device simulation in MEMS+ gives effective width for different acceleration values in the range we intend to operate. The transistor behavior for all the width values is then simulated in TCAD. These LUTs are extracted and used to model SGFET, which is used in Spectre. The mechanical quantities on different nodes, like acceleration and width, are represented as voltages on the corresponding nets in the circuit schematic. For instance, the width of the SGFET is encoded on a net with a conversion of 1 V/μm, which means W = 5 μm is represented as a voltage of 5 V, which is interpreted in Verilog-A to access the LUT and generates corresponding outputs.

3.1.2 Actuator

An actuator is needed in closed-loop operation. A comb-drive actuator with 160 fingers on each side with air-gap of 6 μm between each finger, is used in our design. When an external acceleration displaces proof mass, voltage is applied across the actuator to generate reverse electrostatic forces to counterbalance the external stimulus and keep the gate in a nominal position. Figure 3.3 is the image of a differentially driven comb-drive actuator. The actuator's force is a product of the common mode (V_b) and differential voltage applied (V).

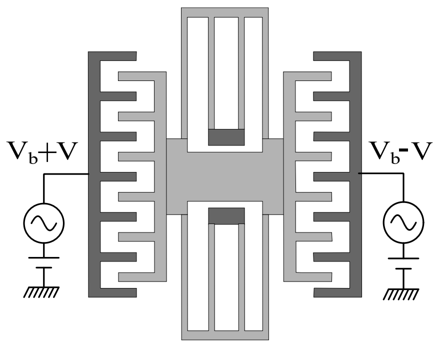


Figure 3.3: Differentially driven comb-drive actuator.

LUT-based modeling technique is also used to model this structure. The electromechanical simulation of the comb-drive actuator is done in MEMS+. The displacement in proof mass when voltage is applied across it, is analyzed and extracted as LUTs. The actuator is modeled using Verilog-A with LUTs from MEMS+ and used in Spectre simulations.

3.2 Differential amplifier design

A simple current-biased differential amplifier circuit with differentially arranged IP-SGFETs as the input pair as shown in Figure 3.4 interfaces the IP-SGFET sensor [44]. LUTs of transistor characteristics and electromechanical behavior of the MEMS device is used with Verilog-A as described in Section 3.1 is used for modeling and simulation. The current source loads are

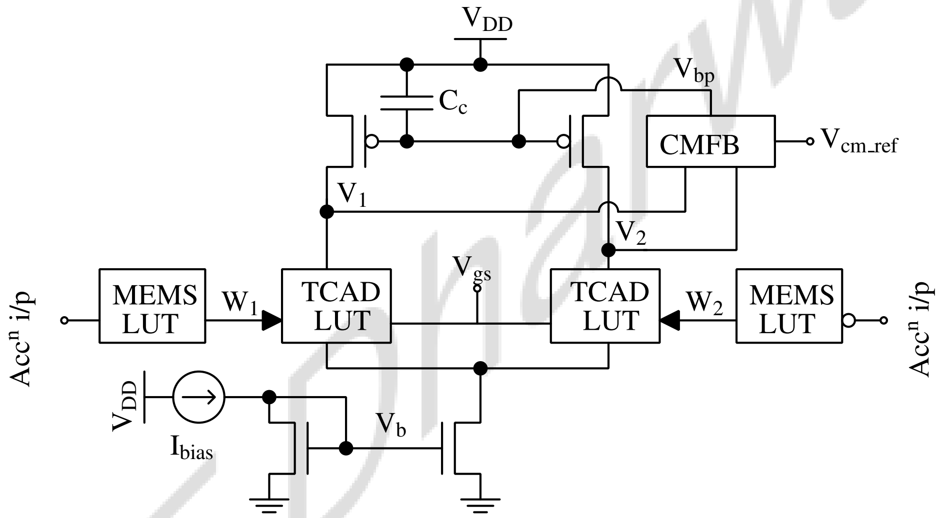


Figure 3.4: Schematic of fully-differential amplifier comprising LUTs of IP-SGFET modeling the device behaviour. The current source loads are biased using a CMFB circuit, shown in Figure 3.5 to sense and set the common-mode voltage.

biased using a CMFB circuit, shown in Figure 3.5 to sense and set the common-mode voltage. Complementary source follower buffers are used as a common-mode detector [47].

The working principle is straight-forward. The drain current changes when the gate is displaced by external acceleration. The differential output voltage of the amplifier is proportional to the changes in drain current.

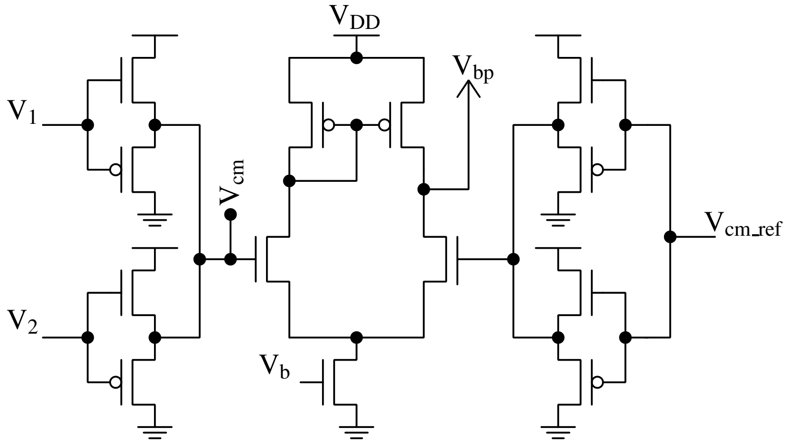


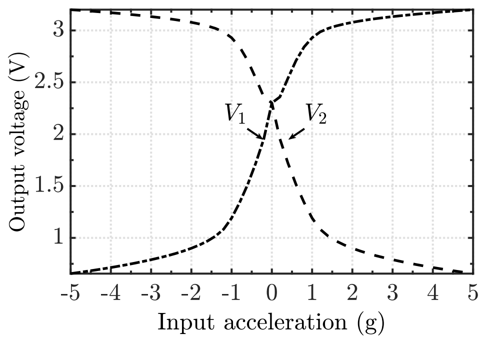
Figure 3.5: Schematic of the common-mode detector and common-mode error amplifier used in the IP-SGFET front-end fully differential amplifier circuit.

The small signal gain (sensitivity) of this differential amplifier stage is governed by,

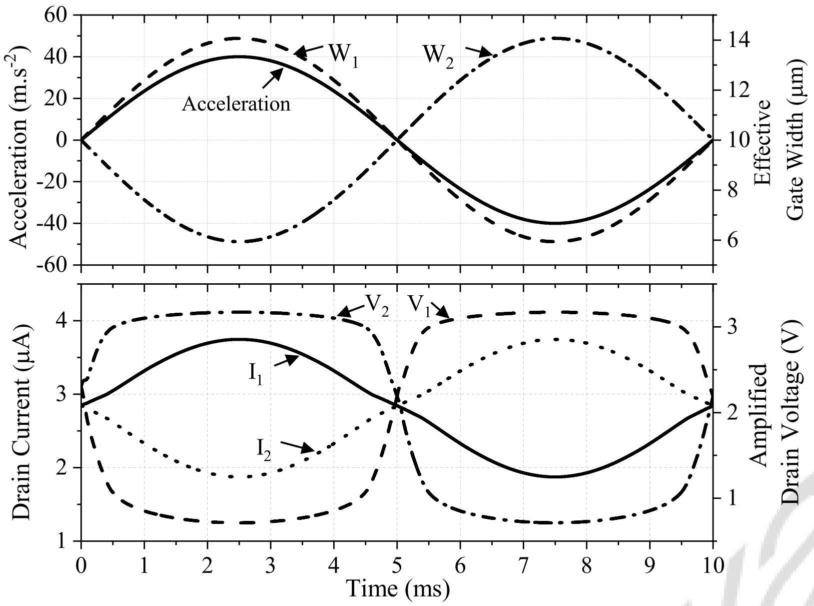
$$\text{Gain} = S_M \times S_T \times r_o \qquad \text{(V/g)} \tag{3.2}$$

where, S_M is the sensitivity of the of the mechanical structure (ΔW/g), S_T is the sensitivity of the transistor ($\Delta I/\Delta W$) and r_o is the output impedance of the fully differential amplifier circuit.

The circuits are designed in UMC180 nm technology at a supply of 3.3 V and the simulated results are presented below. Figure 3.6(a) shows the DC simulation and Figure 3.6(b) shows the transient simulation response of the circuit in Figure 3.4. However, the output voltage saturates due to the high gain. The small signal gain is found to be 927 mV/g. This is the first stage of the system.



(a) DC simulation of IP-SGFET based accelerometer in open loop for acceleration of $\pm$ 5g, for circuit shown in Figure 3.4. Output saturates due to high gain.



(b) Transient simulation of the IP-SGFET based accelerometer in open loop for acceleration of $\pm$ 4 g at 100 Hz, for the circuit in Figure 3.4.

Figure 3.6: Simulated response of differential amplifier interfacing IP-SGFET device.

3.3 Charge pump and digital logic circuit design

In addition to differential amplifier with IP-SGFET input pair discussed in Section 3.2, closed-loop consists of accelerometer with actuator and dynamically reconfigurable charge pump as shown in Figure 3.1. High-voltage differential signals drive the actuator, establishing the force necessary to keep the proof mass in the desired location. CCCPs are used to generate the high voltages.

3.3.1 Cross coupled charge pumps

CCCPs are implemented using two cross-coupled inverters and coupling capacitors discussed in [43]. Figure 3.7 shows schematic of cross-coupled charge pump used. The gates of the transistors are precisely controlled by non-overlapping clock signals and boosted by the coupling capacitors on the counter path as shown in Figure 3.11. The output voltage of charge pump is governed by 3.3.

$$V_{\text{out}} = V_{\text{in}} + N\left(\frac{C_o}{C_o + C_p}V_\phi\right) \tag{3.3}$$

where, V_{out} is the output voltage of the charge pump, V_{in} is the input applied, C_o is the coupling capacitor, C_p is the parasitic capacitance in parallel with coupling capacitor, N is the number of stages cascaded and V_ϕ is the amplitude of clock signal.

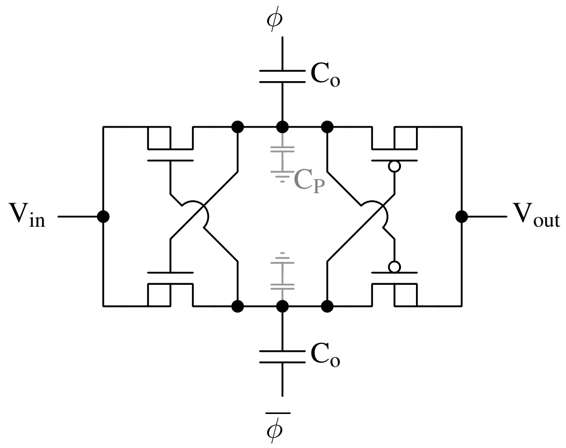


Figure 3.7: Schematic of cross-coupled charge pump. The substrates are connected to sources of respective transistors eliminating body effect. ϕ and $\overline{\phi}$ are non-overlapping clock signals.

The charge pump is designed to operate at a frequency of 50 MHz and the clock signals are generated by standard NAND latch based generator shown in Figure 3.8 [36].

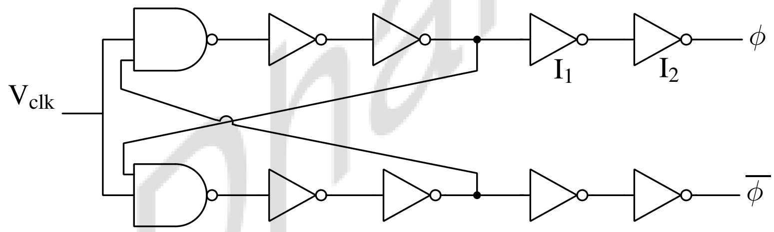


Figure 3.8: Schematic of non-overlapping clock signal generator. The clock signals, ϕ and $\overline{\phi}$ that drive the charge pump

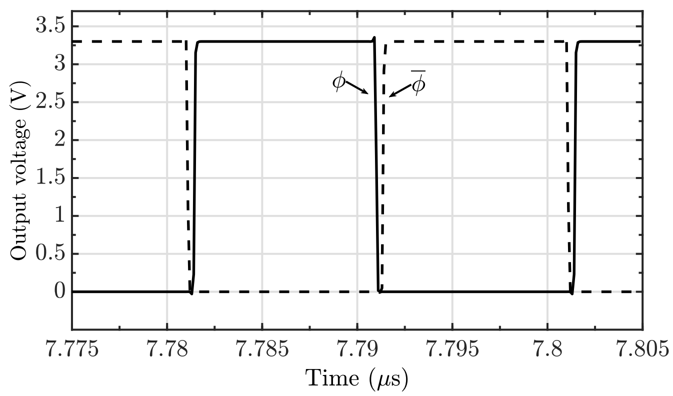
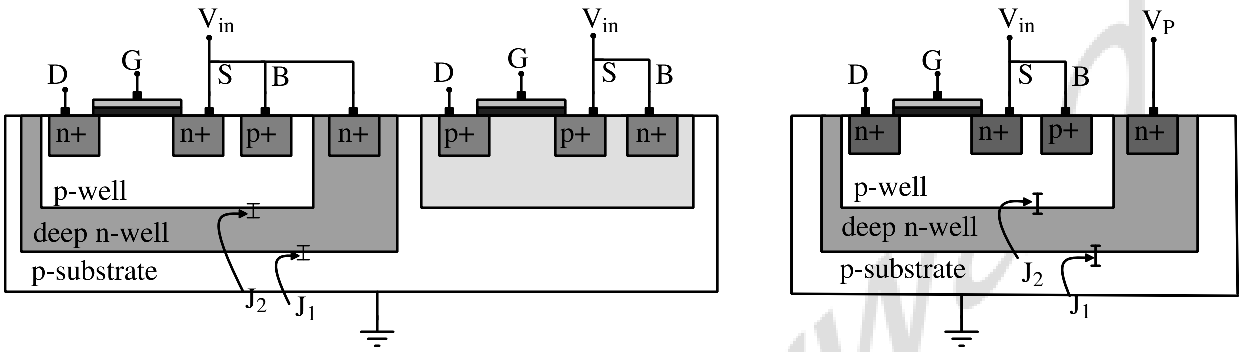


Figure 3.9: Simulated output signals of non-overlapping clock signal generator shown in Figure 3.8.

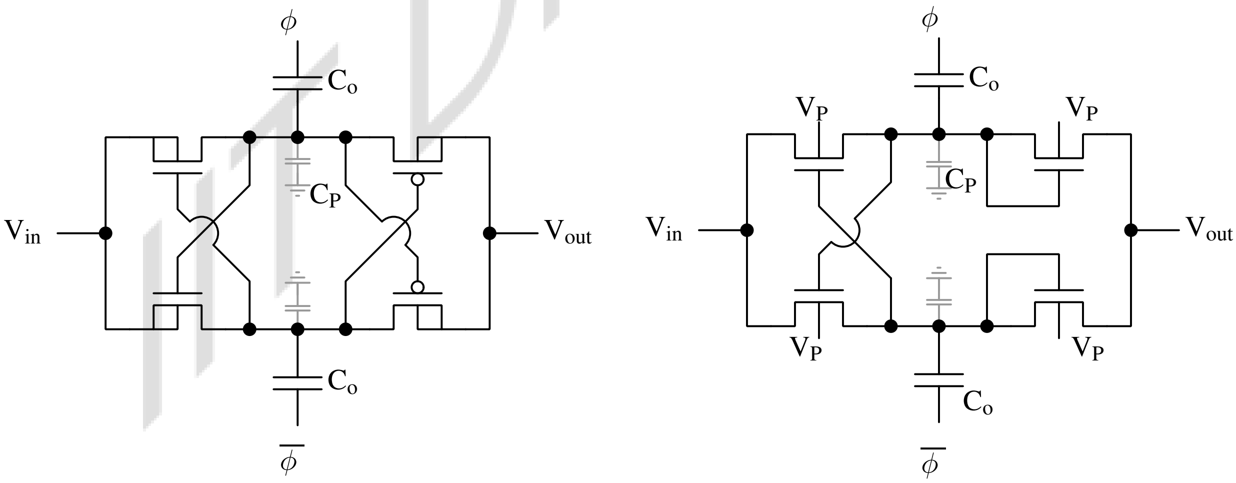
Multiple stages of these charge pumps can be cascaded to generate higher voltages. However, the voltage tolerance level of the CMOS substrate, which is determined by the diode junction breakdown voltages(n-well/p-substrate junction(J_1 in Figure 3.10(a) [36]) of transistors limits the maximum voltage that can be generated by cascading multiple stages of the charge pump. The circuits are designed in UMC180 nm technology, which allows the generation of a maximum voltage of 12 V.



(a) CP-I

(b) CP-II

Figure 3.10: Cross-section of transistors in two types of cross coupled charge pump showing the source and substrate connections in CP-I and CP-II. Junction J_1 : n-well/p-substrate junction, J_2 : n+/p-well junction.



(a) CP-I

(b) CP-II

Figure 3.11: Schematic of two types of cross coupled charge pump used in generating high voltages. (a) CP-I uses triple well transistors. (b) CP-II uses only triple-well NMOS transistors whose substrates are connected to higher potential which is tapped from the outputs of preceding stages. ϕ and $\overline{\phi}$ are non-overlapping clock signals.

To extend the output voltage range further, Cross-coupled Charge Pump type-II (CP-II) can be used. The deep n-well can be connected to a higher potential, as shown in Figure 3.11(b). To avoid the breakdown of junctions in PMOS switches. NMOS diode-connected switches are used instead of PMOS, eliminating the need to generate the voltage to drive the substrate of PMOS. This extends the output voltage range by the breakdown potential of n+/p-well(J_2 shown in Figure 3.10(b) up to 17 V. The limitation of all NMOS-based charge pump stages (CP-II) is that the conversion ratio is lesser than CP-I due to the forward voltage drop across diode-connected NMOS.

Polysilicon P-I-N diode-based Dickson charge pumps can further extend the maximum voltage that the charge pumps can generate. The diodes are fabricated in the polysilicon layer as reported in [48]. Now, the maximum voltage is limited by the dielectric breakdown limit of the polysilicon, which depends on the thickness of the polysilicon layer. The thickness of the polysilicon is 40 nm, and the breakdown voltage is around 400 V in UMC180 nm technology. Appendix A discusses device structure and characterization.

Reconfigurability of charge pumps

In addition to high voltage generation, there is a need to control the voltage generated. The output voltage generated should be continous and proportional to the input acceleration. For this dynamically reconfigurable charge pump is used. The output voltage of charge pump is governed by 3.4 as discussed earlier.

$$V_{\text{out}} = V_{\text{in}} + N\left(\frac{C_o}{C_o + C_p} V_\phi\right) \tag{3.4}$$

The output voltage can be controlled in mutliple ways:

1. By varying the number of active stages, N as reported in [49].
2. By varying value of coupling capacitor, C_o or efficiency of each charge pump.
3. By varying the amplitude to clock signals, V_ϕ driving the coupling capacitor.

Varying number of active stages

It should be noted that the charge pumps contribute to the signal chain when they are active, i.e., clock signals are applied. When V_ϕ is zero, i.e., the clock is disconnected, the output node of the charge pump charges up to the input voltage level through leakage currents. This can

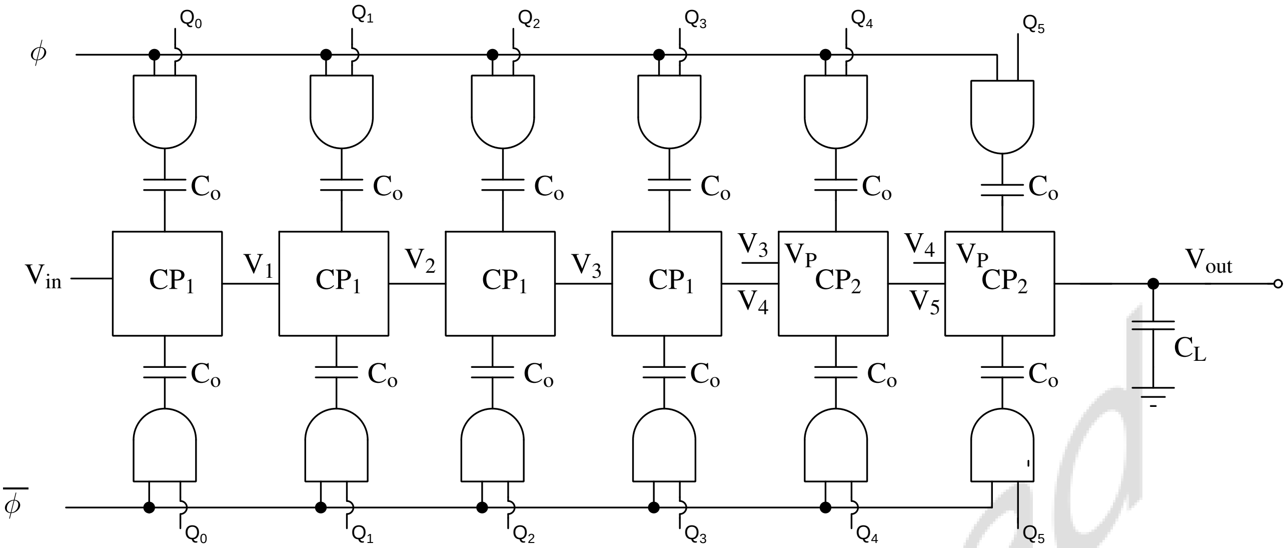


Figure 3.12: Schematic of reconfigurable charge pump with digitally controlled active stages. The digital word, Q controls the number of active stages. When the digital word is low, the clock signals are disconnected, which disables that particular stage in the cascaded structure.

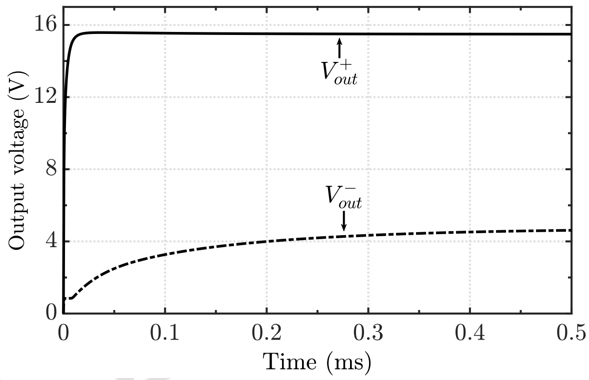


Figure 3.13: Plot showing the simulated results for output voltage of reconfigurable charge pump. Number of active stages are varied to control the output voltage generated. Input voltage,V_{in}=1.7 V, maximum output voltage, V_{max}=16.8 V with latency=212 ns, and minimum output voltage, V_{min}=2.5 V with latency=1.2 ms.

be used to control the output voltage. Based on the output voltage level needed, the number of active stages of the cascaded charge pump structure can be set using the digital word, Q, shown in Figure 3.12. The clock signals are connected when Q is high. Figure 3.13 shows the simulated results at the highest and the least value of N. The value of Q is decided by the digital block based on the instantaneous acceleration. The various blocks of digital control circuitry and design is discussed later.

In the worst case, when one stage is enabled, the output node takes up to 1.2 ms to settle to its final value, as it has to charge all the intermediate nodes(V_1 to V_5) and V_{out} to the desired level only through leakage currents. Simulated results are shown in Figure 3.13. The closed-loop system is designed for a bandwidth of 100 Hz, and this latency in feedback is not desirable for our application as it might lead to instability in the system.

Charge pump with variable efficiency

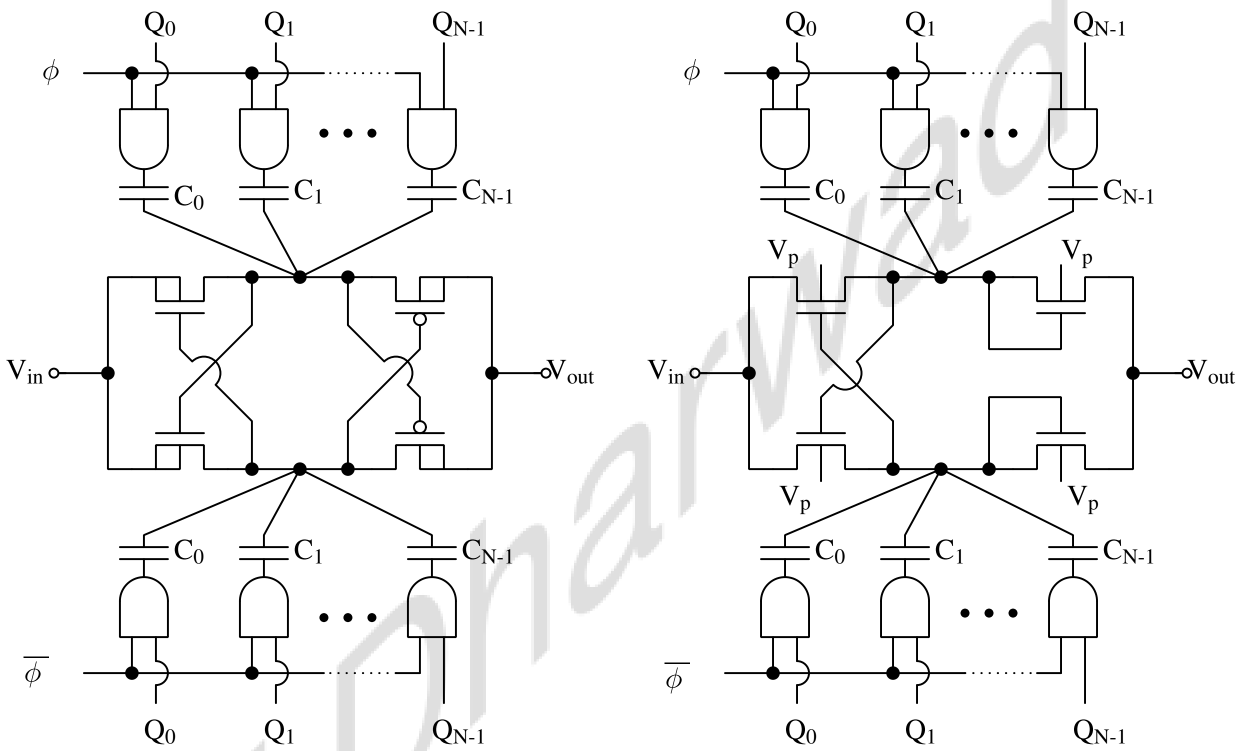


Figure 3.14: Schematic of charge pump CP-I and CP-II with digitally controlled efficiency. The digital word, Q is a thermometer code which controls the output of charge pump by varying the number of capacitors that are coupled with clock signals.

Other way to generate the continous controlled output voltage is to vary the efficiency of each stage of the charge pump, i.e., varying the ratio $(\frac{C_\text{o}}{C_\text{o}+C_\text{p}})$ in 3.3. The parasitic capacitance is fixed. So, the control over final output voltage is achieved by varying C_o, i.e., the value of capacitor coupled with the clock signal. To realize this, C_o in Figure 3.11 is split into mutliple capacitors and the number of capacitors coupled with the clock signal is varied by setting the value of digital word, Q as shown in Figure 3.14. In addition to this, at any given time, none of the stages of the cascaded charge pump structure are completely disconnected from the signal chain. It is connected to the signal chain with very low efficiency, so charge pumps wont be

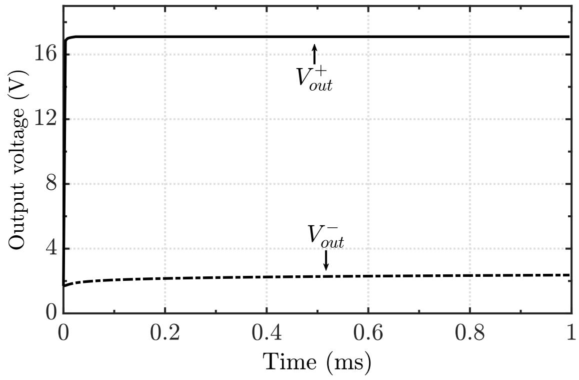


Figure 3.15: Plot showing the simulated results for output voltage of reconfigurable charge pump. The number of capacitors coupled to the clock is varied to control the output voltage generated. Input voltage,V_{in}=2.1 V, maximum output voltage, V_{max}=15.56 V with latency=430 ns, and with input voltage,V_{in}=1.3 V minimum output voltage, V_{min}=4.5 V with latency=70μs.

conducting through leakage current, thus improving the latency.

In our design, the value of parasitic capacitance, C_{p} , is approximately 500 fF. The value of the capacitor coupled with clock C_{o} varies between 250 fF and 1.45 pF. The final output voltage is given by,

$$V_{\text{out}} = V_{\text{in}} + 6\eta V_{\text{clk}} \tag{3.5}$$

The value of η depends on the ratio $\frac{C_{\text{o}}}{C_{\text{o}}+C_{\text{p}}}$. An output voltage ranging from 5.38 V to 15.62 V, which is needed to counterbalance an acceleration range of $\pm$4g, is generated by cascading 4 stages of CP-I and 2 stages CP-II.

The simulated results are shown in Figure 3.15. The variations in latency are within the tolerable range(at least efficiency the range is around a few mirco-seconds).

Discharge stage and Common-mode feedback

CCCP does not allow reverse charge flow, i.e., charge flow is unidirectional from input to output. So, when the input acceleration reduces, there is no discharge path for the output node, as the load is capacitive. MOS based discharge stage added at the output node of the charge pump, facilitates the discharge without leading to the breakdown of junctions when input acceleration reduces. The discharge stage configuration in Figure 3.16 is similar to the one discussed in [49].

It consists of a transistor stack with gates of the transistors controlled by intermediate stage outputs of the charge pump, and substrates of the transistors in this path are connected to elevated voltages, preventing junction breakdown as shown in Figure 3.16.

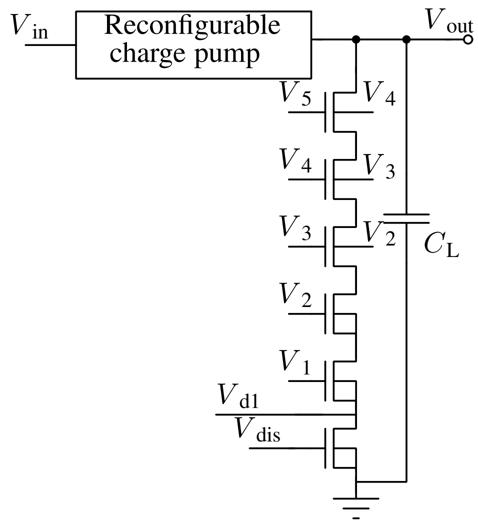


Figure 3.16: Schematic of discharge stage. The gate voltages of the transistors are taken from the outputs of intermediate stages of the charge pump.

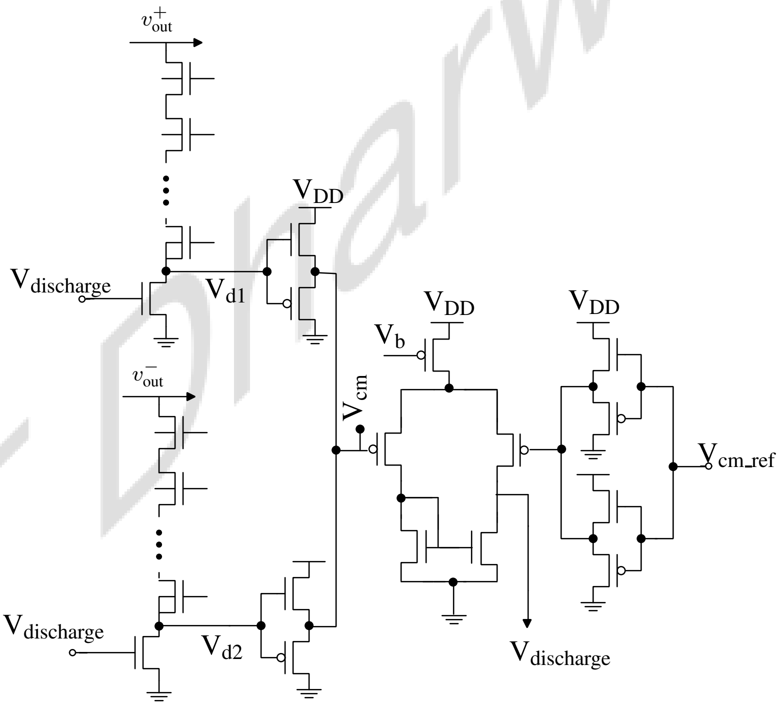


Figure 3.17: Schematic of the common mode detector and common mode error amplifier used as CMFB circuit for monitoring and setting the common mode of two charge pumps' differential output.

The maximum actuator output voltage for counterbalancing an acceleration of ±4 g is approximately 15.62 V (10.5 ± 5.12 V), and the accelerometer is designed for a bandwidth of 100 Hz. Hence, the maximum slope of the charge pump output is about 6300 V/s. Based on the input capacitance of the actuator (approx. 1 pF), the discharge current required for correct operation when the charge pump voltage is 25 nA.

It is difficult to detect common mode of the high-voltage outputs of the charge pumps. In this design, the CMFB circuit compares the common mode of drain voltages of the two current sources of the respective discharge stages of the two charge pumps, and compares it with the reference as shown in Figure 3.17. The reference voltage is fixed at design time using a voltage divider from the 3.3 V supply [44].

Digital control circuit

Figure 3.18 shows various blocks of the digital control circuit, which sets the digital word, Q. It comprises a window comparator, a 5-bit Full Adder (FA), and D Flip Flop (DFF). The con-

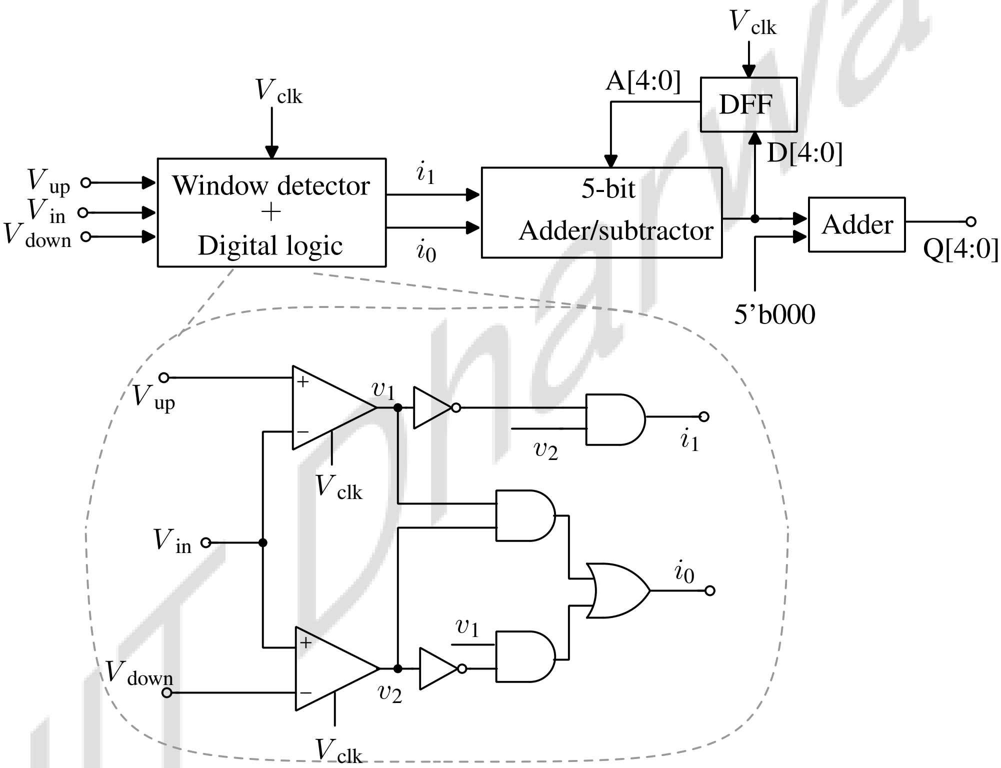


Figure 3.18: Block diagram of digital control logic. V_{up} and V_{down} are the pre-defined threshold against which the input is compared. The 5-bit digital word,Q generated by this block controls the efficiency of the charge pump by varying the number of capacitors that should be coupled with the clock signal.

troller output is given as input to the window comparator, which compares it with pre-defined threshold values. The output of the window comparator acts like control signals and decides if the efficiency should be incremented or decremented. The adder/subtractor circuit takes current Q value and control signals as input and based on control signals, increments/decrements value of Q. The digital circuit output, Q, through a 5:16 thermometer decoder, drives the coupling capacitors through AND gates. Each block of the control circuit is discussed in detail in this section.

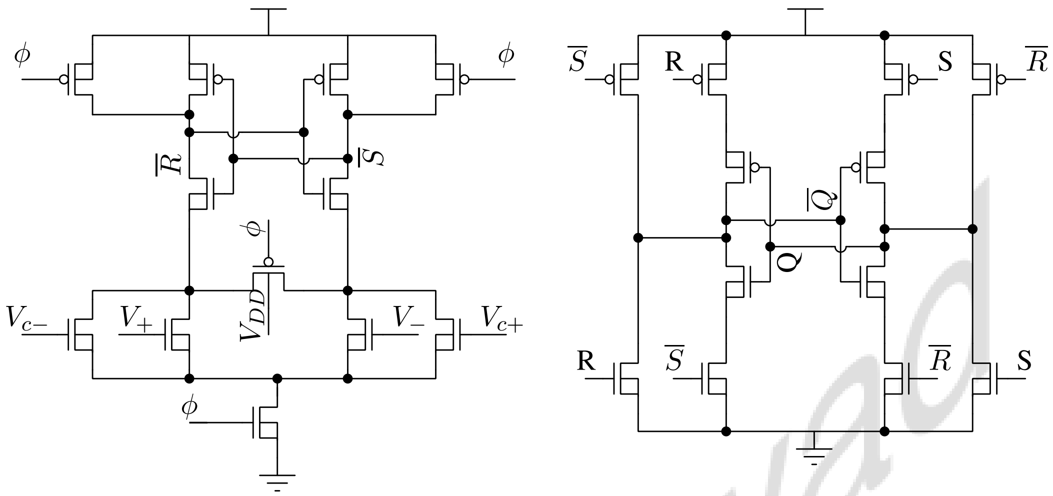


Figure 3.19: Schematic of latch-based comparator used in window detector circuit used in digital control logic shown in Fig. 3.18.

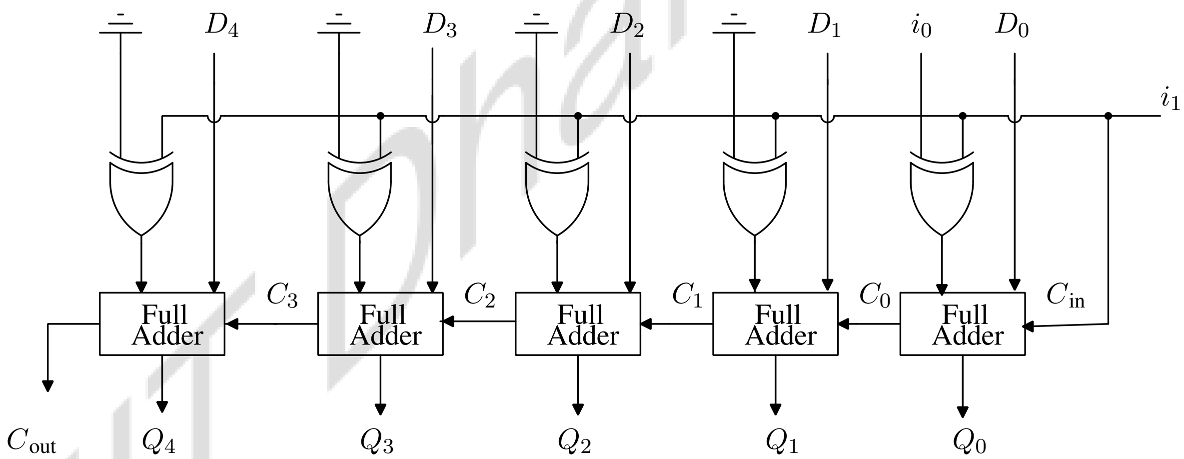


Figure 3.20: Schematic of adder/subtractor block used in digital control logic shown in Fig. 3.18. C is the carry, D is the input and Q is the output.

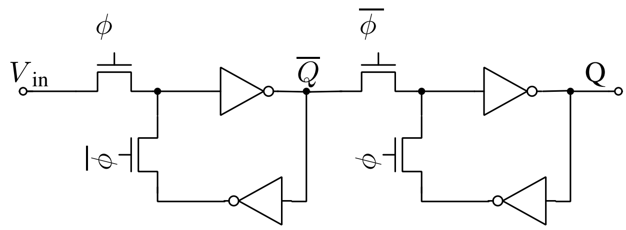


Figure 3.21: Schematic of D flip-flop(DFF) block in digital control logic shown in Fig. 3.18.

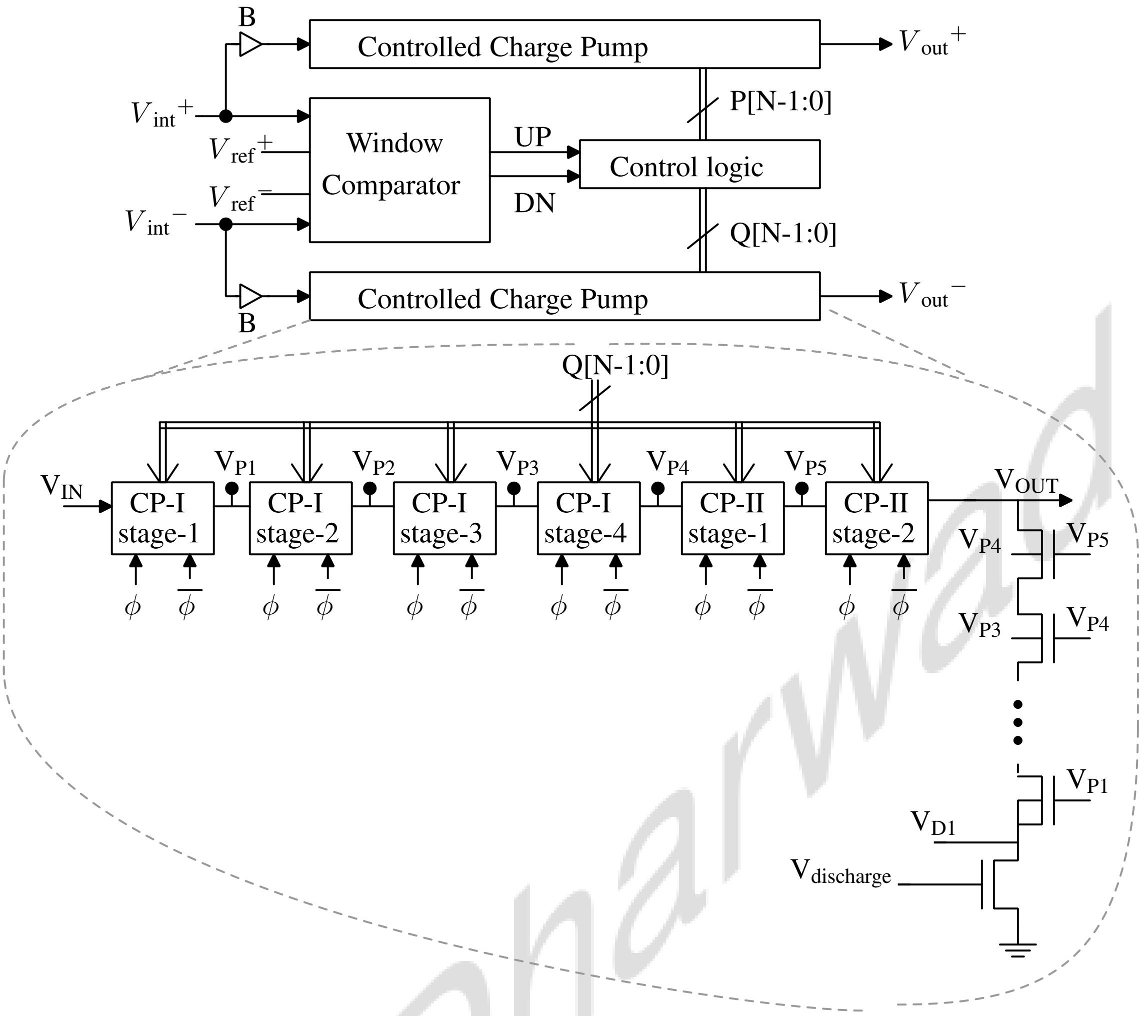


Figure 3.22: Block diagram of reconfigurable charge pump with discharge stage along with the control logic. The schematic of subpumps CP-1 and CP-2 are shown in Fig. 3.11.

The window detector consists of two comparators and a combinational logic, that compares the input with two predefined inputs and generates control signals that are given to adder block as shown in Figure 3.18. The comparator is a strong arm latch, and the schematic is shown in Figure 3.19. Figure 3.20 is a schematic of adder/subtractor circuit. A D flip-flop feeds the current input back to the adder block, which is incremented/decremented based on the control inputs. Figure 3.21 shows the schematic of the D flip-flop. At the common mode, the input to the digital block is at 1.7 V and Q = 5'b00100, resulting in the output voltage of 10.5 V from the charge pumps.

Figure 3.22 is a block diagram of digital block with charge pump. Two charge pumps generate the differential signals that drive the actuator. Complementary source follower buffers shown in Figure 3.5 are added between charge pump and integrator output to prevent coupling between the first stage of output and integrator. Table 3.1 summarizes the working of digital control logic:

Input to window detector	Output of digital logic	Adder/subtractor circuit operation
$V_{in}>V_{up}$	i_1=0, i_0=1	Increments current value of D by 1
$V_{in}<V_{down}$	i_1=1, i_0=1	Decrements current value of D by 1
$V_{down}<V_{in}<V_{down}$	i_1=0, i_0=0	No change in D

Table 3.1: Table showing the outputs of different blocks present in digital logic shown in Figure 3.18 that controls the efficiency of charge pump.

3.4 Closed-Loop simulation

3.4.1 Stability of closed loop

Figure 3.23 shows complete schematic of CLIP-SGFET based accelerometer. As discussed earlier, the first stage of the closed loop system is differential amplifier with IP-SGFETs as input pair [44]. The MEMS+ LUT has the behavior of both the suspended gate and the actuator. It takes external acceleration and the feedback voltage as input and gives the final width of IP-SGFET as output. This effective width is given as input to TCAD LUT of the device as shown in Figure 3.23. The input capacitance of the actuator is modeled as a capacitor, C_{L}. The second stage is a Gm-C integrator. The integrator output is used as input to the charge pump (and its control logic), generating the required high actuation voltages to drive the actuator.

The dominant pole of the system at V_{int}^{+}, V_{int}^{-}, is controlled by the integral controller. The pole is at 1.33 MHz. The second pole is due to the load capacitor, C_L(=1 pF) of the charge pump. The first stage of the system also has a pole due to parasitics at nodes V_1 and V_2 in Figure 3.23, which is at 11 GHz. These poles are far from the dominant pole. The charge pump in the feedback loop introduces a delay, thereby adding a zero to the system. The maximum delay is 70 μs.

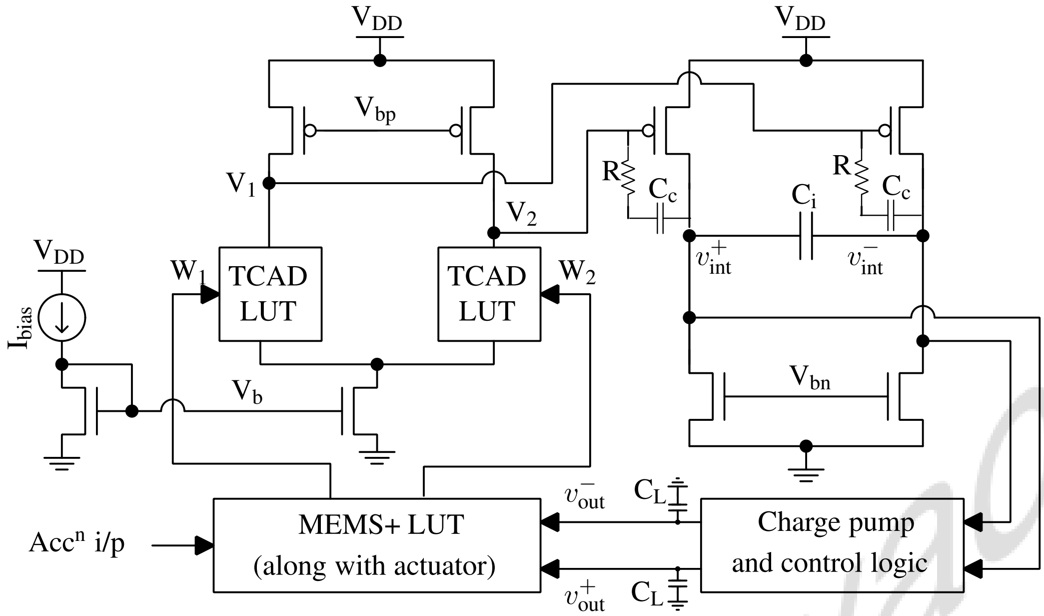


Figure 3.23: Complete schematic of the CLIP-SGFET accelerometer. Local common mode feedback loops are not shown .

As the system involves force feedback, LUTs, switched capacitor circuits, and other analog circuits, it is challenging to perform linear circuit analysis to estimate the system stability. Therefore, the Gm-C integrator capacitor, which controls the dominant pole of the system, is chosen by analyzing the system in time domain. Transient response of the circuit to a step input is observed, and integrating capacitor, C_i is set, ensuring that the system is overdamped and does not exhibit any ringing. Based on this, we estimate that the phase margin is at least $68°$.

3.4.2 Simulated results

Figure 3.24 shows the simulated results of the closed-loop accelerometer for a step-input of 2g. The charge pump output waveform shows the startup between 0 and 0.025 ms which shows that the system is damped and stable.

Figure 3.25 shows the simulated results of the closed-loop accelerometer for a sine input. The spikes in the width and charge pump outputs occur when the charge pump efficiency changes. The charge pump efficiency and gate position change when the integrator output exceeds the window comparator thresholds. Integrator outputs change when the negative feedback loop corrects for this excess voltage, which results in a spike. The amplitude of this spike corresponds to $\Delta\eta V_\phi$ = 0.67 V, where $\Delta\eta$ is the step change in the efficiency. The width of the spike is 30 μs, which corresponds to less than 0.3% of the period of the highest frequency of

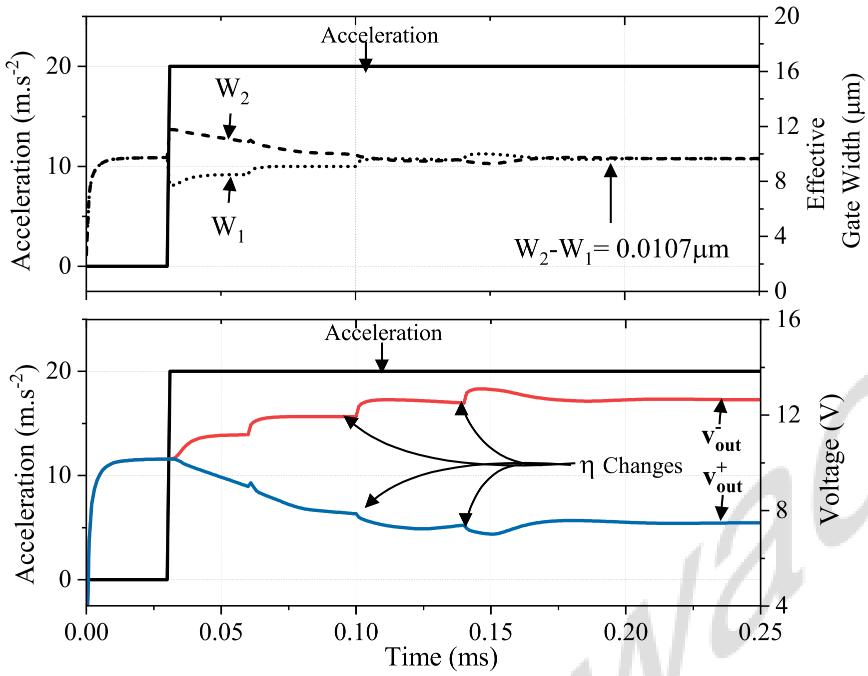


Figure 3.24: Transient simulation results of CLIP-SGFET based accelerometer in closed loop shown in Fig. 3.23 for a step input of 2g at 0.025 ms. The steady-state error is 10.7×10^{-3} g.

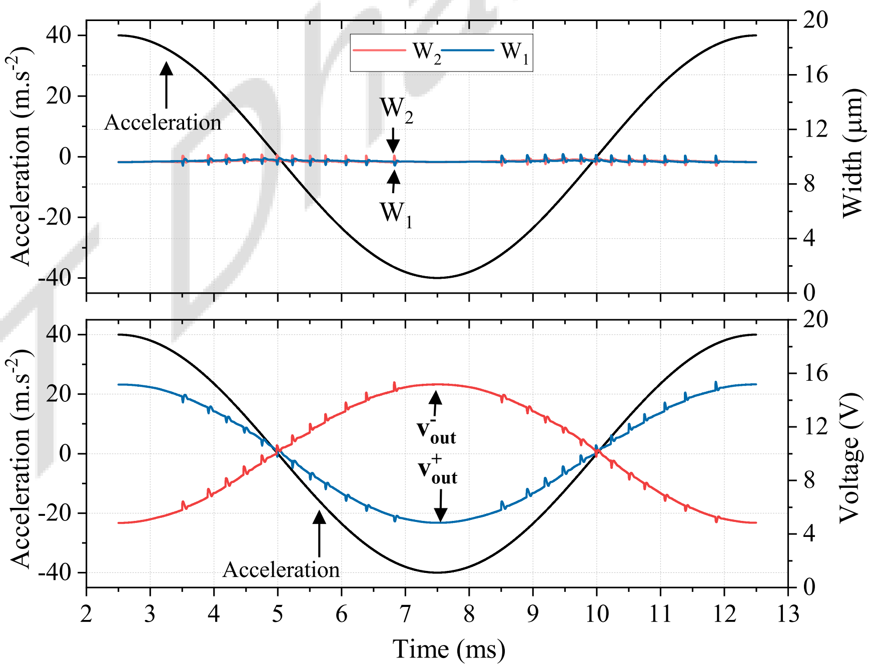


Figure 3.25: Transient simulation results of CLIP-SGFET based accelerometer in closed loop shown in Fig. 3.23 for a sine input at 4g, frequency 100Hz.

expected acceleration (100 Hz). It is outside the bandwidth. The noise due to ripple at 50 MHz is also attenuated due to the limited bandwidth of the mechanical structure and the integrator.

Chapter 4

Design improvements in reconfigurable charge pump

This chapter discusses the limitations in the design of reconfigurable charge pump design.

1. The latency of the charge pump varies over a wide range from 430 ns to 70 μs. The closed-loop should be designed to operate in stable conditions over that entire range.

2. The effective resistance of the MOS-based discharge stage used is not constant. It varies because the voltages driving these transistors are outputs of intermediate charge pump stages. These values vary with input. In addition to that, it is different for both charge pumps at any given instant. This leads to inaccuracy in common-mode measurement.

This chapter discusses circuit level design techniques to improve the performance of the reconfigurable charge pumpand optimize the system.

4.1 Bootstrap-switch based reconfigurable charge pump

Figure 3.15 shows the simulated results of the charge pump output at different efficiency settings. As mentioned earlier, the latency of the charge pump is not the same for all efficiency settings. The variations in the latency still vary over a wide range from 430 ns to 70 μs. The closed-loop should be designed to operate in stable conditions over that entire range.

Also, while operating in differential mode, one of the charge pumps settles faster than the other, resulting in common-mode shift and inaccurate common-mode measurement as shown in Figure 4.1. Due to this, it is necessary to reduce the latency variations.

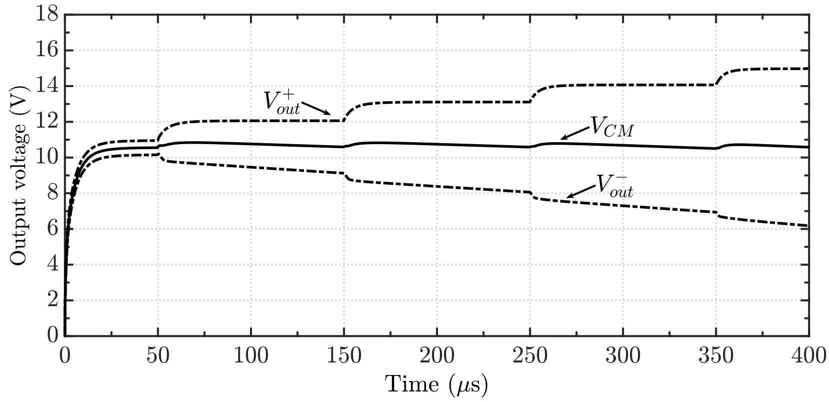
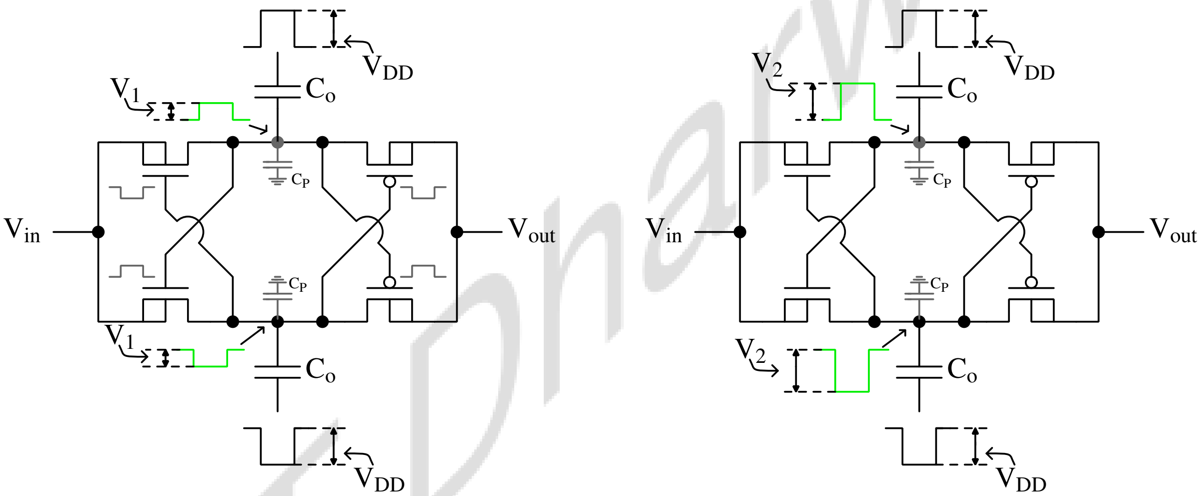


Figure 4.1: Simulated results of charge pump outputs showing the shift in common-mode voltage which is corrected by the common-mode feedback circuit. The charge pump circuit is designed for a common-mode of 10.5 V. As the charge pump with higher efficiency settles before the other charge pump, the common-mode voltage shifts up and settles back to 10.5 V.



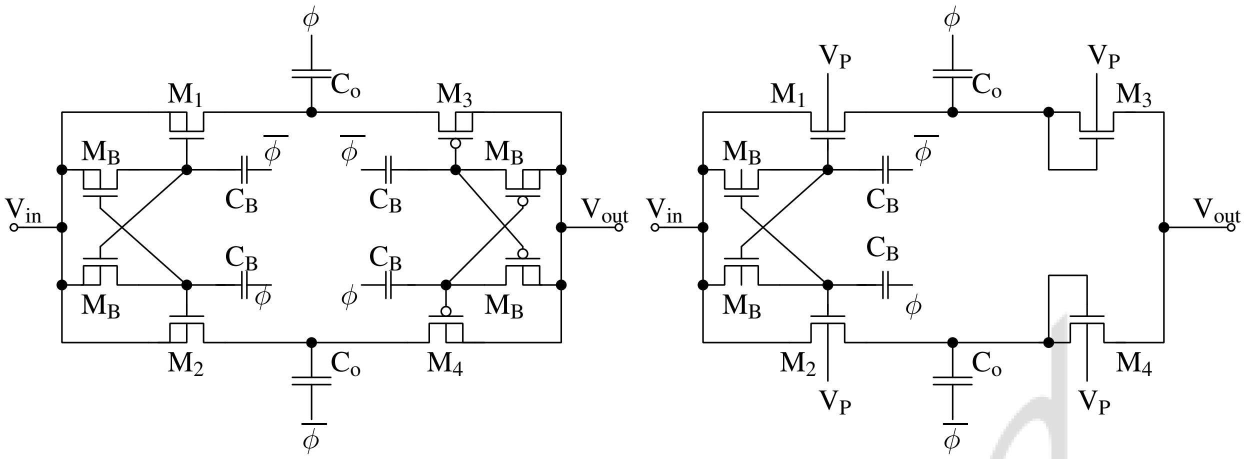
(a) At least efficiency, coupling capacitor, voltage added by each stage of CP-I is V_1 = 0.5 V. Here, C_o = 250 fF, Efficiency = η = 15.15%.

(b) At highest efficiency, coupling capacitor, voltage added by each stage of CP-I is V_2 = 2.3 V. Here, C_o = 1.45 pF, Efficiency = η = 69.69%.

Figure 4.2: Schematic of CP-I showing the voltage jump at the intermediate node at least and highest efficiency settings. The jump is not equal which implies that V_{GS} is different. Due to this, on-resistance also different and hence the time constant.

The reason for these variations is variation in time constant of the charge pump, which is discussed in detail next. The gate-to-source voltage applied to the gate of the transistors, V_{GS}, is not constant at all efficiency settings. Figure. 4.2 shows that the gate voltage added differs at two efficiency settings.

The value of V_{GS} is the jump seen at intermediate node when clock is applied. It varies



(a) CP1

(b) CP2

Figure 4.3: Schematic of charge pumps with bootstrapped switches.

with the efficiency of the charge pump, changing the on-resistance, $R_{DS(on)}$ of MOSFETs, governed by 4.1. This changes the time constant of the charge pump, thus varying the latency.

$$R_{DS(on)} = \frac{1}{\mu_n C_{ox} \frac{W}{L} \left(V_{GS} - V_{TH}\right)} \tag{4.1}$$

By maintaining V_{GS} constant, these variations in latency can be eliminated. This can be achieved by using bootstrapped switches [50], as shown in Figure 4.3. This circuit design technique decouples the gate and source terminal of the transistors. The gate is directly coupled to clock signals through the capacitor, C_B, and is not controlled by the intermediate node voltage. This way, the on-resistance of the switch is maintained constant.

Results

Figure 4.4 shows the simulated output voltage values at various efficiencies. The small variations are due to the CP - II due to diode-connected NMOS transistors as shown in Figure 4.3. The settling time varies over a very small range of 2.8 μs(approx.). The output voltage range from 1 V(latency=2.8 μs) to 16.8 V(latency=200 ns).

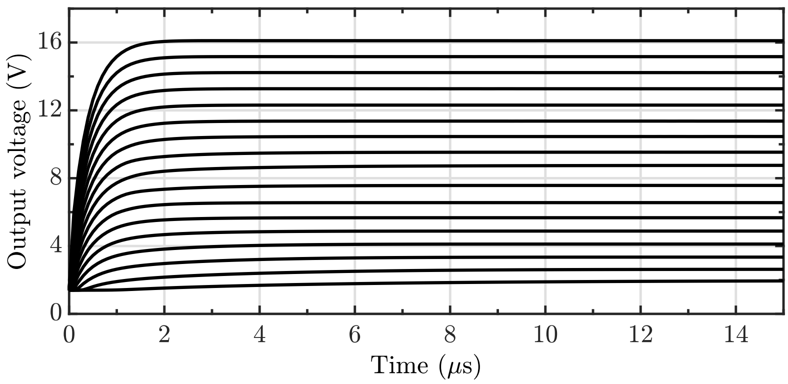


Figure 4.4: Simulated results of output voltage at various efficiency settings for the schematic shown in Fig. 4.3. The small variations are due to the CP - II as they dont have bootstrapping for diode-connected NMOS transistors.

4.2 Discharge stage for charge pump

The conventional discharge stage design discussed in Section 3.3.1 provides a path for the discharge of the output node when the input reduces. The voltages driving gates of these transistors are the intermediate stage outputs of the reconfigurable charge pump, and these voltages change with the input. This results in a change in the on-resistance of each transistor and, thus, the effective resistance of the discharge path. While operating in the differential mode, the effective resistance is not the same for both the charge pumps, which leads to inaccurate common-mode measurement. It is necessary to accurately control the common-mode of the actuator. In this section, Mutli Level Converter (MLC) based discharge stage is proposed as the discharge stage for the charge pump, eliminating these limitations of conventional design.

4.2.1 MLC as discharge stage

MLCs are DC-DC converters containing capacitors and sets of complementary switches that conduct at regular intervals. Figure 4.5 shows a 3-level Flying Capcitor Converter (FCC) with one capacitor and two pairs of complementary switches. High-side switches, S_1, and S_2 complement low-side switches, $\overline{S_1}$, and $\overline{S_2}$. The circuit can be configured to provide multiple output voltage levels varying from 0 to V_{dc}. Hence, the circuit is called a Multi-level converter. This can be achieved with appropriate switching sequence for controlling the switches, which is discussed next.

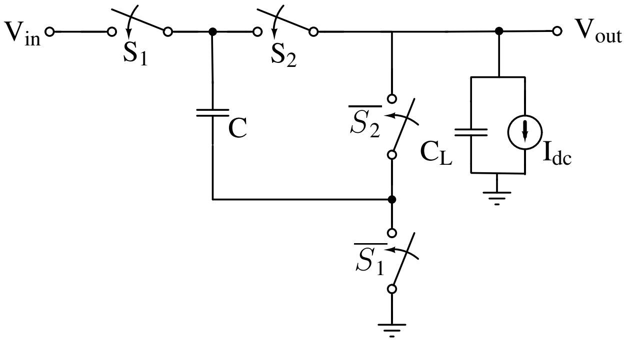
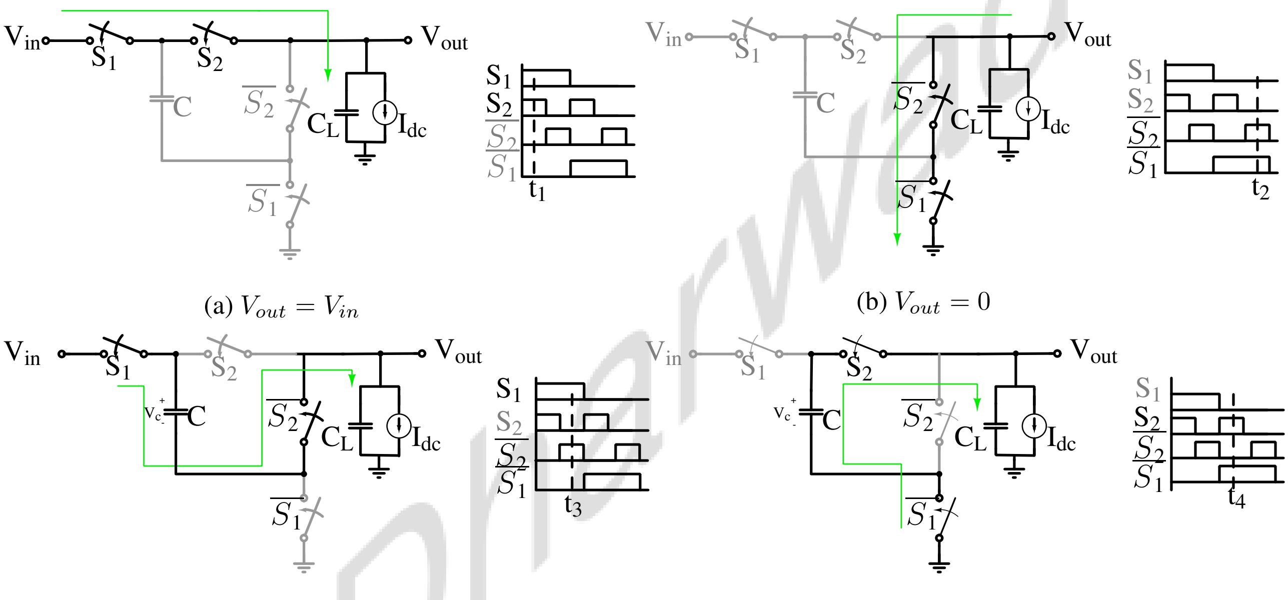


Figure 4.5: Schematic of 3-level Multi-level converter.



(a) $V_{out} = V_{in}$

(b) $V_{out} = 0$

(c) $V_{out} = V_{in} - V_c$

(d) $V_{out} = V_c$

Figure 4.6: Different output levels that can be generated by 3-level flying capacitor multi-level converter and the corresponding control signals for complementary switches.

The main challenge in FCC is achieving and maintaining the right voltage level on the fly capacitor,C. This voltage is useful is setting the output to desired level as shown in Figure 4.6. Various techniques for capacitor voltage balancing are reported in the literature. In this work, the voltage balancing is done by the active balancing through duty cycle compensation [51]. The charge is balanced when switching between the states and the converter generates a desired output voltage as shown in Figure 4.6(c) and Figure 4.6(d). The duty cycle is chosen based on the allowed ripple voltage range on the output. Figure 4.6 shows different output levels 3-level FCC can generate.

By using right switching control sequence, a multi-level converter can also be used to step down an applied input voltage without leading to breakdown of any junctions. This output is

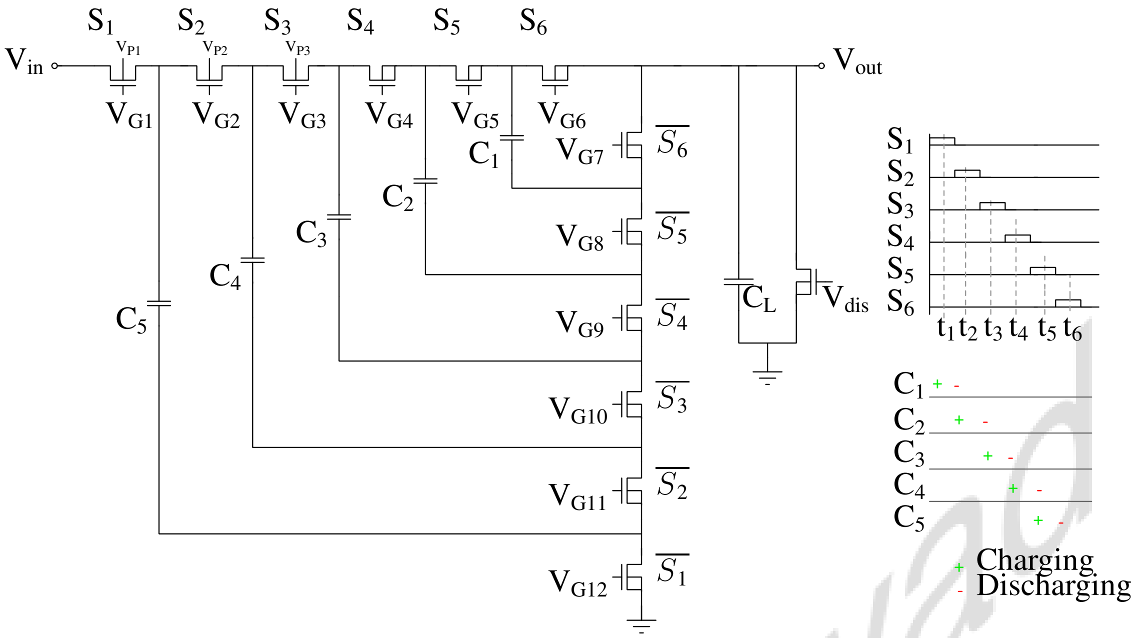


Figure 4.7: Schematic of 7-level flying capacitor multi-level converter with the switching sequence used to obtain the desired voltage level.

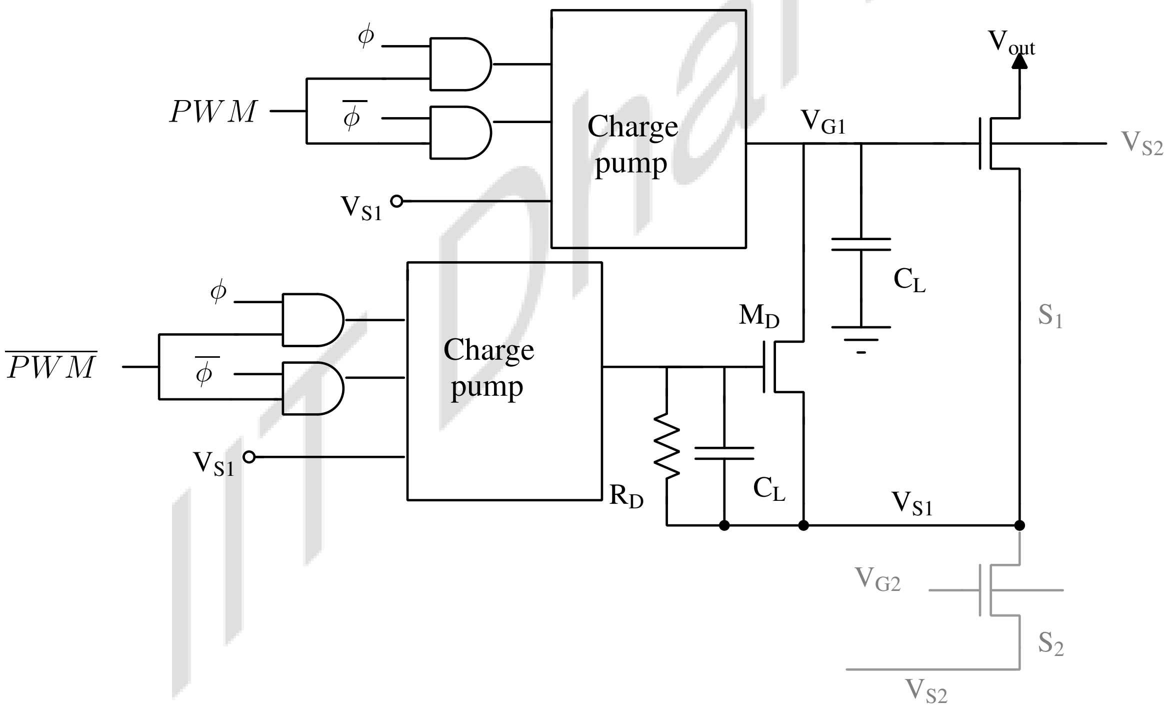


Figure 4.8: Schematic of gate drive for NMOS switches in MLCs.

used in detecting and setting the common mode of the charge pump and all the circuit involved in CMFB operate at the supply of 3.3 V which will be discussed later in this section.

For our application, we design a 7-level converter to step down the output of the charge pump (which goes up to 18 V). This maintains the right voltage difference across the terminals of all the switches without leading to a breakdown and final output is less than supply at which we operate. Figure 4.7 shows the schematic of a 7-level converter with NMOS transistors used as switches.

One significant challenge is generating the gate drive signals of these NMOS switches. Most of the switches are not ground-referenced, so it is necessary to shift the levels of the signals. Most of the work reported in the literature uses bootstrapped methods that generate the supply signals which drive the gate driver IC(like LM5113) [52]. Such methods are unfortunately not suitable for monolithic designs can't be implemented in this design. A compact, efficient design that can drive the gate of transistors is needed. Figure 4.8 shows the schematic of a proposed gate drive circuit. The switching or PWM signal acts as a control signal. The drain potential of the lower switch(source of the higher switch) is boosted by the charge pump, and the output voltage drives the gate of the switch when it needs to be on. When the switch is off, the transistor M_D is turned on, connecting the source and gate of the transistor S_1 making $V_{GS} = 0$.

4.2.2 Discharge stage and Common-mode feedback design

As two charge pumps are used to generate differential high-voltage signals, common mode feedback is necessary and it is difficult to set output common-mode at design time.

A common-mode feedback circuit is necessary. It is difficult to detect common mode of the high-voltage outputs of the charge pumps. In this design, the CMFB circuit compares the common mode of output of multi-level converters of the two charge pumps, compares it with the reference and adjusts the gate voltage as shown in Figure 4.9. The reference voltage is fixed at design time using a voltage divider from the 3.3 V supply.

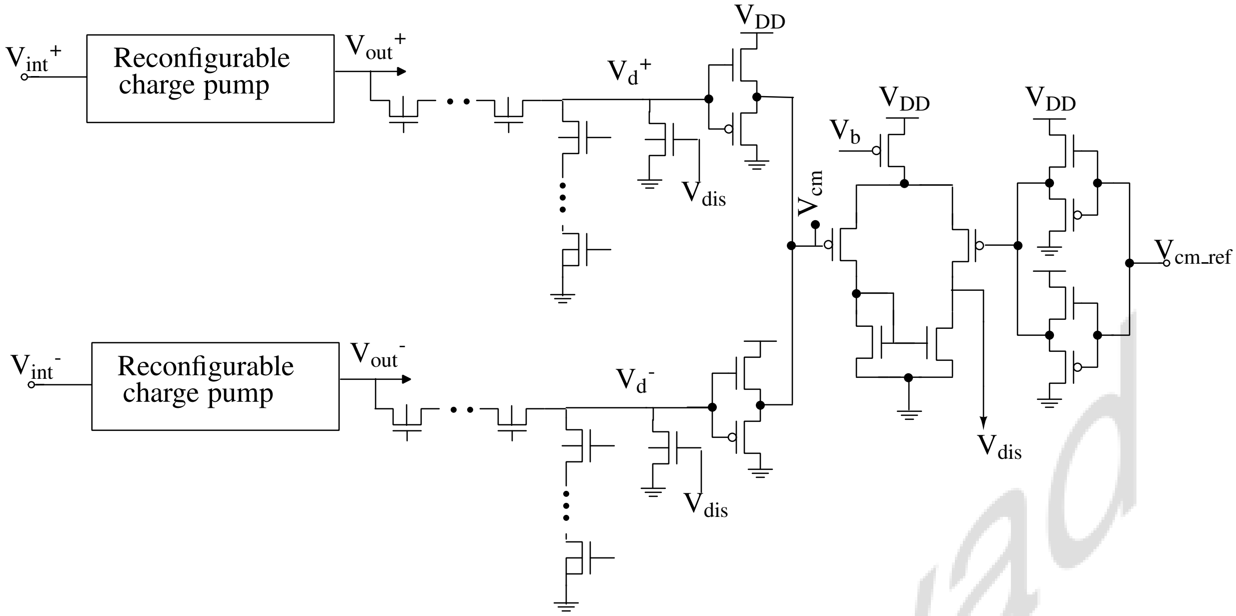
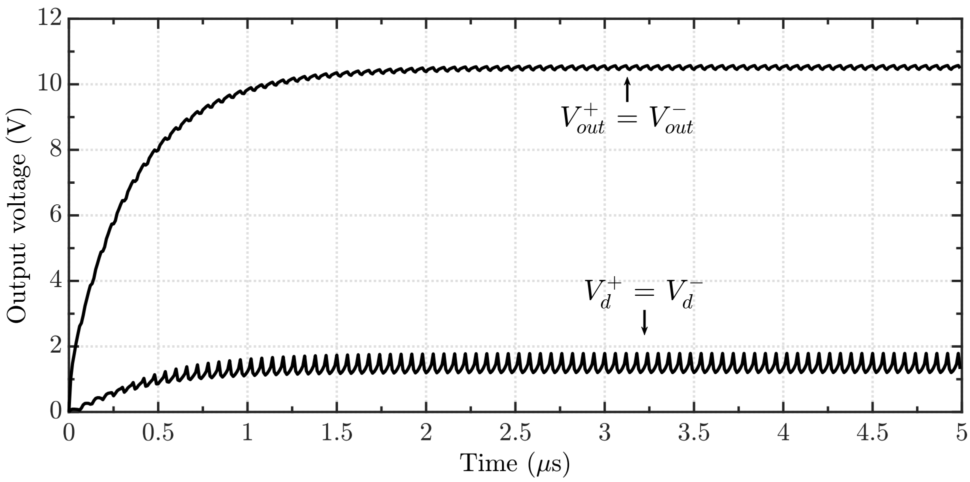


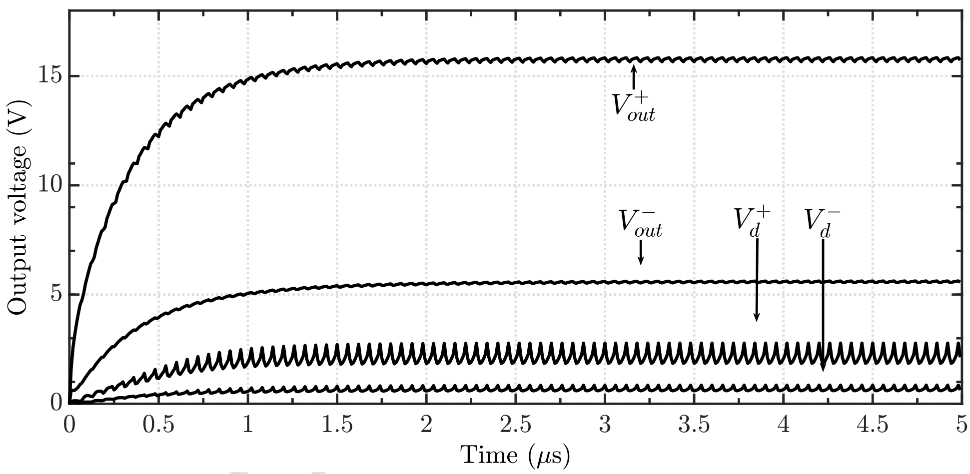
Figure 4.9: CMFB circuit for monitoring and setting the common mode. The average of the outputs of the multi-level converters used as discharge stage for charge pumps is compared with the reference and the gate voltage of transistor at the output node is set adjusting the current flowing through the converter.

4.2.3 Simulated results

Figure 4.10 shows simulated results of charge pump with multi-level converter at common-mode and differential efficiency setting. V_{out}^+, V_{out}^- are charge pump outputs and V_d^+, V_d^- are discharge stage (multi-level converter) outputs.



(a) At common-mode, $V_{out}^+ = V_{out}^- = 10.52\ V$ and $V_d^+ = V_d^- = 1.39\ V$.



(b) With differential efficiency setting, $V_{out}^{+} = 15.77\,V, V_{out}^{-} = 5.58\,V$ and $V_{d}^{+} = 2.16\,V, V_{d}^{-} = 0.67\,V$.

Figure 4.10: Simulated results of digitally reconfigurable charge pump with MLC based discharge stage.

Chapter 5

Accurate device and circuit modeling techniques for optimizing CLIP-SGFET based accelerometer system

The following are the limitations in measurement and implementation of CLIP SGFET-based accelerometer system offering scope of improvement in measurement, simulation accuracy and performance:

1. The LUT for actuator used in closed-loop simulation of the system does not consider the transient behavior of the actuator.

2. The input capacitance of the actuator is modeled as a constant 1 pF capacitor. An accurate model for the actuator considering the variations in input capacitance with external acceleration and feedback voltage is needed.

3. Very high simulation time as the complete closed-loop system has to carried out for a long interval (as it is designed for bandwidth of 100Hz) with very small sampling intervals. Small sampling intervals are due to the high frequency clock at which the charge pump operates (50 MHz).

5.1 Issues in accuracy of measurement

5.1.1 Actuator model

As discussed in section 3.1.2, actuator is modeled by doing electromechanical simulation in MEMS+. Voltage is applied across the combs and the displacement in proof mass is measured.

The issues in the existing actuator model are as mentioned below:

1. The LUT generated won't capture transient behaviour of actuator.

2. The input capacitance is modelled as 1 pF capacitance, but it depends on the external acceleration on gate and voltage applied across it.

Due to the inaccuracy in actuator model, the stability analysis performed on the closed-loop is not accurate. In this section, we propose a better modeling technique for the actuator. A circuit equivalent for the actuator is presented which is designed using passive elements. To verify the reliability of the design, AC response of circuit is compared with actuator structure.

AC analysis of the actuator structure is done in MEMS+ for a DC operating point of $10.5\ V$. Bode plot for the analysis is extracted. The plot obtained has a peak at 505 Hz(resonant frequency of the actuator), as shown in Figure 5.1.

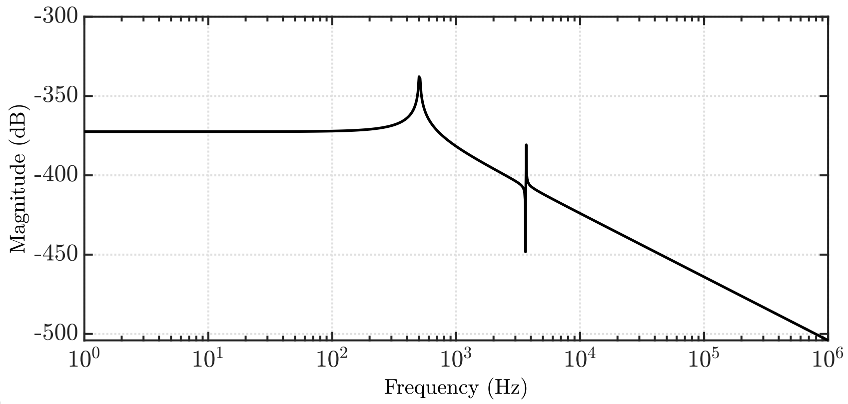


Figure 5.1: AC response of actuator structure

Figure 5.2 shown block diagram of the equivalent circuit. It consists of a resonator in parallel with low-pass filter. The AC responses of the actuator and the passive circuit are matched, verifying the accuracy of the circuit model of the actuator. The simulated results of the AC response of the actuator block and the equivalent circuit in Figure 5.2 are presented in Figure 5.3. This model can be used instead of LUT-based model in closed-loop simulations.

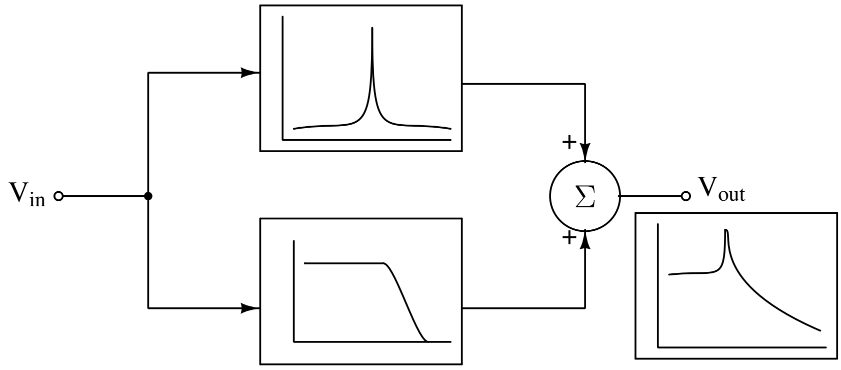


Figure 5.2: Block diagram of circuit modeling the actuator structure. Actuator is modeled by connecting a resonator with resonant frequency of 505.18 Hz in parallel to a second-order LPF.

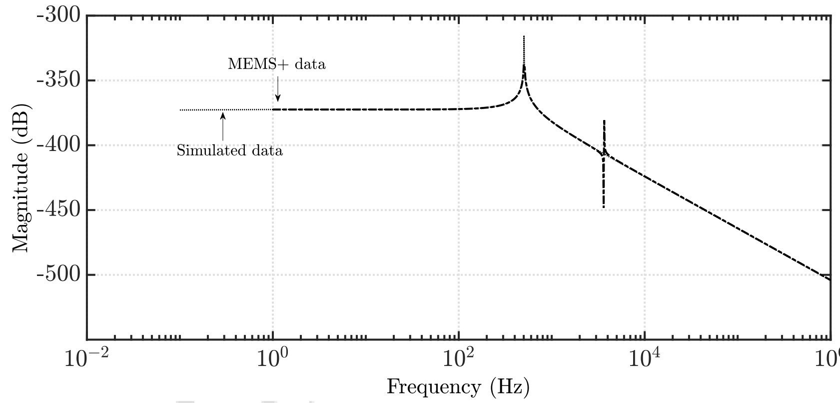


Figure 5.3: Simulated AC response of actuator structure and circuit model.

This input capacitance of the actuator is modeled as a 1 pF capacitor shown in Figure 3.23, and the closed-loop design and simulation were done. But the effective capacitance depends on the overlap area which is dependent on input acceleration and the feedback voltage from charge pump. These variations also need to be captured. To achieve this, DC simulation of the actuator is performed in MEMS+. The electrical drive voltage applied to the actuator is varied from 0 to 6.4 V with a common mode of 10.5 V (and step size of 0.5 V) for an external acceleration range of $\pm$ 5g, and the input capacitance values are extracted as LUT. The input capacitance can be modeled using Verilog-A with LUT which can be used in Spectre simulations, which is part of future work.

5.2 Simulation acceleration

The accelerometer system is designed for an input acceleration bandwidth of about 100 Hz. So, closed-loop simulation must be carried out for at least 20 ms to obtain reliable results. The sampling rate is very high due to the high operating frequency of the charge pump. Due to this, simulation must carried out for a long time with a very small sampling interval, thus resulting in a very high simulation time.

This section proposes techniques to model the reconfigurable charge pump accurately with the discharge stage and the clock signal generator, which can be used in the closed-loop simulation. This will reduce the simulation time of the system significantly. The entire system can be simulated and optimized with the using equivalent model of the charge pump.

5.2.1 Black box model for charge pump

It is difficult to accurately model the charge pump using linear circuit elements as it is nonlinear due to the switching elements and a high-frequency clock. So, a black box method is efficient and accurate.

Neural networks can be used to model the charge pump. Neural Network (NN) recognize and learn patterns. It can be trained to approximate any relation without any knowledge regarding the internal working. They can be trained to learn the behavior of the charge pump and predict the output for new inputs. It consists of different layers which learn different features/parameters of the applied input. A network consisting of input, output, and hidden layers is called a multi-layer network. Inputs are applied to input layer. The neural network modifies it's parameters to fit complex, non-linear function which related the given input and output. It uses the training examples which are labeled to train.

Network Architecture

In this work, a 3-layer Time Delayed Neural Network (TDNN) is trained to predict the output voltage. TDNN is a multi-layer, feedforward neural network with a delay element associated with input layer. Figure 5.4 shows the structure of TDNN (adapted from [53]).

The structure consists of input layer, one hidden layer and an output layer. The input consists of time-series data and finite responses of the output are captured. This is the labeled data used in training the model. Each input has a connection to output from the units below

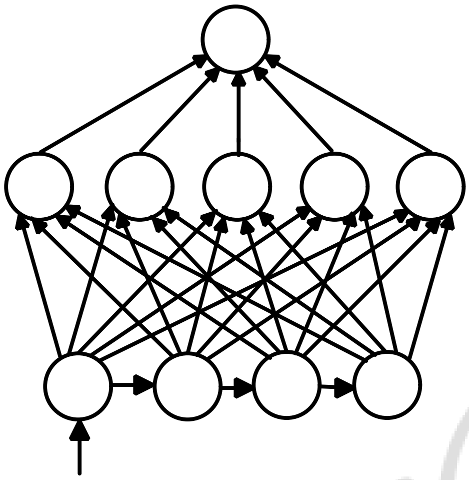


Figure 5.4: Structure of Artificial Neural Network.

and also past(time-delayed) versions of the same units. The output can be described using the following equation:

$$\hat{y}(t) = F(x(t), x(t-1), x(t-2), ...x(t-d)) \tag{5.1}$$

The number of nodes in the input layer depends on the settling time of the circuit and the rate at which signals are sampled. The dataset used in training is discussed in detail in the next section. The final number of nodes in the input, hidden, and output layers are 600, 16, and 1, respectively, with a total of 9633 parameters to be trained. A series of simulations were done on a defined dataset, and the trained model was tested.

Dataset

The dataset consists of the input voltage to the charge pump, the decimal equivalent of the 5-bit digital code controlling the charge pump efficiency, and the charge pump output voltage. The simulated results are sampled at a frequency of 100 MHz. Time-shifted copies of each input step are previously generated, stored as Comma Seprated Values (CSV), and loaded as network inputs while training. The simulated results of the bootstrap-switch-based charge pump were sampled until the output settled (about 2.5μs), resulting in 300 samples of each input. The final input dataset has 300 samples of two input signals, which results in a 600-neuron deep input layer.

Training

Training a neural network means finding appropriate weights and bias values of the neural network. Training is done to learn the behaviour of schematic shown in Figure 4.9. The goal is to arrive at the point of least error between the predicted and actual value. At first, the network parameters are randomly chosen which are later updated to reduce the error. The Figure 5.5 describes the process of neural network training.

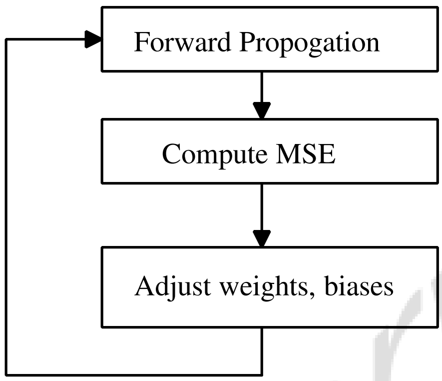


Figure 5.5: Block diagram explaining process of training neural network

The difference between this predicted and actual output is called the loss given by the 5.3. The loss function of the network is monitored to decide when the network training should stop to avoid over-fitting the data. The output of every node is given by the 5.2. The loss value is back-propogated updating the weights and bias parameters of the network.

$$\hat{y}(t) = \sigma(b_i + \sum w_{ij}x_i) \tag{5.2}$$

where, σ is the activation function, Rectified Linear Unit (ReLU), w_{ij} is the weight of connection between i^{th} node is current layer to j^{th} node in next layer, x_i is the input to i^{th} perceptron in input layer, b_i is the bias.

$$MSE_i = (y_i - \hat{y}_i)^2 \tag{5.3}$$

where, $y(t)$ is the expected output and $\hat{y}(t)$ is the output of neural network. The computed cost function is back-propagated, adjusting the network parameters of the network to reduce Mean Squared Error (MSE). It computes the gradient of the loss function with respect to the network weights. The adjustment depends on the weights contribution of the final error, given by 5.4.

$$
\begin{aligned}
w_i &= w_i - \left(\alpha \times \frac{\partial C}{\partial w_{ij}}\right) \\
b &\Rightarrow b - \left(\alpha \times \frac{\partial C}{\partial b_i}\right)
\end{aligned}
\tag{5.4}
$$

In the current work, the training was done for 6000 samples(time-shifted versions of 10 input steps) for 250 epochs. The model is evaluated to verify the performance of the trained network. The python codes are pushed onto git: `https://github.com/GangaKM/Charge-pump`

Prediction and Results

The network performance was tested on different set of time-shifted input data and the error was within 5%.

Verilog-A model

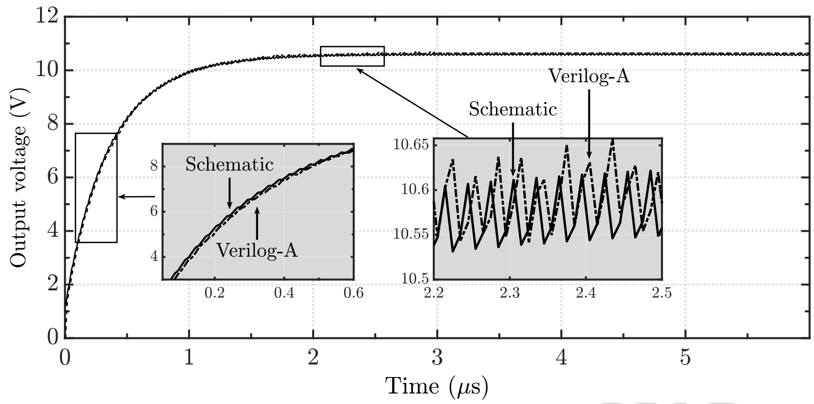
The weights and biases of the trained neural network are extracted after training, and are converted to Verilog-A arrays. As Verilog-A doesnt support 2-D arrays, the weights matrix of first layer is stored in 16 1-D matrices of size 600 each. $W_{1_1}, W_{1_2}, .., W_{1_{16}}$ are 1-D arrays of size 600, x is the input to network of size $(600, 1)$, O_1 is output of hidden layer, W_2 is the weight matrix of hidden layer of size $(16, 1)$, b_1 and b_2 are bias matrices of size $(16, 1)$ and $(1, 1)$ respectively.

$$
out_1 = \begin{bmatrix} O_1 \\ O_1 \\ \vdots \\ \vdots \\ O_{15} \\ O_{16} \end{bmatrix} = \sigma \left\{ \left(\begin{bmatrix} W_{1_1} \\ W_{1_2} \\ \vdots \\ \vdots \\ W_{1_{15}} \\ W_{1_{16}} \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_{599} \\ x_{600} \end{bmatrix} \right) + \begin{bmatrix} b_{1_1} \\ b_{1_2} \\ \vdots \\ \vdots \\ b_{1_{15}} \\ b_{1_{16}} \end{bmatrix} \right\}
$$

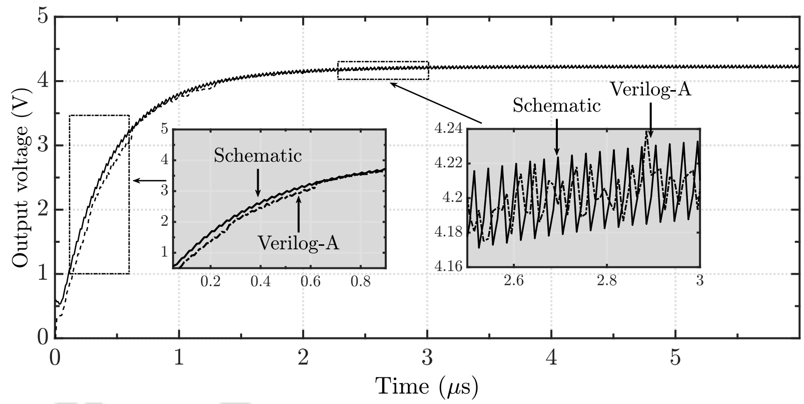
$$
output = \sigma \left\{ \begin{bmatrix} O_1 \\ O_1 \\ \vdots \\ \vdots \\ O_{15} \\ O_{16} \end{bmatrix} \times \begin{bmatrix} W_{2_1} & W_{2_2} & \cdots & \cdots W_{2_{15}} & W_{2_{16}} \end{bmatrix} + b_2 \right\}
$$

$$\text{where,}\sigma = \left\{\begin{matrix} x\text{ , if } x > 0 \\ 0\text{ , if } x = 0 \end{matrix}\right\}$$

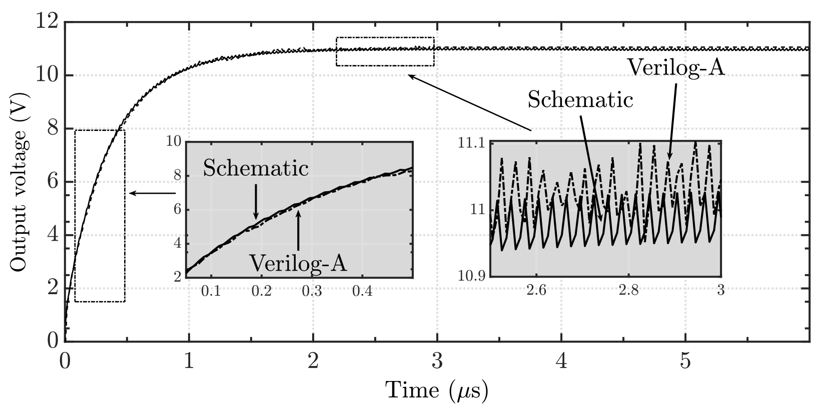
$output$ is the final output of network after passing through the non-linear activation function, σ which is defined above. The final output for each input sample is plotted in Cadence and verified against the circuit output. The simulated results are presented in Figure 5.6.



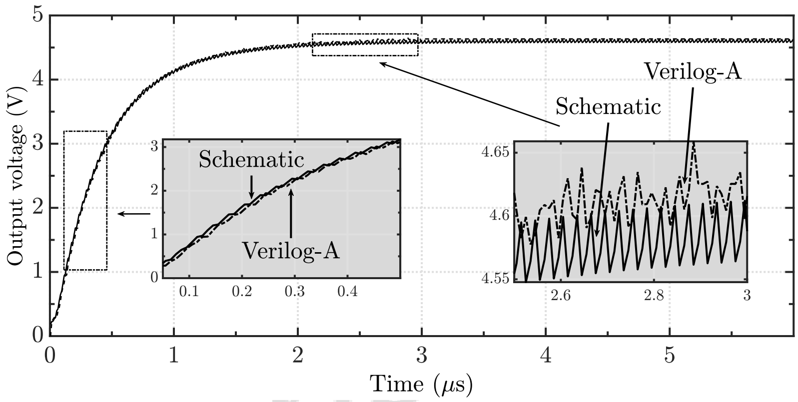
(a) V_{in} = 1.9 V, η = 34.84%



(b) V_{in} = 1 V, η = 15.15%



(c) V_{in} = 2.3 V , η = 34.84%



(d) V_{in} = 1.5 V, η = 15.15%

Figure 5.6: Simulated results of the charge pump circuit, black box and Verilog-A model of the dynamically reconfigurable charge pump. The results are presented in this figure. The practical of charge pump was simulated in Cadence. The efficiency and input to the charge pump were taken and a neural network was trained. The weights and biases of the network were extracted to build a Verilog-A model of the charge pump to improve the simulation speed of the system in Cadence Spectre.

Chapter 6

Design and simulation using open-source CAD Tools

Our proposal titled 'Design of Analog Front End (AFE) circuits for Closed-Loop IP-SGFET based accelerometer.' was accepted and was selected to be incorporated into the 2023 Universalization of IC Design from CASS (UNIC-CASS)(CASS-sponsored fabrication runs via the Efabless chipIgnite) program by IEEE Circuits And Systems Society (IEEE-CASS). It aimed to design and test our proposed closed-loop accelerometer design.

This chapter discusses the circuits we intend to fabricate and test in this program, shown in Figure 6.1. SGFETs, along with actuator, are fabricated separately. The input to the AFE comes from an IP-SGFET-based sensor that is external to the chip, as shown in Figure 6.1. The final read-out of the circuit is the acceleration, which can be computed from a measurement of the integrator output voltage along with the charge pump digital control word. The system was designed in Skywater130 nm technology using open-source CAD tools. A brief overview of these tools and PDK used in design and simulation is given, and presents post-layout simulation results. Finally, a test plan for the fabricated chip to verify open-loop, reconfigurable charge pump, and closed-loop operation is also discussed.

6.1 Overview of open-source tools

6.1.1 Design Tools

XSchem was used for schematic capture. XSchem is an open-source program for electronic design, creating electronic circuits, and visualizing an electronic circuit's various elements and connections. It is compatible with Unix-like systems. The software has some standard symbols and components that can be used in designing or integrated with other PDKs. We use Skywater's 130 nm PDK, known for its open access, in designing all the circuits involved in this work.

Simulation

The simulations are done in Ngspice. It can run various simulations on the circuits like transient, AC, DC, etc. It adheres to the SPICE standard, which defines commands and models for simulating circuits.

6.1.2 Layout tools

Magic is an open-source layout editor tool used to edit and draw the arrangements of different components on the chip. Magic includes Design Rule Check (DRC) features to help ensure that the layout adheres to specified manufacturing rules and guidelines. Magic also allows us to extract the netlist of the layout, which can be used to verify the post-layout functionality of the design.

Layout v/s Schematic (LVS) check

LVS is a crucial step in Integrated Circuit (IC) design flow. It ensures that the physical layout of the design matches the schematic representation. LVS was perfomed using netgen. Netgen takes both schematic and layout netlists as input and compares them with each other. The report generated due to the LVS check highlights any discrepancies between the schematic and layout, such as missing or extra connections, mismatches in component sizes, and other issues.

Post-layout simulation

A post-layout simulation can be performed in XSchem using the extracted SPICE file of the layout. The Simulation Program with Integrated Circuit Emphasis (SPICE) netlist can be exported into XSchem, and it is simulated with various input combinations and simulation conditions(like AC, DC, Transient, etc.) to verify the functionality of the design.

6.1.3 Tools for digital circuit design

OpenLANE is an open-source design tool that supports Register Transfer Level (RTL) to Graphic Design System II (GDSII) design flow, allowing designers to take their digital designs from the high-level description to the final layout suitable for fabrication. It uses a set a tools like yosis for synthesis, OpenROAD for place and route and Magic for layout, to perform various steps involved in digital circut design flow. Different stages of a digital IC design are discussed next.

Different stages of digital IC design

Digital IC design includes stages like synthesis, floorplanning, etc., each of which is crucial.

1. **RTL Design:** High-level description of the purpose and functionality of the design using Hardware Description Language (HDL) like Verilog. There are multiple abstraction levels: Behavioural, Gate, Transistor, etc. Functional verification is performed to verify correctness.

2. **Synthesis:** Generation of gate-level netlist using tools like Yosys followed by technology mapping.

3. **Floorplanning:** Physical layout planning of the IC, allocating space for different functional blocks, and ensuring proper connectivity and power signal distribution.

4. **Place and Route:** Placing the logic cells and connecting various metal rules adhering to design rules considering timing, area, and other crucial parameters.

5. **Clock tree synthesis:** To ensure efficient, reliable, and low-skew clock distribution in design.

6. **Physical Verification:** This step involves performing DRC and LVS checks on the design.

7. **Static Timing Analysis (STA):** STA is performed to verify the timing requirements of the design, like setup and hold time.

8. **Tapeout:** Preparing design for fabrication, including generation of GDSII files.

6.2 Circuits involved in design of CLIP-SGFET based accelerometer

This section discusses the design of different blocks in the shaded area in Figure 6.1, a closed-loop accelerometer with IP-SGFET, and an actuator in a feedback loop. The **shaded area** is the analog front end we intend to fabricate and test.

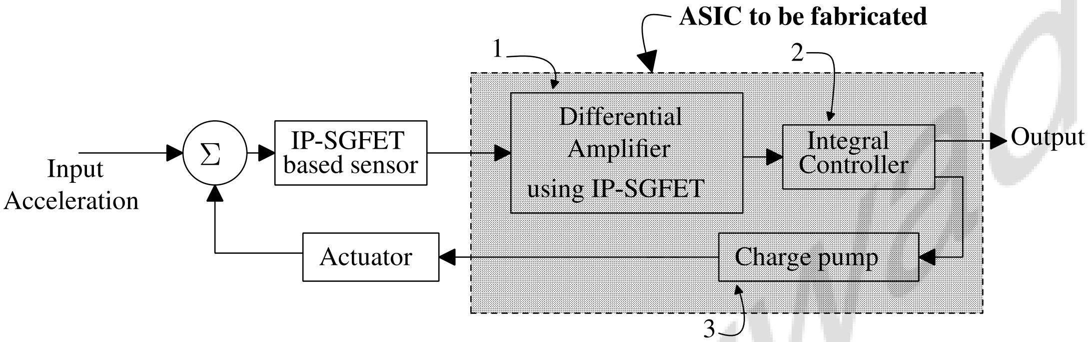


Figure 6.1: Block diagram of closed-loop accelerometer system with IP-SGFET and actuator in feedback driven by charge pump. The different blocks of the system include: (1) Differential amplifier with IP-SGFET as input pair. The complete schematic is shown in Figure. 6.2. (2) Second stage is Gm-C controller, schematic shown in Figure. 6.3. (3) Charge pump with digital control logic, shown in Figure. 3.23.

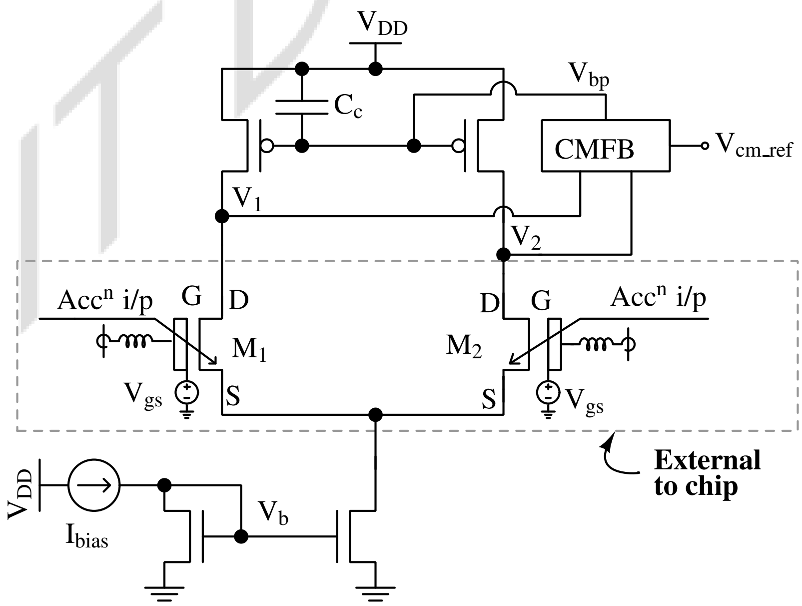


Figure 6.2: Schematic of fully-differential amplifier comprising IP-SGFET as input pair. The IP-SGFET based sensor is external to the chip.

The behavior of SGFETs is modeled using NMOS transistors for simulation purposes. These are used in the design of differential amplifier based interface circuit for SGFETs shown in Figure 6.2. The output signals of this differential amplifier are input to the charge pump and digital control block. The integrator and digital control logic, which sets the efficiency of the charge pump, were modified to optimize the design, which will be discussed in this section. The design of the rest of the circuits and various parameters considered in designing were discussed in Chapter 3.

6.2.1 Integral control

The design was modified by adding a tail current source to set the output common-mode level precisely and maintain all the transistors in saturation. The schematic of the modified controller is shown in Figure 6.3.

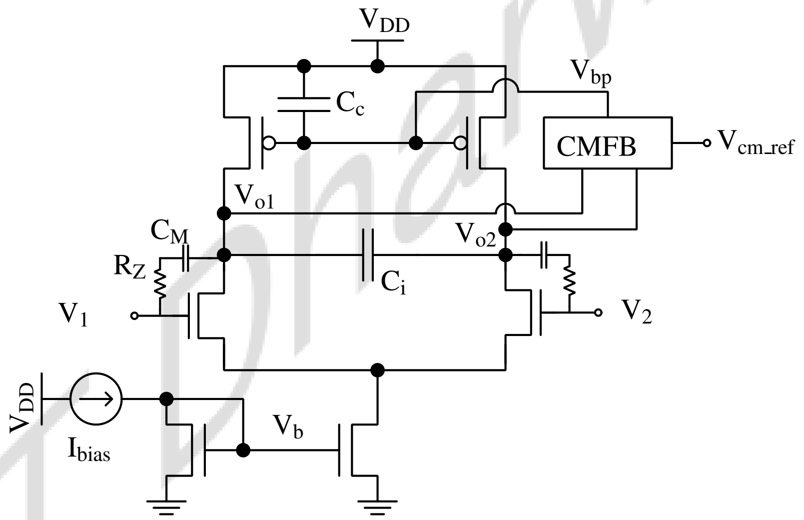


Figure 6.3: Schematic of integral control block. Here, M_{M} is the miller capacitor and R_{Z} is the zero-cancelling resistor. V_1 and V_2 are inputs to the controller from differential amplifier block. The signals, V_{o1} and V_{o2} are outputs.

The simulated results of the integrator are shown in Figure 6.4. The output saturates due to high gain of controller.

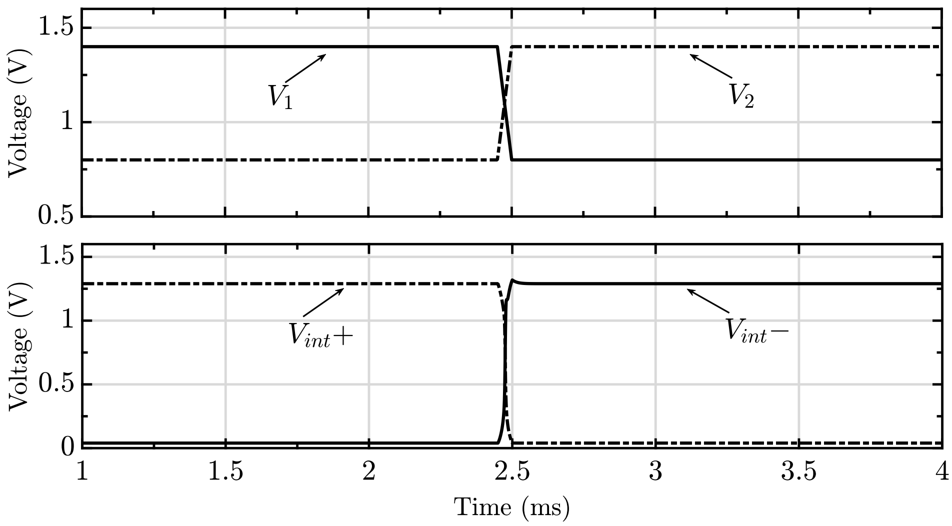


Figure 6.4: Transient simulation results of integral control shown in Figure 6.3. v_{in1}, v_{in2} are inputs and v_{o1}, v_{o2} are outputs to integrator. The output saturates due to high gain.

6.2.2 Digital control block

The digital control logic is needed for setting the efficiency of the charge pump by varying the number of capacitors coupled with the clock signal. The design for control logic was discussed in Chapter 3. An optimized design for the digital block is proposed, shown in Figure 6.5. This design eliminates the need for a binary-to-digital thermometer code converter circuit. This is achieved by using bi-directional shift registers. Various blocks in digital control logic as shown in Figure 6.5. The shift registers are populated with 1s when efficiency is to be incremented, as shown in Figure 6.7. V_{up} and V_{down} are the upper and lower thresholds to the window comparator which is 1.1 V and 0.7 V respectively are generated by the circuit shown in Figure 6.6.

Shift registers

The shift register used in digital control logic is bi-directional. A bi-directional shift register is a sequential circuit that can shift data leftwards or rightwards based on the control signal polarity(here, shift is the control signal). The inputs to the bi-directional shift register are connected to supply and ground. Based on the polarity of the shift signal, the 8-bit register is populated with 1s or 0s. The scan chain is included in the design to ease the testing process. The signals in blue are part of the scan chain used to test the working of DFFs in the shift register.

OpenLane tool provides an end-to-end automated digital design flow, covering various stages of IC design, including synthesis, place and route, etc, which is discussed in detail in

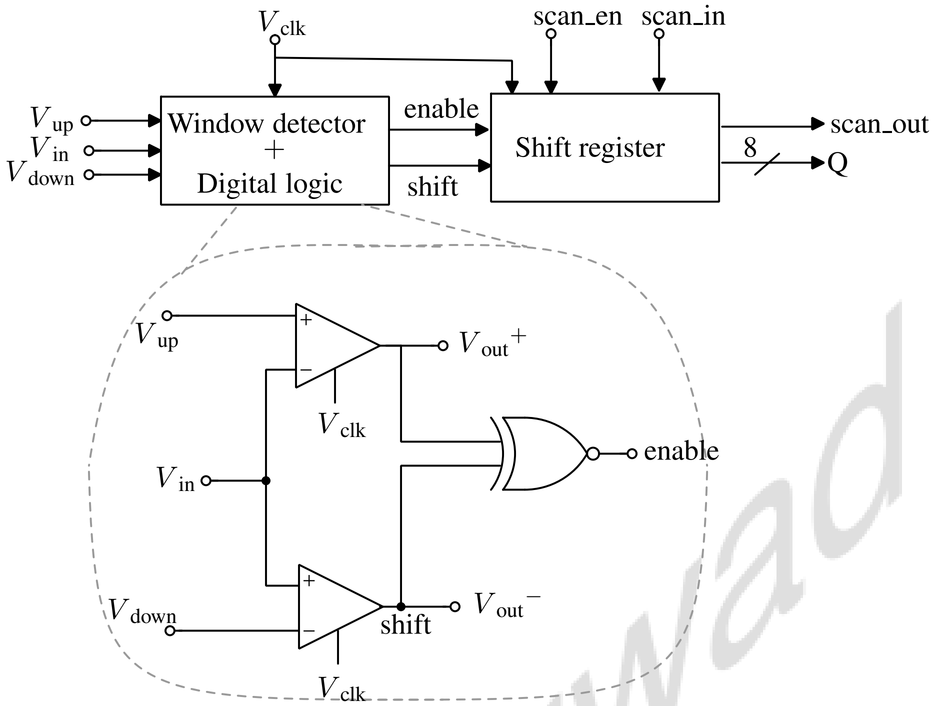


Figure 6.5: Block diagram of digital control logic. The 8-bit digital word, Q generated by this block controls the efficiency of the charge pump by varying the number of capacitors coupled with clock.

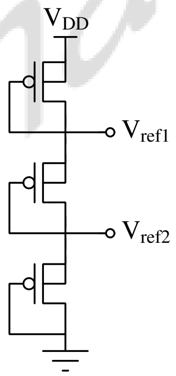


Figure 6.6: Schematic of PMOS based reference voltage generator circuit. V_{ref1} and V_{ref2} are reference voltage levels(1.1 V and 0.7 V) used as reference in CMFB and window comparator thresholds.

the next section. The design of shift registers with scan-chain in digital control logic shown in Figure 6.5 can be automated using OpenLANE. A Verilog code specifying the functions and behavior of the design at the behavioral level is simulated using iVerilog to verify the functionality. OpenLane generates the final layout and GDS. The netlist of the generated layout can be extracted to perform post-layout simulation in XSchem. The Verilog code is pushed to GitHub : `https://github.com/GangaKM/CLIP-SAFE`

Figure 6.7 shows the block diagram of RTL generated by OpenLane for the Verilog code.

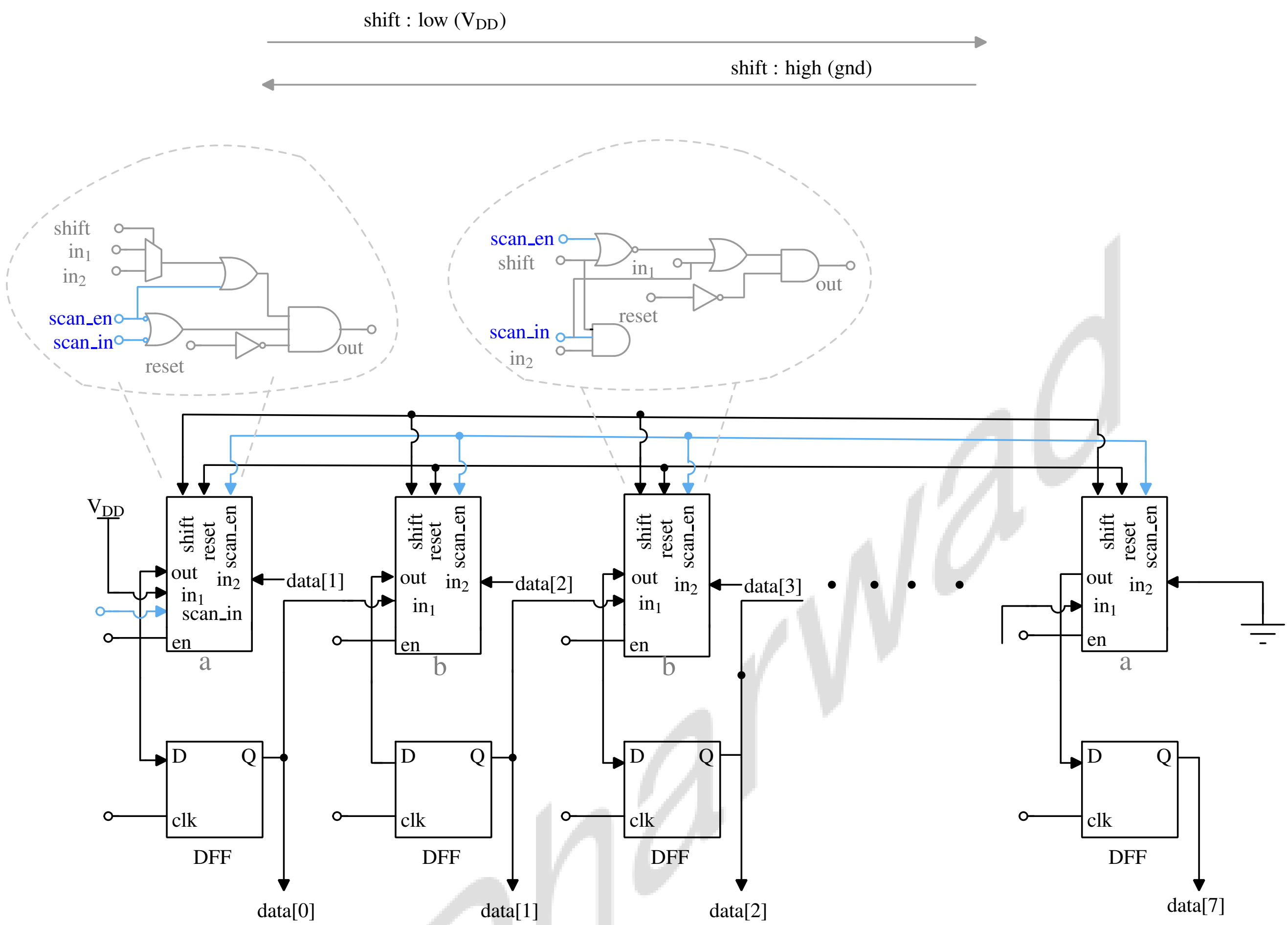


Figure 6.7: Schematic of shift register block. When shift is high, 0 is populated and when shift is low (decreasing value of Q), 1 is populated in shift register (increasing value of Q). Blue signals are part of scan chain. Input to scan chain is given to pin scan_in and MSB(data[7]) is output of scan chain.

6.3 Post layout simulation

The image of layout designed using SkyWater 130nm triple-well process is shown in Figure 6.8. It consumes an active area of 1.2 mm x 1.18mm. All the layout files can be accessed here : `https://github.com/GangaKM/CLIP-SAFE`

Post-layout simulation is performed to verify the design after the physical layout has been completed. It is performed in XSchem by applying inputs to layout netlist. The physical layout is converted to netlist along with parasitic elements using layout extraction tools. This section includes the post-layout simulation results of various blocks in the CLIP-SGFET-based accelerometer, designed using SkyWater 130nm technology.

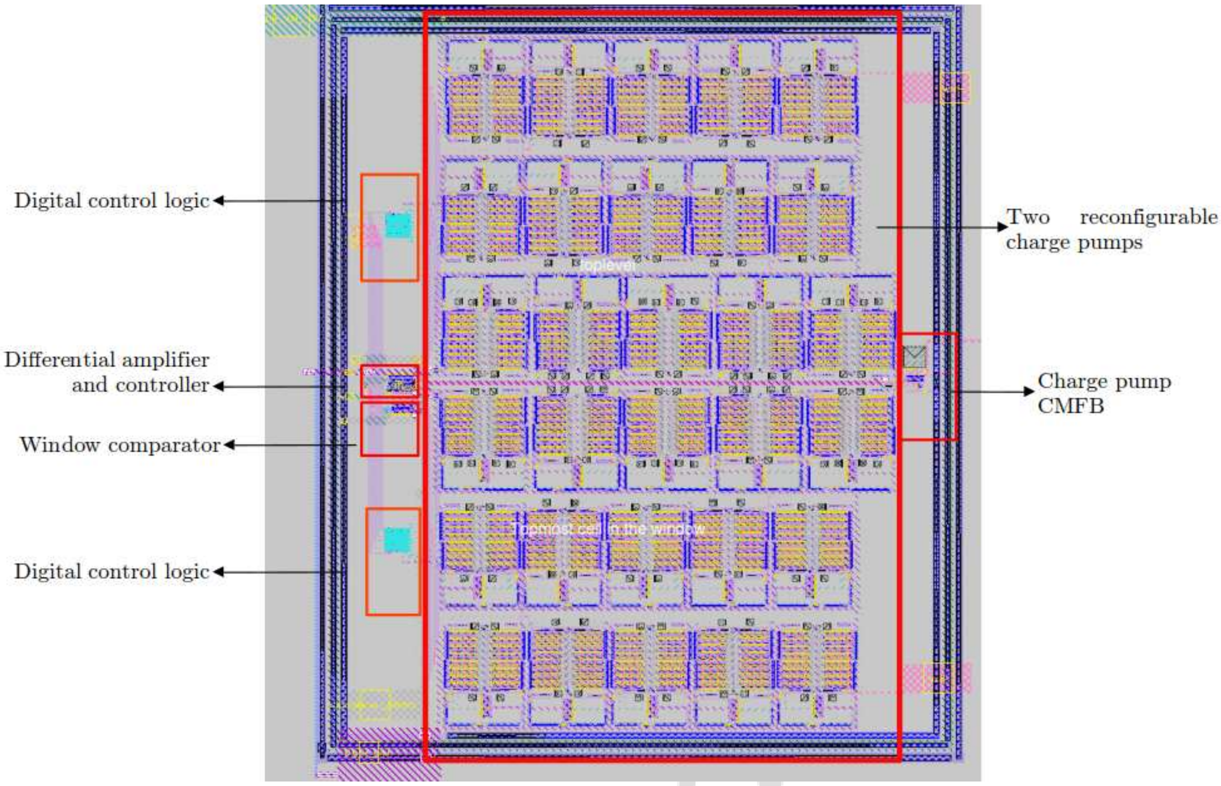


Figure 6.8: Layout of the proposed accelerometer with all the sub-circuits marked on it.

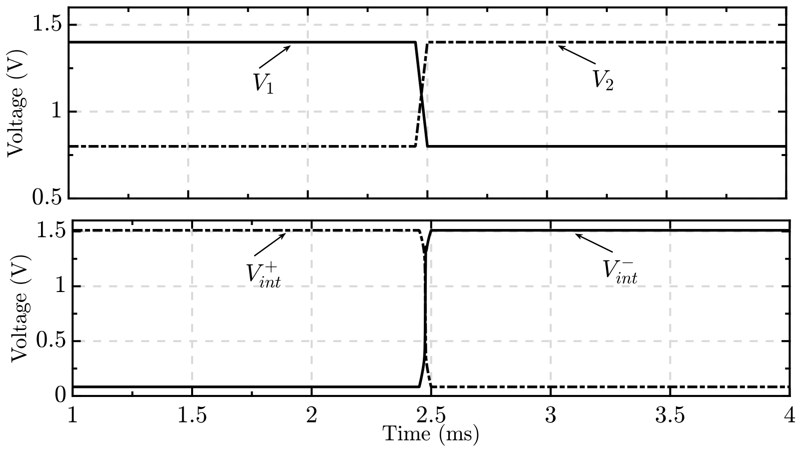
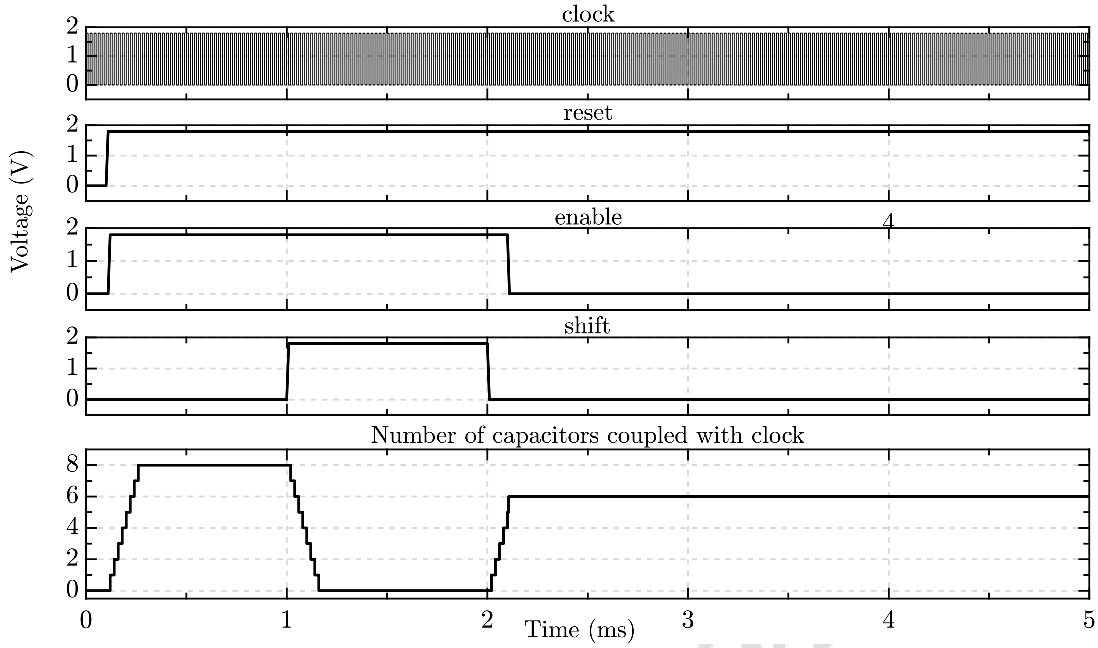


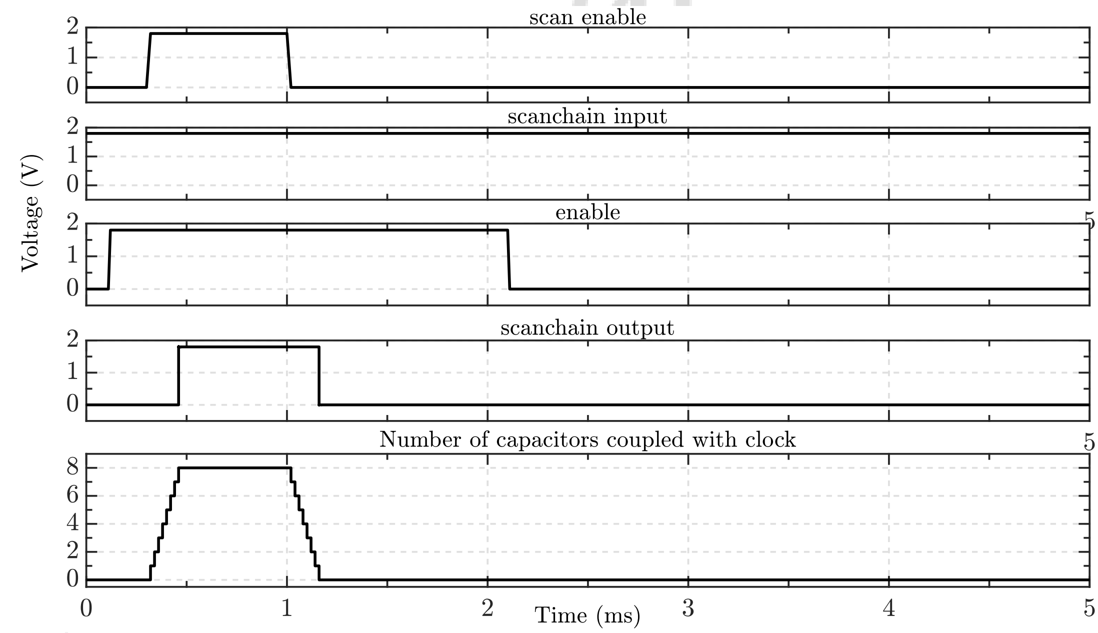
Figure 6.9: Post-layout simulation results of IP-SGFET based accelerometer in open-loop for acceleration of ± 4g at 100 Hz, for circuit in Figure 6.4 designed using SkyWater 130nm technology. The output saturates due to high voltage gain.

Figure 6.9 shows post layout simulation results of integrator schematic shown in Figure 6.3.

Figure 6.10(a) and (b) shows the post-layout simulation of the bi-directional shift register and shift register in scan-mode respectively. The layout was generated through OpenLane. The plot shows various input combinations and the corresponding output signals.



(a) Post-layout simulation results of shift-registers shown in Figure 6.7.



(b) Post-layout simulation results of shift-registers in scan-mode shown in Figure 6.7.

Figure 6.10: Post layout simulation results of digital control logic

The output signals are 0 when reset is low (active low signal). The circuit is enabled, the input (1) shifts towards the right when the shift signal is low and the left (0) when the shift signal is high. The bits are stored and shift operation is disabled when enable = 0. The working of the scan chain is also verified. It operates in scan-mode when scan enable =1.

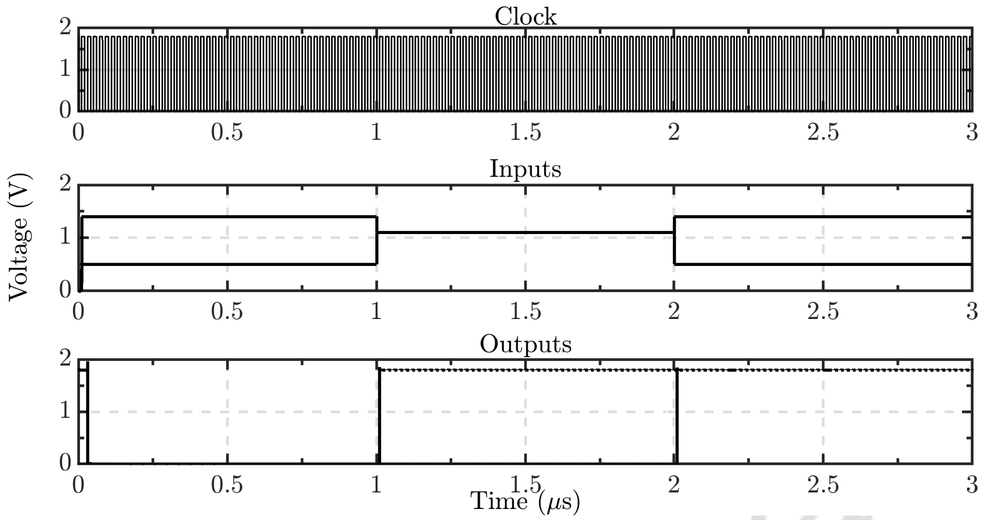


Figure 6.11: Post layout simulation results of window comparator circuit shown in Figure 6.5. Pre-defined threshold levels are V_{up} = 1.1 V and V_{up} = 0.7 V.

Figure 6.11 shows post layout simulation results of window comparator and digital logic block shown in Figure 6.5. The input signal is compared with pre-defined threshold levels that are generated by the voltage divider shown in Figure 6.6.

Post layout simulation of clock generator used to drive charge pump is shown in Figure 6.12.

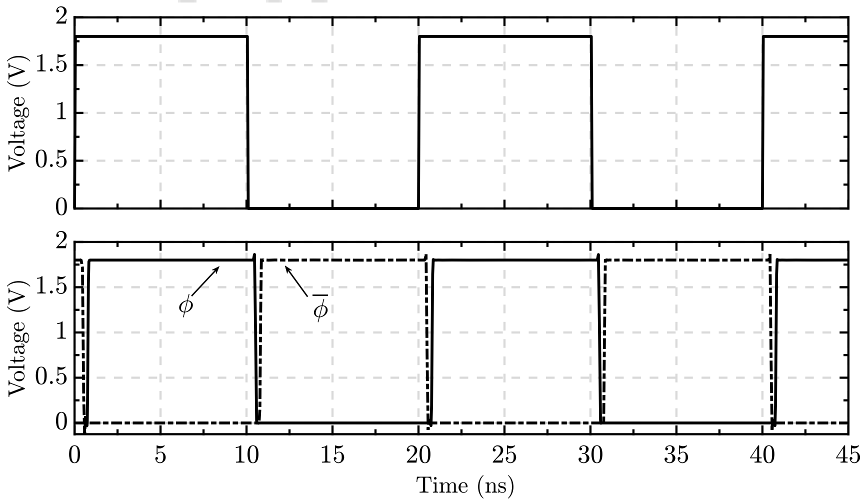
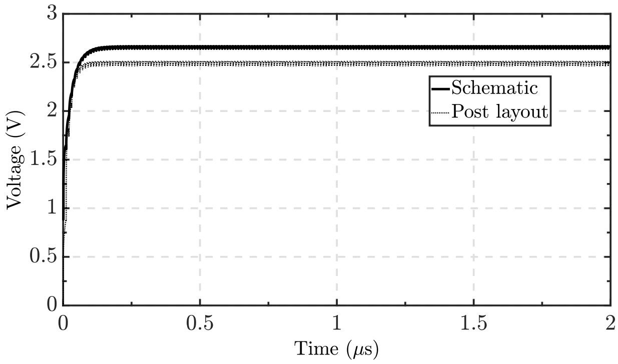
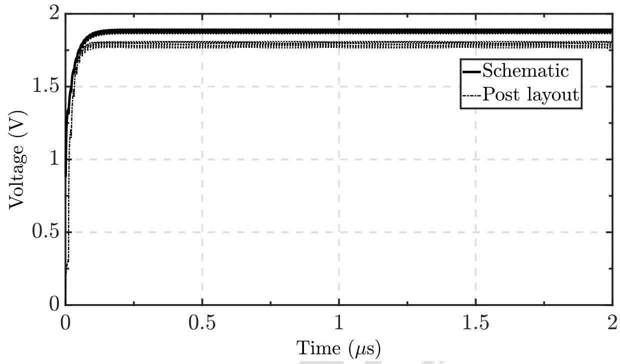


Figure 6.12: Post layout simulation results of the non-overlapping clock generator.

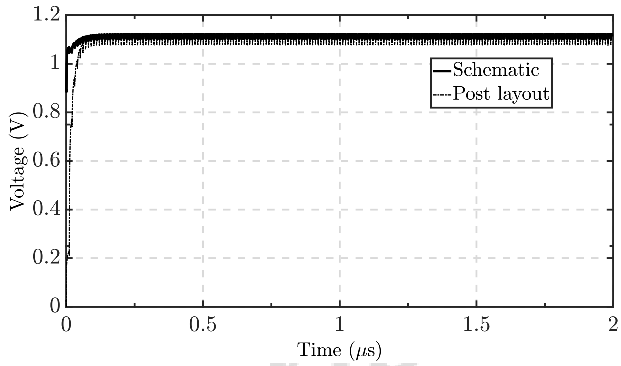
Figures 6.13 and 6.14 show the post-layout simulation results and the schematic simulation results of CP-I and CP-II shown in Figure 3.11.



(a)

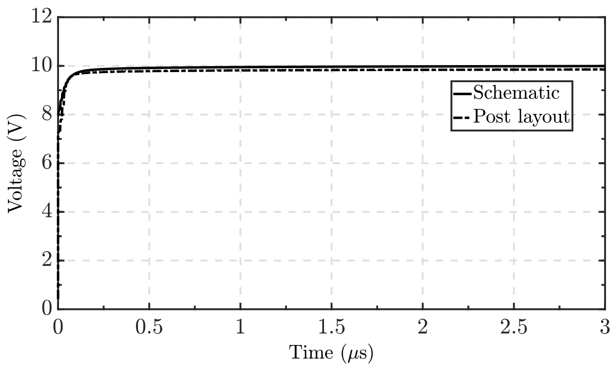


(b)

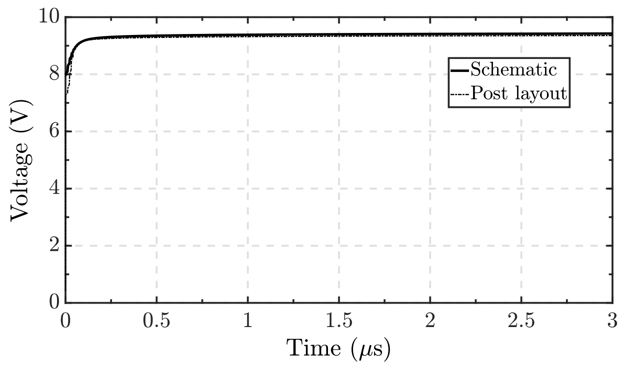


(c)

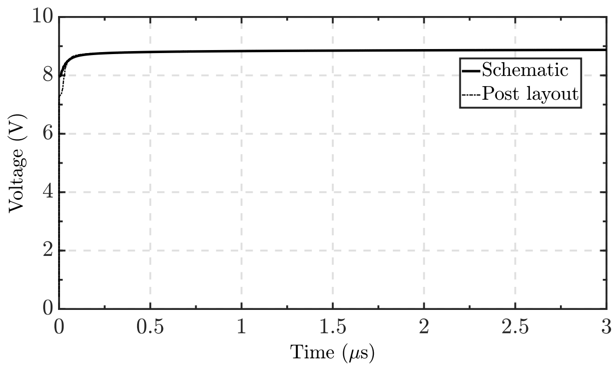
Figure 6.13: Post-layout simulation results of charge pump CP-I shown in Figure 3.11 at different efficiency settings designed in SkyWater 130nm technology at (a) highest, (b) common-mode and (c) least efficiency for an input voltage of 0.9 V.



(a)



(b)



(c)

Figure 6.14: Post-layout simulation results of charge pump CP-II shown in Figure 3.11 at different efficiency settings designed in SkyWater 130nm technology at (a) highest, (b) common-mode and (c) least efficiency for an input voltage of 9 V.

6.4 Test plan and expected outcome

The target application is to use these circuits with an IP-SGFET-based sensor and electrostatic actuator in feedback.

1. We intend to design and test the differential amplifier that interfaces the IP-SGFETs. A provision will be made in the chip to verify the output voltage of the open-loop as shown in Figure 6.15. $V_{\text{int+}}$ and $V_{\text{int-}}$ are outputs of integrator. Output of first test set-up is common-mode voltage as R_{L} kills the gain. This verifies the common-mode level of the circuit. While operating in closed-loop, C_{i} is the integrating capacitor that can be connected externally and adjusted to ensure stability.

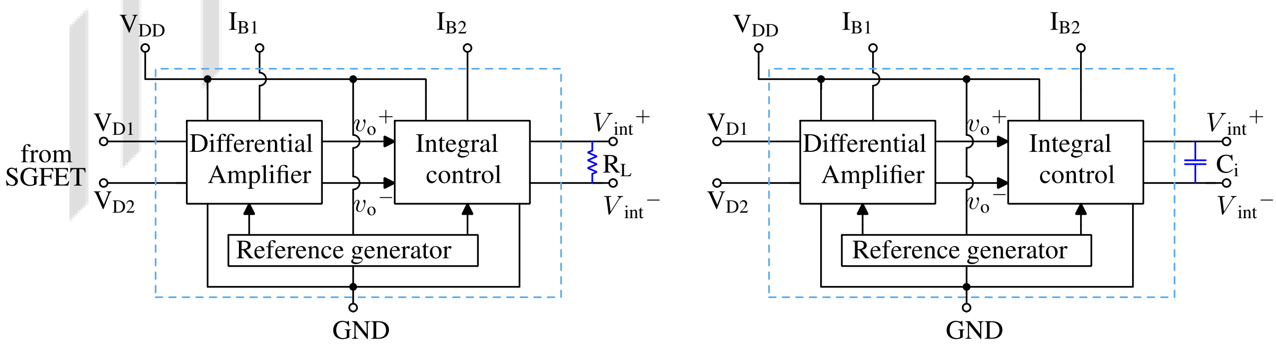


Figure 6.15: Block diagram showing test plan to verify open loop operation of CLIP-SGFET based accelerometer. The pins V_{D1} and V_{D2} are connected to the drains of the SGFET that is external to chip. The reference generator block is shown in Figure 6.6 generates reference voltage for CMFB.

2. Design and test a continuous, digitally reconfigurable charge pump that is needed to drive the electrostatic actuator. The working of the shift register is verified using scan chain. To test the charge pump, the following provisions are made.

   The differential outputs of charge pump are taken out. The common mode feedback circuit for the differential amplifier is inside the chip, but the discharge stage is external as shown in Figure 6.16. The stack of transistors or mutli-level converters were used to design discharge stage to facilitate voltage step down without breakdown of any junctions. But, a very high resistance can also be used to facilitate discharge. The CMFB signal is connected to gate of transistors, M_1/M_2 to control the current. The value of R_{D} is determined by the value of discharge current.

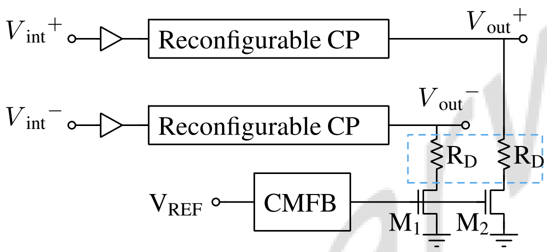


Figure 6.16: Block diagram of test set-up for verifying charge pump output voltage. The resistor, R_{D} along with transistor, M_1/M_2 facilitate discharge of output node. R_{D} is external to the chip.

Chapter 7

Conclusions and Future work

7.1 Conclusions

This work discusses active SGFET-based sensors as a superior alternative to conventional passive capacitive accelerometers and the LUT-based modeling method to design and simulate the device. It discusses design of interface circuits and the simulated results were presented. The designed system has an open-loop sensitivity of 927 mV/g at a supply voltage of 3.3 V.

The closed-loop consisting of an actuator and a reconfigurable high-voltage charge pump is designed for a dynamic sensing range of $\pm$ 4 g operating at a maximum frequency of 100 Hz, and the simulated results were presented. The closed-loop designs uses a differential actuator which requires two reconfigurable charge pumps for generating the actuator drive. The high voltage differential output from the two charge pumps require constant current discharge stages and a common-mode feedback circuit for maintaining accuracy of the closed-loop system. Multi-level converter based discharge stages were designed for the same. Thorough analysis of the reconfigurable charge pump and the discharge stage was done and the findings were discussed. Following this, circuit level design optimization techniques, for the reconfigurable charge pump were discussed, and the simulated results were presented.

Post layout simulation results of all the proposed designs were shown and also the test plan to verify the functionality of the chip was presented in the report.

7.2 Future work

A look-up table-based modeling technique is used to model the behavior of SGFET and the actuator. This device model proposed in this work does not capture the parasitics and noise performance of the SGFET. Better modeling techniques to model the device should be explored.

An actuator model that captures the frequency-dependent behavior of the actuator was presented in this work. The closed-loop operation should be verified and optimized using the new actuator and better device models. The neural network or black box model for a charge pump proposed can be used to reduce the simulation time. After optimization, the model can be replaced with the charge pump, and a final simulation can be performed to verify the results. Quantitative analysis of the system across all PVT corners to be done.

The post-layout simulation results of the designs were presented. These are designed to be implemented and tested to verify the performance of the CLIP-SGFET accelerometer.

P-I-N diodes were implemented and tested to verify the breakdown voltage levels. The results were discussed in the report. These P-I-N diodes can be used to design the charge pumps and explored to generate higher voltages to further extend the maximum voltage range that the charge pump can generate. These will be present in the feedback path. So, techniques to improve the latency of P-I-N diode-based charge pumps are to be explored.

Appendix A

P-I-N diodes for charge pumps with output exceeding substrate junction breakdown voltages

As discussed earlier, the breakdown voltage of n-well/p-substrate junction diode limits the output voltage of the charge pump(around 12 V in UMC180 nm technology), shown in Figure 3.10 (a). To generate a voltage beyond 12 V in UMC180 nm, the deep n-well is connected to a higher potential, V_P in CP-II (all NMOS stage), as shown in the Figure 3.10 (b), extending the output voltage range of the charge pump to a maximum of 17 V. However, the n+/p-well junction diode limits the output voltage.

To further extend the output voltage range, P-I-N diodes can be used instead of relying on the reverse junction. P-I-N diodes are fabricated in polysilicon layer as reported in [48, 54]. Now the maximum voltage is limited by the dielectric breakdown limit of the polysilicon, which depends on the thickness of the polysilicon layer. The thickness of the polysilicon is 40 nm and breakdown voltage is around 400 V in UMC180 nm technology. The charge pump is based on the conventional Dickson charge pump but with P-I-N diodes and capacitors as shown in Figure A.1.

A.1 P-I-N Diode

P-I-N diode is a semiconductor diode which has an undoped intrinsic material sandwiched between the P-type and N-type semiconductor. The symbol of P-I-N diode is shown below in

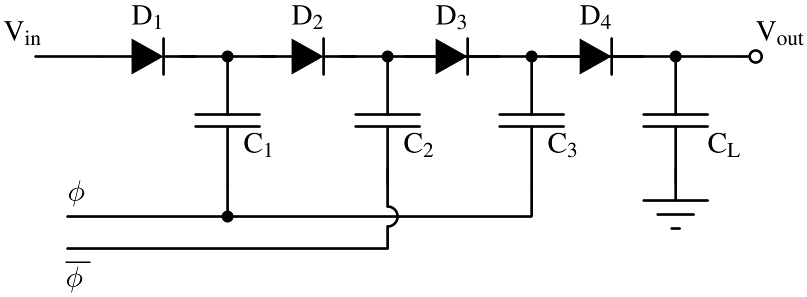


Figure A.1: Schematic of conventional Dickson charge pump which uses P-I-N diodes and capacitors. ϕ and $\overline{\phi}$ are non-overlapping clock signals.

Figure A.2. It has 3 regions, namely

- **P**-type region
- **I** region
- **N**-type region

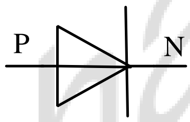


Figure A.2: Symbol of P-I-N diode

Due to the presence of intrinsic layer between the P-type and N-type semiconductor, the distance between the doped regions increases, capacitance reduces and the diode imparts ability of tolerate high reverse voltages without leading to breakdown.

A.1.1 Device structure

The diodes are fabricated in standard UMC180 nm triple-well process and image of layout is shown in Fig. A.3. The die consumes an active area of 1.11 mm x 1.17 mm. Figure A.4 (a) and (b) shows the top-view and device structure of P-I-N diode, respectively. The device is a polysilicon diode with intrinsic polysilicon doped with N-type and P-type materials to form diodes. D signifies the thickness of polysilicon layer where N-type and P-type polysilicon regions are present and L distance between P-type and N-type polysilicon. In these diodes, the thickness of the oxide,D is 40 nm. The spacing between the P-type and N-type regions, L various between 0.3 μm to 0.7 μm, as specified in Table A.2.

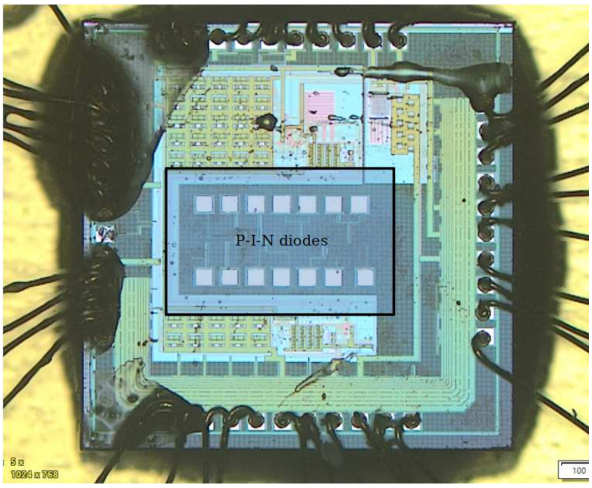
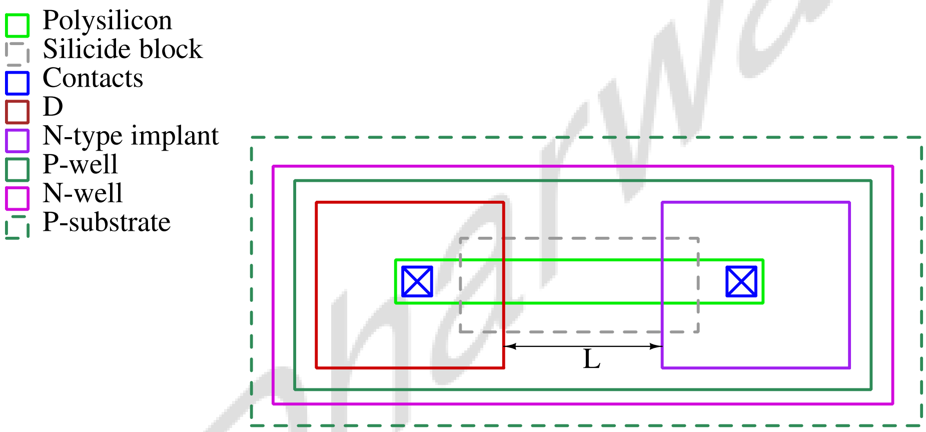
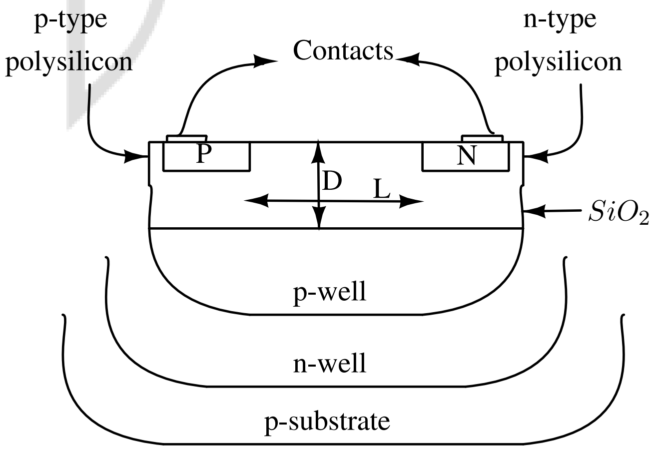


Figure A.3: Die micrograph of the IC with position of diodes marked on it.



(a) Top-view of P-I-N diode.



(b) Cross-section of P-I-N diode

Figure A.4: Device structure of P-I-N diode.

The inputs were applied to measure various junction and dielectric breakdown potential. The deep n-well layer forms two junctions, p-well/n-well junction and n-well/p-substrate junction as shown in Figure A.4 (b). The p-well and n-well can be shorted preventing forward bias of that junction diode. A reverse voltage was applied across the terminals of diode formed by the n-well/p-substrate junction and breakdown of the junction was observed at 15.1 V. The dielectric breakdown voltage depends on the dielectric strength and thickness of the oxide. Dielectic strength of gate oxide layer is around 10 MV/cm. The breakdown voltage was calculated for this oxide layer and it is around 400 V.

A.1.2 Pin descriptions

The diodes fabricated in UMC180 technology and packaged in a DIL-18 package. Top view of the IC is shown in Figure A.5(Image not to scale). The pin description of the IC is given in Table A.1.

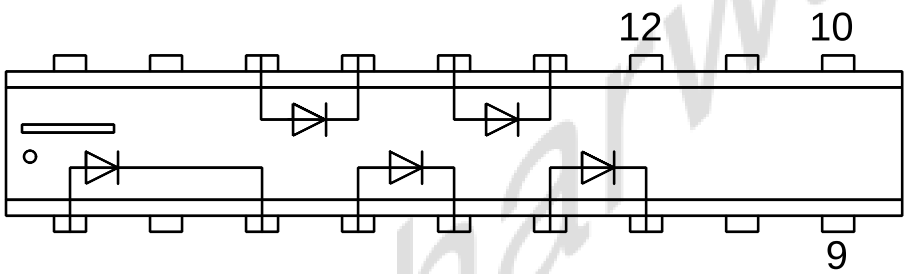


Figure A.5: P-I-N diode packaging.

A total of 5 P-I-N diodes are present in each IC. Table A.1 specifies the names and functions of the pins of IC in Figure A.5.

The details on P-type, N-type polysilicon material and length of intrinsic region are discussed in Table A.2.

Pin no.	Pin name	Description
1	IN1	Input of Diode-1
2	NC	No internal connection
3	OUT1	Output of Diode-1
4	IN2	Input of Diode-2
5	OUT2	Output of Diode-2
6	IN3	Input of Diode-3
7	OUT3	Output of Diode-3
8	NC	No internal connection
9	N-WELL	Deep n-well
10	SUBSTRATE	P-substrate
11	NC	No internal connection
12	P-WELL	P-well
13	OUT4	Output of Diode-4
14	IN4	Input of Diode-4
15	OUT5	Output of Diode-5
16	IN5	Input of Diode-5
17	NC	No internal connection
18	NC	No internal connection

Table A.1: Pin description of IC shown in Fig. A.5.

A.2 Diode characterization

P-I-N diode obeys the Shockley diode current equation governed by Eq. A.1.

$$I_D = I_S \left(e^{\frac{qV_D}{\eta kT}} - 1 \right) \tag{A.1}$$

where, q is the charge of electron, k is Boltzmann constant, T is temparature in Kelvin and I_D and V_D are diode current and voltage, respectively. .

A diode allows current flow in one direction. A positive voltage is applied to the p-region and a negative voltage to the n-region of the diode, this reduces the potential barrier. It is called the forward bias condition. The majority carriers from the p-region diffuse over and reach the n-side through the intrinsic region and electrons from the n-side move over to the p-side of the

Diode	p-type material	n-type material	Length of intrinsic layer
1	n-	n+	0.3 μm
2	p+	p-	0.3 μm
3	p	n	0.3 μm
4	p	n	0.5 μm
5	p	n	0.7 μm

Table A.2: Details on p-type, n-type materials and intrinsic layer lengths of diodes shown in IC in Fig. A.5.

junction. This process continues as long as the voltage is applied. Thus, in this mode, the diode carries a large current. The test set-up to characterize the diode in forward bias is shown in Figure A.6. A voltage ranging from 0 V to 4 V with current limit set to 50 mA, is applied across the terminals of the diode and current flowing through the diode is measured.

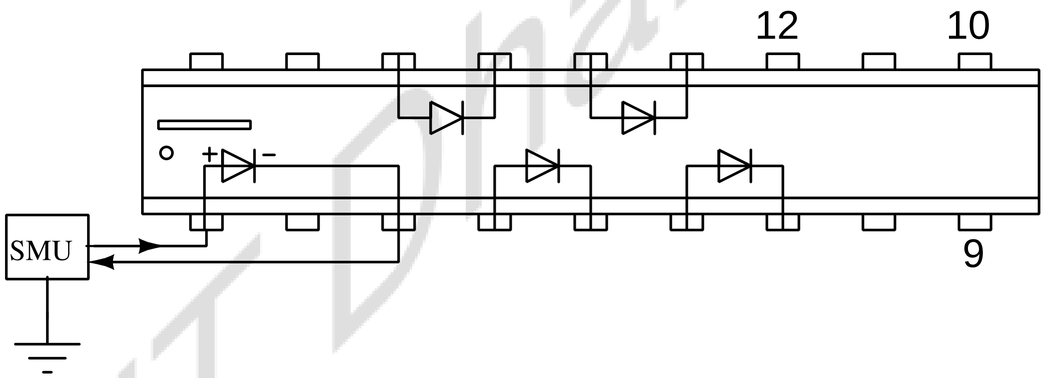


Figure A.6: Test set-up to characterize diode behaviour in forward bias condition.

When the diode is in reverse bias (n region of diode is at higher potential), barriers increases offering a very high resistance. As external voltage is placed across the diode with the same polarity as the built-in potential/barrier potential, the depletion zone continues to act as an insulator, preventing any significant electric current flow. Thus, a very small current called reverse saturation current flows through the diode when it is in reverse bias condition. As the intrinsic region is wide, the capacitance is low. The testing set-up is shown in Figure A.7. A voltage ranging from 0 V to -4 V is applied across the terminals of the diode and current flowing through the diode is measured.

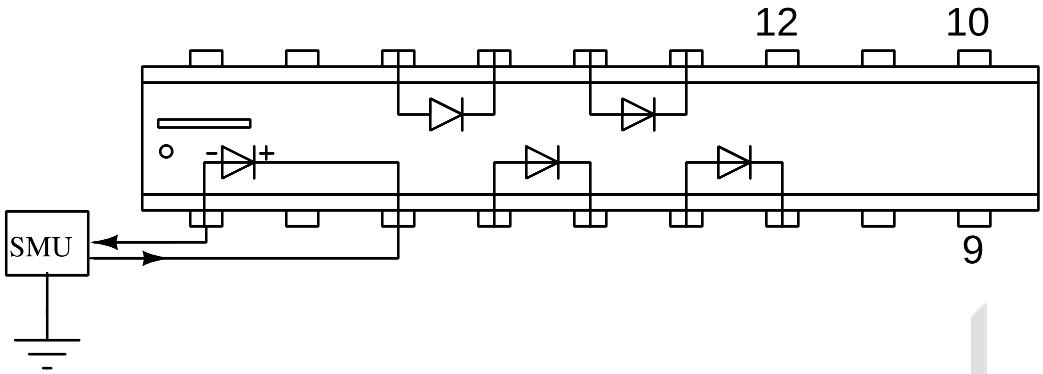
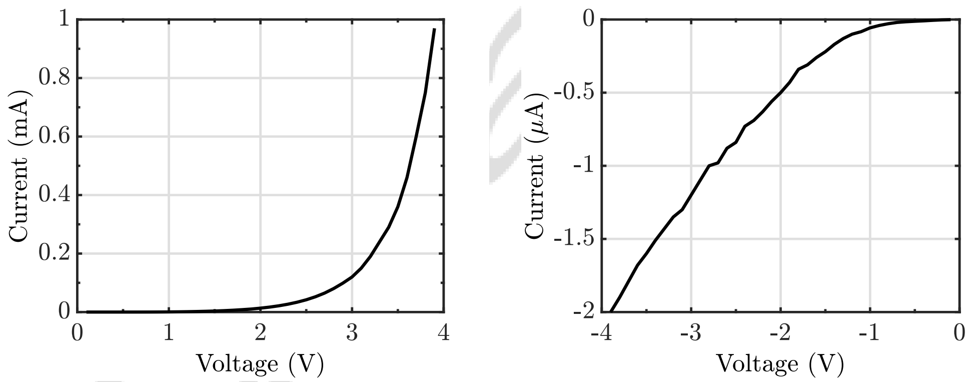


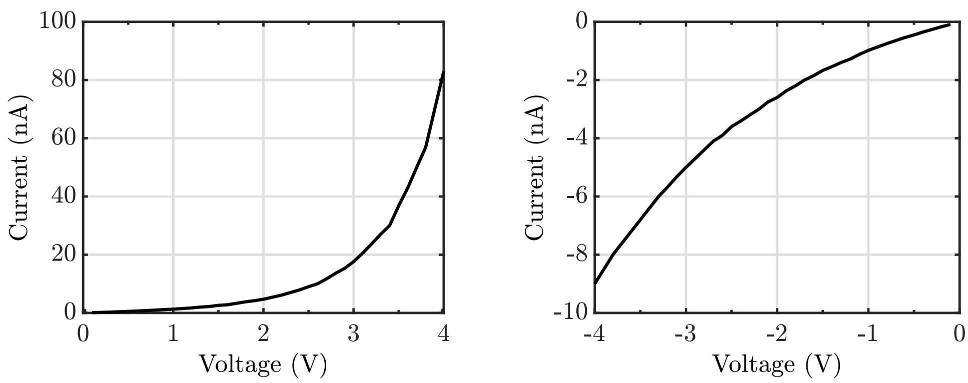
Figure A.7: Test set-up to characterize diode behaviour in forward bias condition.

A.2.1 Measured results

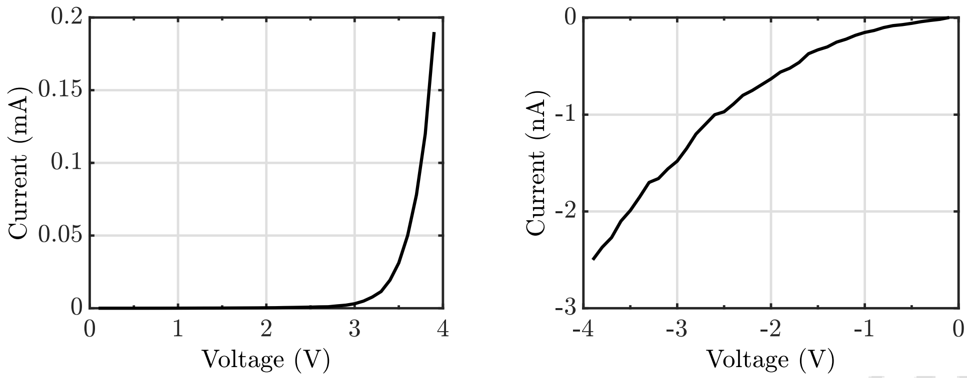
Figure A.6, Figure A.7 shows circuit diagram of test circuit. A voltage ranging -4 V to 4 V with a current limit of 50 mA, is applied across the terminals of the diode and the current flowing through the diode is measured. The I-V characteristics of the diodes are shown in graphs below.



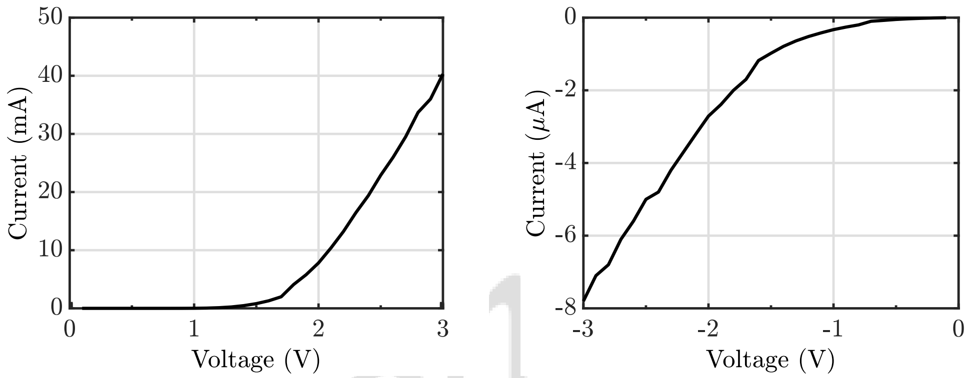
(a) Forward and reverse characteristics of diode-1 between pins 1 & 2.



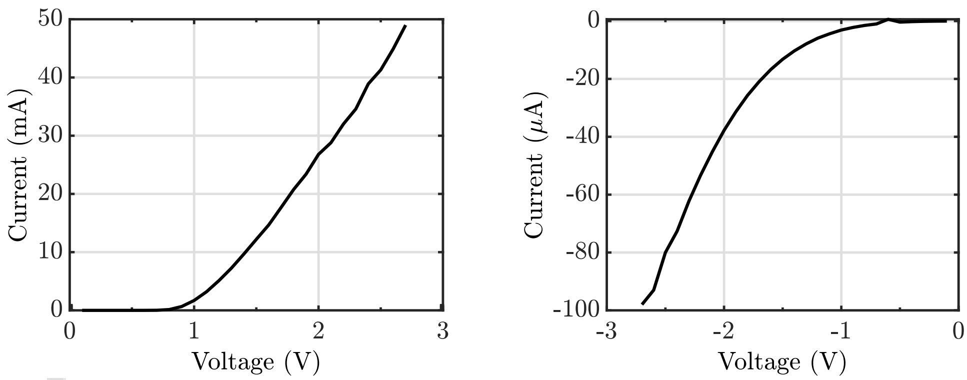
(b) Forward and reverse characteristics of diode-2 between pins 4 & 5.



(c) Forward and reverse characteristics of diode-3 between pins 6 & 7.



(d) Forward and reverse characteristics of diode-4 between pins 13 & 14.

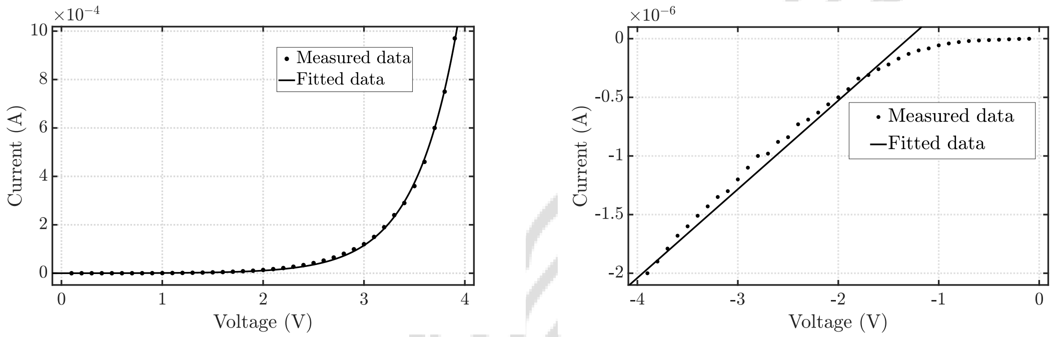


(e) Forward and reverse characteristics of diode-5 between pins 15 & 16.

Figure A.8: Measured I-V characteristics for polysilicon P-I-N diodes shown in Figure A.3. Forward characteristics have an exponential behaviour and reverse characteristics are similar to leaky resistor(straight line), obeying the Shockley diode current equation.

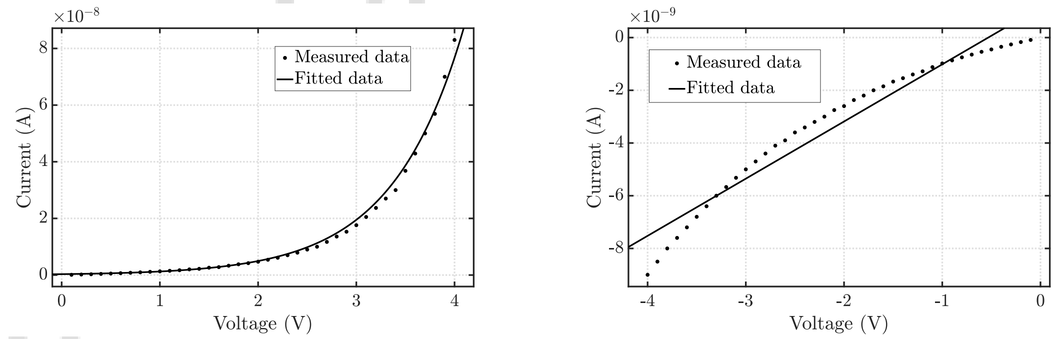
A.2.2 Curve fitting

Curve fitting is performed to analyze the measured current and voltage values of P-I-N diodes and verify the closeness of measured results to the expected behavior governed by Eq. A.1. The curve fitting tool of MATLAB is used to fit the measured results of diode current-voltage behaviour as algebraic equation, infer the measured values and obtain forward current and reverse current approximation for measured I-V characteristics of the diodes. Forward current is approximated as an exponential curve, $y = ae^{bx}$, with x,y being measured voltage and current values respectively and reverse current is approximated as a straight line with intercept(a linear fit), $y = a + bx$ can be done as there is only leakage current flowing through it. The plots below show the measured data and curve fit.



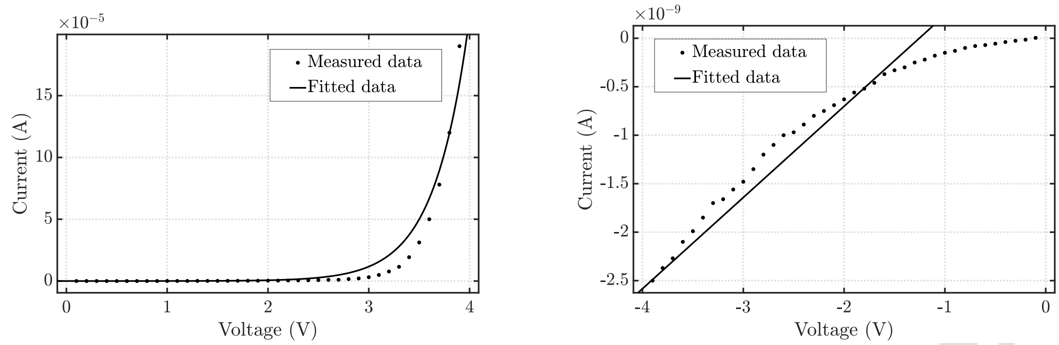
(a) Forward current approximation : $I_D = (9.784 \times 10^{-8})e^{2.356V_D}$

Reverse current approximation : $I_D = 7.561 \times 10^{-7}V_D + (9.846 \times 10^{-7})$.



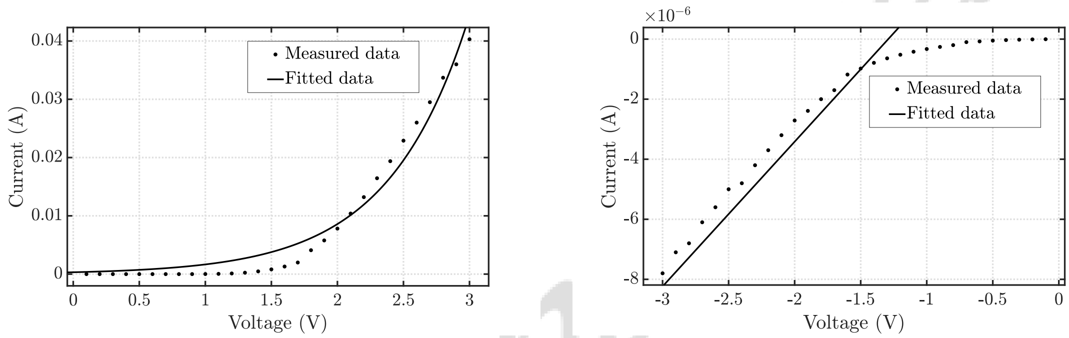
(b) Forward current approximation : $I_D = (3.202 \times 10^{-10})e^{1.37V_D}$

Reverse current approximation : $I_D = 2.45 \times 10^{-9}V_D + (1.72 \times 10^{-9})$.



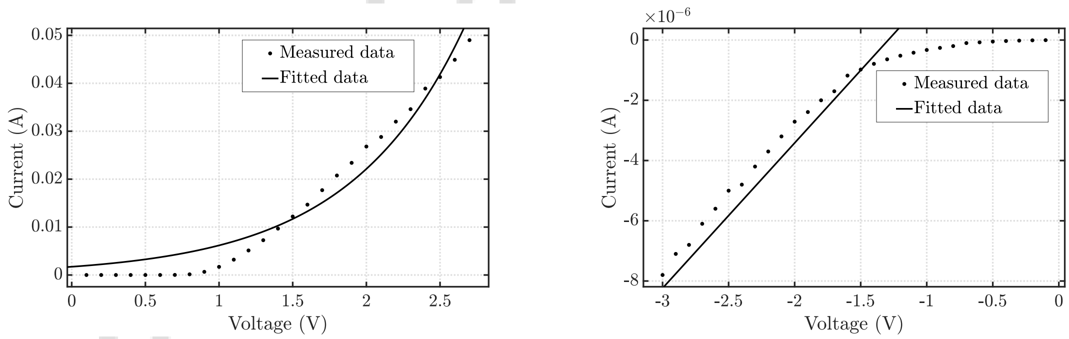
(c) Forward current approximation : $I_D = (1.906 \times 10^{-9})e^{2.907V_D}$

Reverse current approximation : $I_D = 8.57 \times 10^{-9}V_D + (1.18 \times 10^{-9})$.



(d) Forward current approximation : $I_D = (3.214 \times 10^{-4})e^{1.643V_D}$

Reverse current approximation : $I_D = 4.82 \times 10^{-6}V_D + (6.23 \times 10^{-7})$.



(e) Forward current approximation : $I_D = (1.725 \times 10^{-3})e^{1.276V_D}$

Reverse current approximation : $I_D = 6.709 \times 10^{-5}V_D + (8.218 \times 10^{-5})$.

Figure A.9: Plots showing forward and reverse current approximation for the measured I-V characteristics of P-I-N diodes. The approximation was done using MATLAB curve fitting tool. The results obtained verify the diode behaviour.

A.2.3 Evaluation of fit

After curve fitting, one way to evaluate the goodness of the fit is through visual evaluation. The curve-fitting toolbox of MATLAB offers methods to evaluate the goodness of fit. The Table A.3 contains the details on RMS error of the fit.

Diode	Biasing	RMSE value
1	Forward	5.427×10^{-6}
	Reverse	3.281×10^{-7}
2	Forward	1.602×10^{-9}
	Reverse	6.842×10^{-10}
3	Forward	8.594×10^{-6}
	Reverse	4.422×10^{-10}
4	Forward	2.171×10^{-3}
	Reverse	2.355×10^{-6}
5	Forward	3.362×10^{-3}
	Reverse	3.169×10^{-5}

Table A.3: RMS error of curve fitting.

A.3 Summary

The junction and dielectric breakdown voltages of P-I-N diodes fabricated in UMC180 nm technology were measured. Also, the forward and reverse behavior of the diode was measured, and closeness to the expected behavior was verified.

References

[1] S. Senturia, *Microsystem Design*, 2002.

[2] X. Zhang, Z. Li, S. Zou, G. Wang, H. Liu, Z. Cheng, L. Tu, and S. Yang, "Noise Analysis and Optimization of Front-End Circuit in Ultra-High Precision Capacitive Transducer," *IEEE Sensors Journal*, vol. 22, no. 13, pp. 13 005–13 011, 2022.

[3] P. Martha, N. Kadayinti, and V. Seena, "A CMOS-MEMS Accelerometer With U-Channel Suspended Gate SOI FET," *IEEE Sensors Journal*, vol. 21, no. 9, pp. 10 465–10 472, 2021.

[4] N. Abelé, "Design and fabrication of suspended-gate MOSFETs for MEMS resonator, switch and memory applications," EPFL, Tech. Rep., 2007.

[5] I.-H. Song and P. K. Ajmera, "Differential laterally movable gate FETs (LMGFETs) as a position sensor," in *Smart Structures and Materials 2004: Smart Electronics, MEMS, BioMEMS, and Nanotechnology*, V. K. Varadan, Ed., vol. 5389, International Society for Optics and Photonics. SPIE, 2004, pp. 267 – 273. [Online]. Available: https://doi.org/10.1117/12.539782

[6] C. Senft, P. Iskra, and I. Eisele, "Theory and application of suspended gate FET gas sensors," *Solid State Gas Sensors-Industrial Application*, pp. 79–112, 2012.

[7] K. Tamersit, M. Kotti, and M. Fakhfakh, "A new pressure microsensor based on dual-gate graphene field-effect transistor with a vertically movable top-gate: Proposal, analysis, and optimization," *AEU - International Journal of Electronics and Communications*, vol. 124, p. 153346, 2020. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1434841120313984

[8] W. Gao, C. Jia, Z. Jiang, X. Zhou, L. Zhao, and D. Sun, “The Design and Analysis of a Novel Micro Force Sensor Based on Depletion Type Movable Gate Field Effect Transistor,” *Journal of Microelectromechanical Systems*, vol. 28, no. 2, pp. 298–310, 2019.

[9] S. Aoyagi, M. Suzuki, J. Kogure, T. Kong, R. Taguchi, T. Takahashi, S. Yokoyama, and H. Tokunaga, “Accelerometer using MOSFET with movable gate electrode: Electroplating thick nickel proof mass on flexible Parylene beam for enhancing sensitivity,” in *2011 16th International Solid-State Sensors, Actuators and Microsystems Conference*, 2011, pp. 2030–2033.

[10] P. Martha, N. Kadayinti, and V. Seena, “A CMOS-MEMS Accelerometer With U-Channel Suspended Gate SOI FET,” *IEEE Sensors Journal*, vol. 21, no. 9, pp. 10 465–10 472, 2021.

[11] ——, “CMOS-MEMS Accelerometer With Stepped Suspended Gate FET Array: Design and Analysis,” *IEEE Transactions on Electron Devices*, vol. 68, no. 10, pp. 5133–5141, 2021.

[12] M. Bao, *Analysis and Design Principles of MEMS Devices*, 2005.

[13] J. I. Seeger and S. B. Crary, “Stabilization of electrostatically actuated mechanical devices,” in *Solid State Sensors and Actuators*, vol. 2, 1997, pp. 1133–1136.

[14] I.-H. Song and P. K. Ajmera, “A Laterally Movable Gate Field Effect Transistor,” *Journal of Microelectromechanical Systems*, vol. 18, no. 1, pp. 208–216, 2009.

[15] T. C. Thomas, V. S. Reddy, and V. Seena, “Novel CMOS-MEMS accelerometer with fingered differential LMGFET architecture with linear response,” in *2017 IEEE 12th Nanotechnology Materials and Devices Conference (NMDC)*, 2017, pp. 125–126.

[16] W. Gao, L. Zhao, Z. Jiang, Y. Xia, X. Guo, Z. Zhao, Y. Zhao, and D. Sun, “A novel MEMS force sensor based on laterally movable gate array field effect transistor (LMGAFET),” in *2017 IEEE 12th International Conference on Nano/Micro Engineered and Molecular Systems (NEMS)*. IEEE, 2017, pp. 723–727.

[17] K. M. D. J. e. a. Algamili, A.S., “A Review of Actuation and Sensing Mechanisms in MEMS-Based Sensor Devices,” *Nanoscale Research Letters*, vol. 16, no. 16, 2021.

[18] V. Pachkawade, “Transduction in MEMS/NEMS based actuation and sensing,” *IEEE transactions on Ultrasonics, Ferroelectrics and Frequency control*, 2023.

[19] Q. X. Sijie Yang, “A review on actuation and sensing techniques for MEMS-based microgrippers,” *Journal of Micro-Bio Robotics*, vol. 13, pp. 1–14, 2017.

[20] J.-O. Dennis, A.-Y. Ahmed, and M.-H. Khir, “Fabrication and characterization of a CMOS-MEMS humidity sensor,” *Sensors*, vol. 15, no. 7, pp. 16 674–16 687, 2015.

[21] S. Siahpour, M. M. Zand, and M. Mousavi, “Dynamics and vibrations of particle-sensing MEMS considering thermal and electrostatic actuation,” *Microsystem Technologies*, vol. 24, pp. 1545–1552, 2018.

[22] R. de Oliveira Hansen, M. Mátéfi-Tempfli, R. Safonovs, J. Adam, S. Chemnitz, T. Reimer, B. Wagner, W. Benecke, and S. Mátéfi-Tempfli, “Magnetic films for electromagnetic actuation in MEMS switches,” *Microsystem Technologies*, vol. 24, pp. 1987–1994, 2018.

[23] W.-M. Zhang, H. Yan, Z.-K. Peng, and G. Meng, “Electrostatic pull-in instability in MEMS/NEMS: A review,” *Sensors and Actuators A: Physical*, vol. 214, pp. 187–218, 2014. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924424714002088

[24] A. Anil and R. K. Sharma, “A high efficiency charge pump for low voltage devices,” *International Journal of VLSI Design & Communication Systems*, vol. 3, no. 3, p. 43, 2012.

[25] J. Dickson, “On-chip high-voltage generation in MNOS integrated circuits using an improved voltage multiplier technique,” *IEEE Journal of Solid-State Circuits*, vol. 11, no. 3, pp. 374–378, 1976.

[26] O. Wong, H. Wong, W. Tam, and C. Kok, “A comparative study of charge pumping circuits for flash memory applications,” *Microelectronics Reliability*, vol. 52, no. 4, pp. 670–687, 2012, advances in non-volatile memory technology. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0026271411004495

[27] J.-T. Wu and K.-L. Chang, “MOS charge pumps for low-voltage operation,” *IEEE Journal of Solid-State Circuits*, vol. 33, no. 4, pp. 592–597, 1998.

[28] A. Anil and R. K. Sharma, "A high efficiency charge pump for low voltage devices," *International Journal of VLSI Design & Communication Systems*, vol. 3, no. 3, p. 43, 2012.

[29] N.-X. Huang, M.-S. Shiau, Z.-H. Hsieh, H.-C. Wu, and D.-G. Liu, "Improving the efficiency of mixed-structure charge pumps by the multi-phase technique," in *2010 5th IEEE Conference on Industrial Electronics and Applications*. IEEE, 2010, pp. 358–363.

[30] F. Ueno, T. Inoue, I. Oota, and I. Harada, "Emergency power supply for small computer systems," *1991., IEEE International Sympoisum on Circuits and Systems*, pp. 1065–1068 vol.2, 1991.

[31] C.-Y. Tsui, H. Shao, W.-H. Ki, and F. Su, "Ultra-low voltage power management circuit and computation methodology for energy harvesting applications," in *Asia and South Pacific Conference on Design Automation, 2006.*, 2006, pp. 2 pp.–.

[32] Y. Allasasmeh and S. Gregori, "A performance comparison of Dickson and Fibonacci charge pumps," in *2009 European Conference on Circuit Theory and Design*. IEEE, 2009, pp. 599–602.

[33] L. Gobbi, A. Cabrini, and G. Torelli, "A discussion on exponential-gain charge pump," *2007 18th European Conference on Circuit Theory and Design*, pp. 615–618, 2007.

[34] M. Mihara, Y. Terada, and M. Yamada, "Negative heap pump for low voltage operation flash memory," in *1996 Symposium on VLSI Circuits. Digest of Technical Papers*, 1996, pp. 76–77.

[35] R. Arona, E. Bonizzoni, F. Maloberti, and G. Torelli, "Heap charge pump optimisation by a tapered architecture," in *2005 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2005, pp. 1903–1906 Vol. 2.

[36] A. H. Alameh and F. Nabki, "A 0.13 μm CMOS Dynamically Reconfigurable Charge Pump for Electrostatic MEMS Actuation," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 25, no. 4, pp. 1261–1270, 2017.

[37] F. Pan and T. Samaddar, "Charge pump circuit design," *(No Title)*, 2006.

[38] L. F. Rahman, M. Marufuzzaman, L. Alam, and M. B. Mokhtar, "Design Topologies of a CMOS Charge Pump Circuit for Low Power Applications," *Electronics*, vol. 10, no. 6, 2021. [Online]. Available: https://www.mdpi.com/2079-9292/10/6/676

[39] A. Ballo, A. D. Grasso, and G. Palumbo, "A Review of Charge Pump Topologies for the Power Management of IoT Nodes," *Electronics*, vol. 8, no. 5, 2019. [Online]. Available: https://www.mdpi.com/2079-9292/8/5/480

[40] P. Karuppanan, K. Khan, and S. R. Ghosh, "Dynamic gate and substrate control charge pump circuits: a review," *Analog Integrated Circuits and Signal Processing*, vol. 83, pp. 257–270, 2015.

[41] D.-H. Lee, D. Kim, H.-J. Song, and K.-S. Min, "A modified Dickson charge pump circuit with high efficiency and high output voltage," *IEICE transactions on electronics*, vol. 91, no. 2, pp. 228–231, 2008.

[42] T. Yim, C. Lee, and H. Yoon, "A high speed modified dickson charge pump," in *2021 IEEE International Symposium on Circuits and Systems (ISCAS)*. IEEE, 2021, pp. 1–5.

[43] A. H. Alameh, A. Bouchami, and F. Nabki, "A wide voltage range charge pump in 0.13 μm CMOS for biasing of MEMS resonators," in *2016 IEEE International Conference on Electronics, Circuits and Systems (ICECS)*, 2016, pp. 165–168.

[44] P. Martha, K. M. Ganga, A. Sebastian, V. Seena, and N. Kadayinti, "A Closed-Loop In-Plane Movable Suspended Gate FET (CLIP-SGFET) Sensor With a Dynamically Reconfigurable Charge Pump," *IEEE Sensors Journal*, vol. 22, no. 22, pp. 21 550–21 560, 2022.

[45] V. Seena, S. Anju, and K. Naveen, "Closed Loop In-Plane Movable Suspended Gate FET based Accelerometer and Fabrication Method Thereof," Patent, Oct. 4, 2023, Patent no. 456720 (IP India).

[46] P. Martha, A. Sebastian, V. Seena, and N. Kadayinti, "A Technique for Modeling and Simulating Transistor Based MEMS Sensors," in *2021 IEEE International Symposium on Inertial Sensors and Systems (INERTIAL)*, 2021, pp. 1–2.

[47] M. Daliri and M. Maymandi-Nejad, "A 0.8-V 420 nW CMOS switched-opamp switched-capacitor pacemaker front-end with a new continuous-time CMFB," in *2008 15th IEEE International Conference on Electronics, Circuits and Systems*, 2008, pp. 758–761.

[48] Y. Ismail, H. Lee, S. Pamarti, and C.-K. K. Yang, "A 34 V charge pump in 65nm bulk CMOS technology," in *2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC)*, 2014, pp. 408–409.

[49] A. H. Alameh, A. Robichaud, and F. Nabki, "A reconfigurable charge pump in 0.13 μm CMOS for agile MEMS actuation," in *2014 21st IEEE International Conference on Electronics, Circuits and Systems (ICECS)*, 2014, pp. 670–673.

[50] X. Zeng, X. Zhang, L. Yao, and N. Xue, "A 15.4V Fully-Integrated Energy-Efficient Pulse Generator in Standard 0.18 μmm CMOS," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 68, no. 6, pp. 1812–1816, 2021.

[51] A. Stillwell, E. Candan, and R. C. Pilawa-Podgurski, "Active voltage balancing in flying capacitor multi-level converters with valley current detection and constant effective duty cycle control," *IEEE Transactions on Power Electronics*, vol. 34, no. 11, pp. 11 429–11 441, 2019.

[52] Z. Ye, Y. Lei, W.-C. Liu, P. S. Shenoy, and R. C. N. Pilawa-Podgurski, "Improved Bootstrap Methods for Powering Floating Gate Drivers of Flying Capacitor Multilevel Converters and Hybrid Switched-Capacitor Converters," *IEEE Transactions on Power Electronics*, vol. 35, no. 6, pp. 5965–5977, 2020.

[53] P. Lingras, M. Zhong, and S. Sharma, *Evolutionary Regression and Neural Imputations of Missing Values*, 01 1970, vol. 226, pp. 151–163.

[54] Y. Ismail, H. Lee, S. Pamarti, and C.-K. K. Yang, "A 36-V 49% efficient hybrid charge pump in nanometer-scale bulk CMOS technology," *IEEE Journal of Solid-State Circuits*, vol. 52, no. 3, pp. 781–798, 2017.

List of Publications

[1] Pramod Martha, **K. M. Ganga***, Anju Sebastian , V. Seena, and Naveen Kadayinti, “A Closed-Loop In-Plane Movable Suspended Gate FET (CLIP-SGFET) Sensor With a Dynamically Reconfigurable Charge Pump”, *IEEE Sensors Journal,* 2022, vol. 22, no. 22, pp. 21550-21560. (***Equal contribution as first author)**

[2] Journal paper titled ”A Closed-Loop In-Plane Movable Suspended Gate FET (CLIP-SGFET) Sensor With a Dynamically Reconfigurable Charge Pump” presented at the *IEEE APSCON 2024* conference, under the Journal-Conference Synergy initiative of *IEEE Sensors Council*.

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